

ECSE 4560/ 6560

Modern Communication Systems

Lecture 3: Wireless Systems Overview

Part 2

Ish Jain, Fall 2025



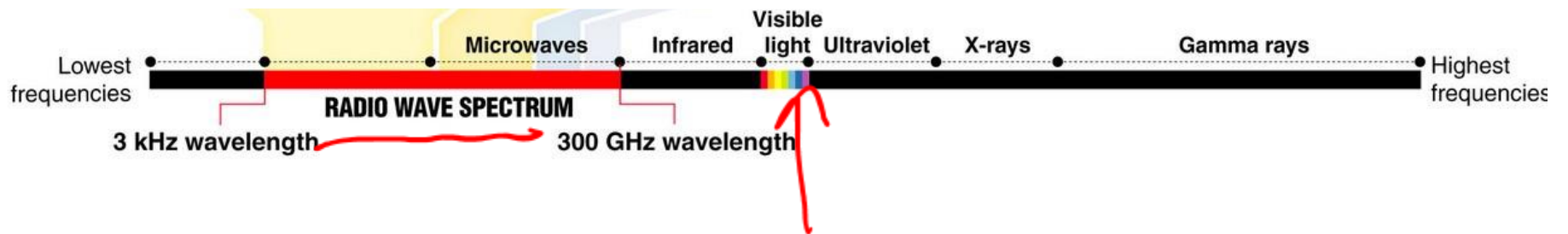
Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- GradeScope for assignment submission (You will be added)
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
 - Single attempt; Don't forget to submit, o/w not graded.
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Capacity**
 - 80% attendance mandatory, In-class participation 5%
- Projects
 - A list of papers will be posted
 - Group formation due 9/12
- HW1/Lab1 to be posted, due Friday 9/19

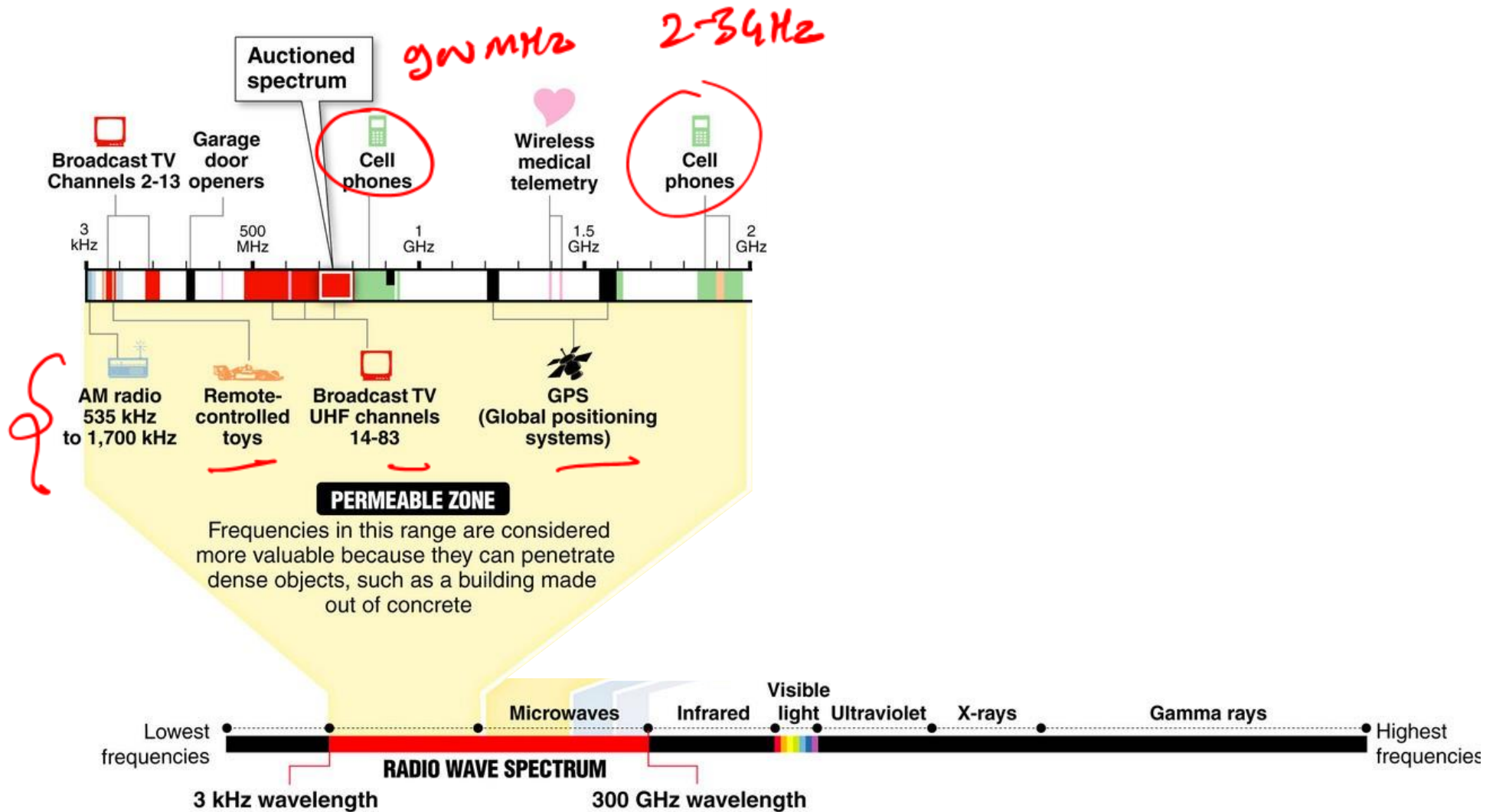
Today's Class: Wireless Overview

- Baseband/Passband Communication
- Up-Conversion and Down conversion
- Modulation
- What is SNR?
- Higher order modulation and its relation to SNR
- BER-SNR curves
- Capacity

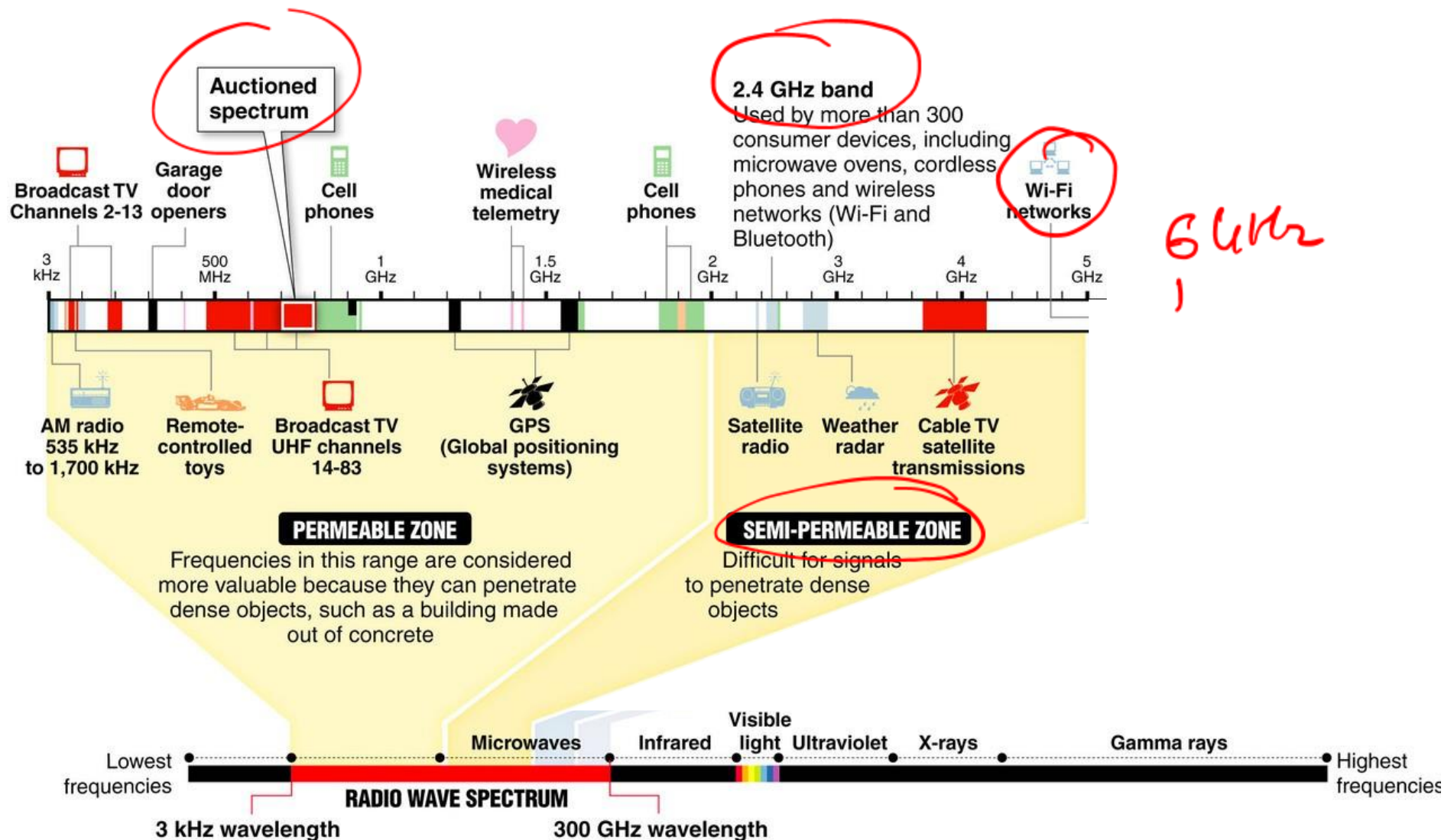
The Wireless Spectrum



The Wireless Spectrum

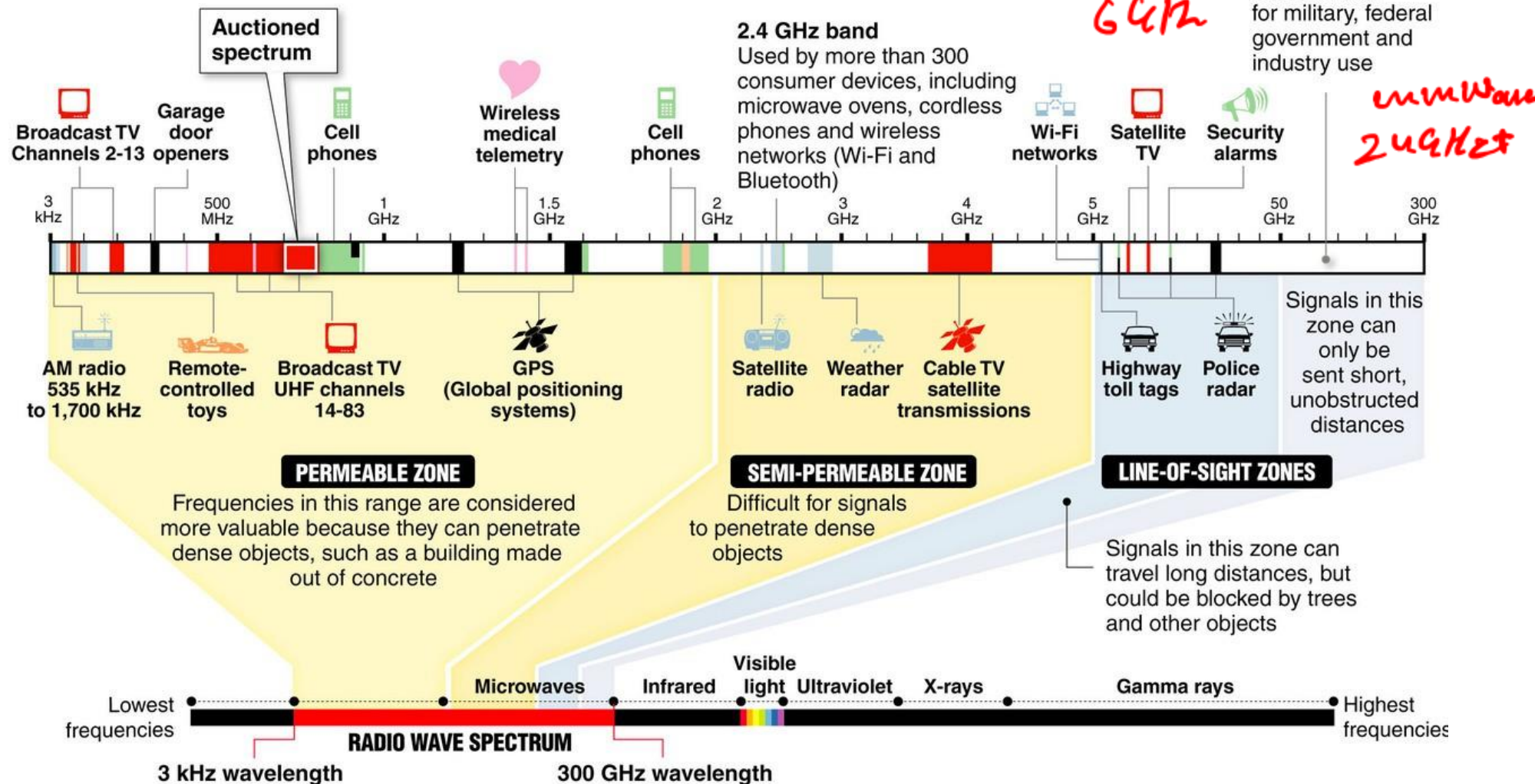


The Wireless Spectrum

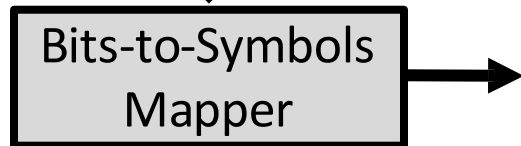
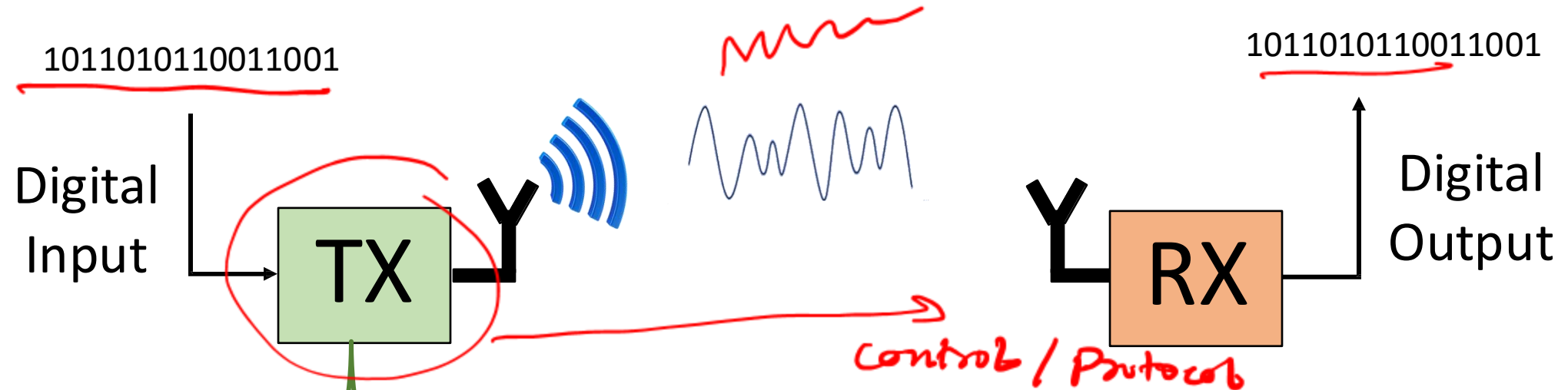


The Wireless Spectrum

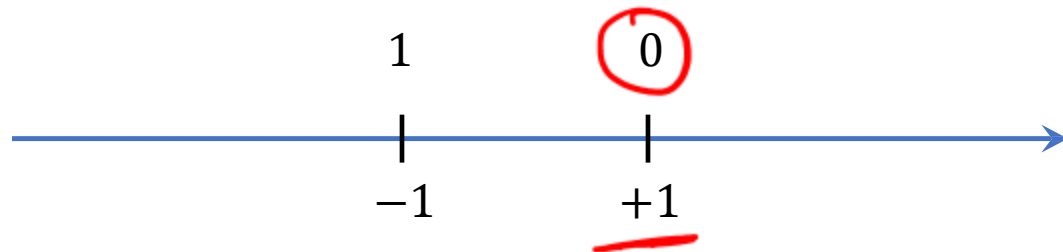
Sub 6 *mid-band 6-24 GHz*
6 GHz



Wireless Communication System



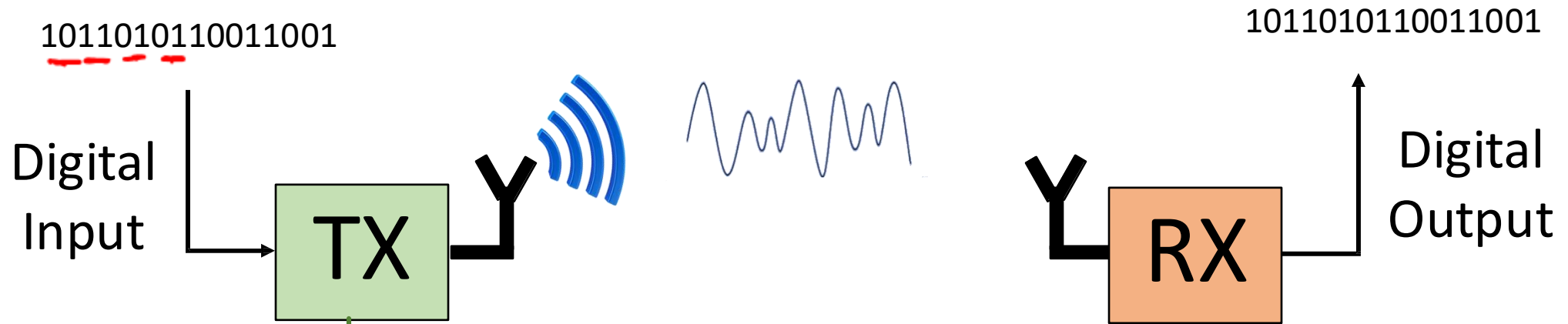
BPSK: 1 bits/symbol



Modulation
(Encoding)

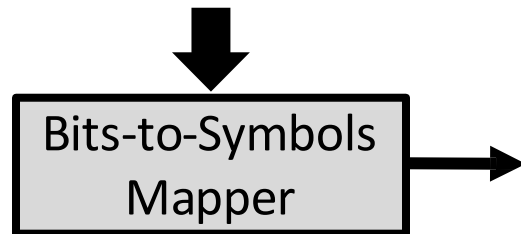


Wireless Communication System

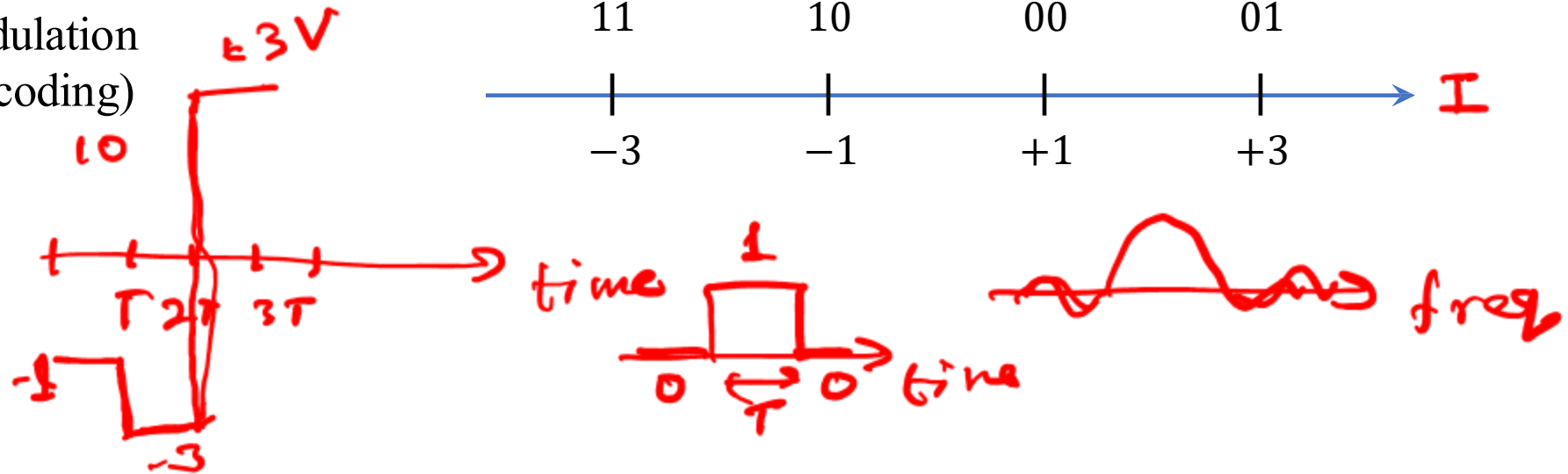


Modulation: Mapping of Bits to Symbols

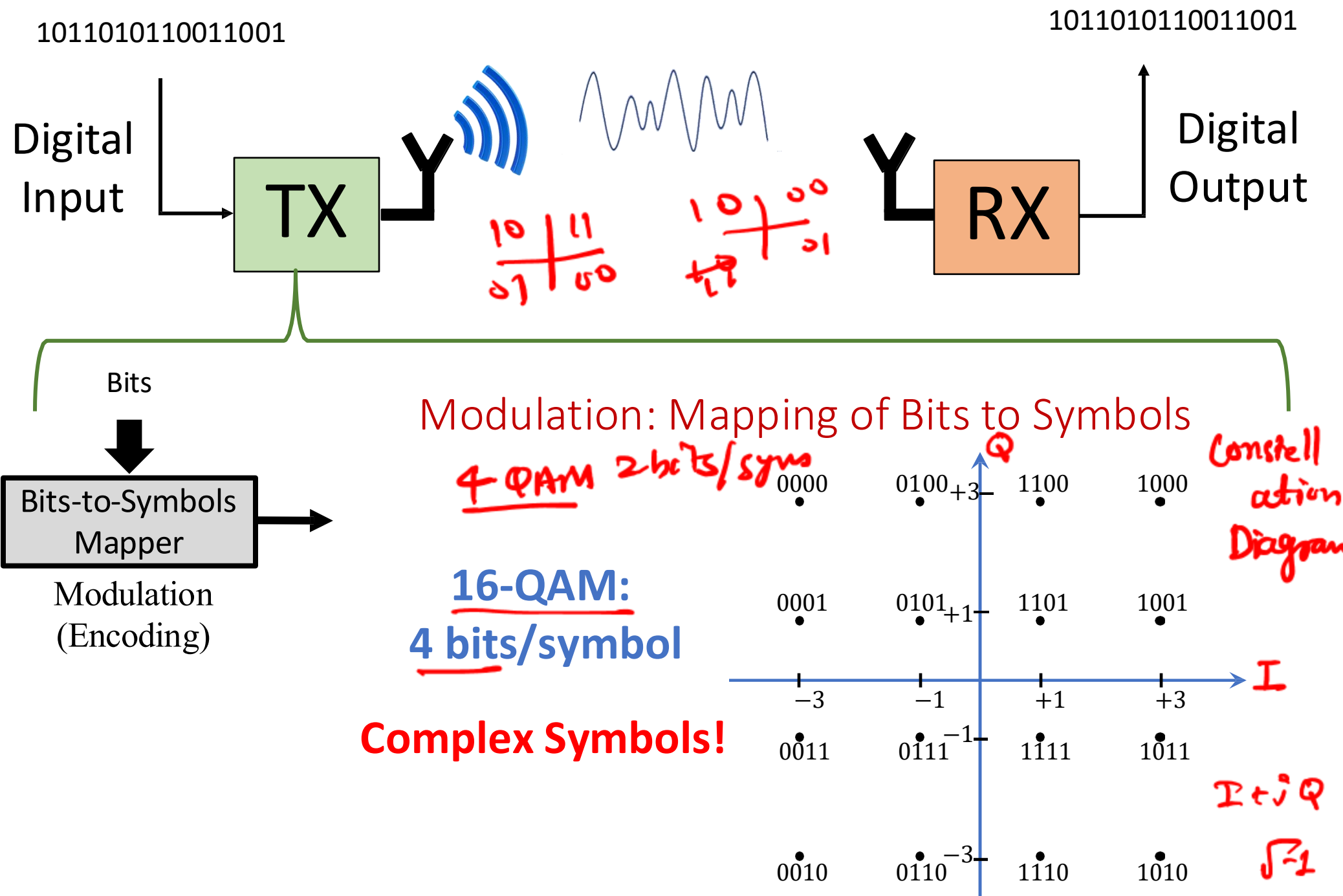
PAM: 2 bits/symbol



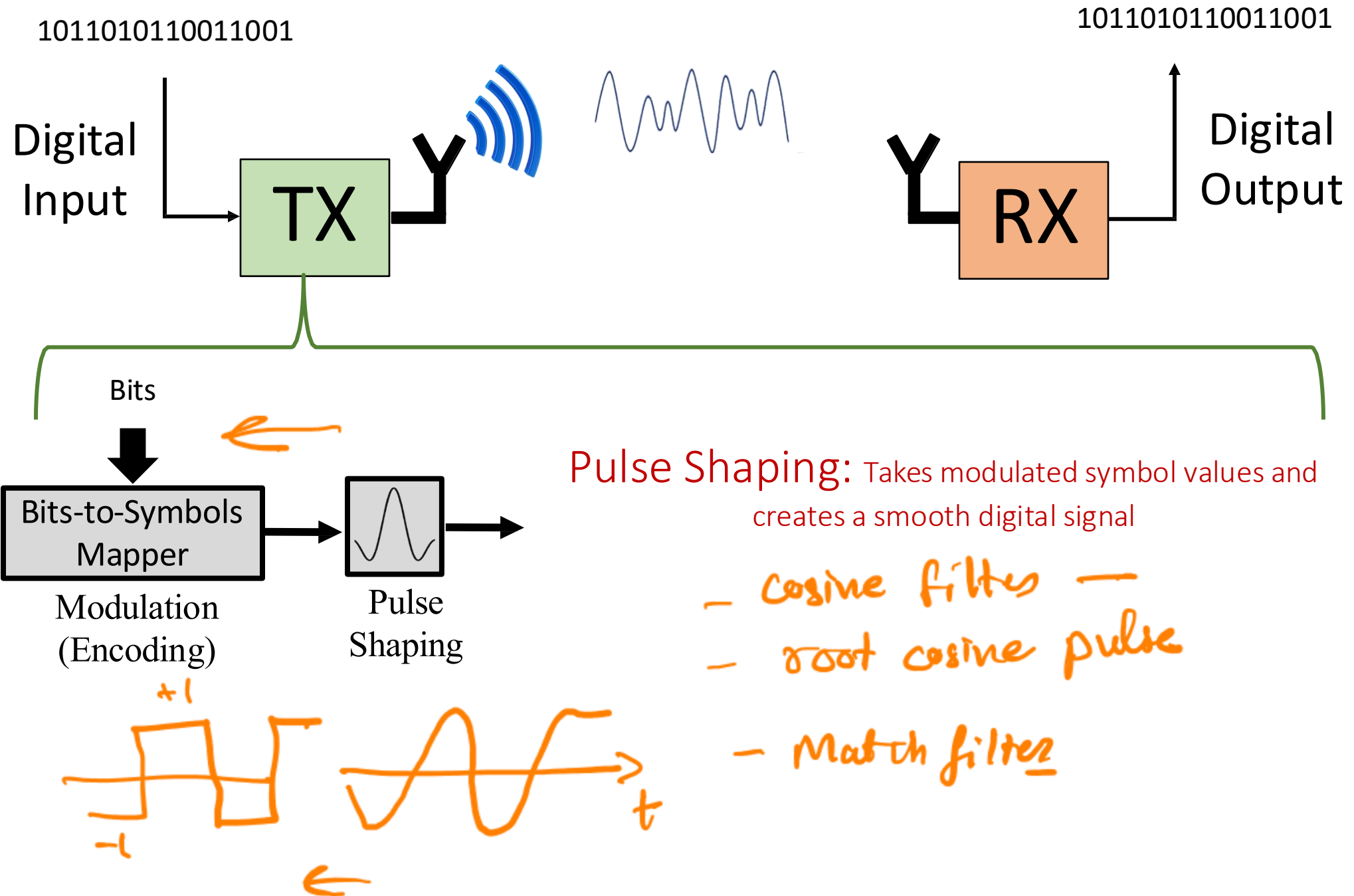
Modulation (Encoding)



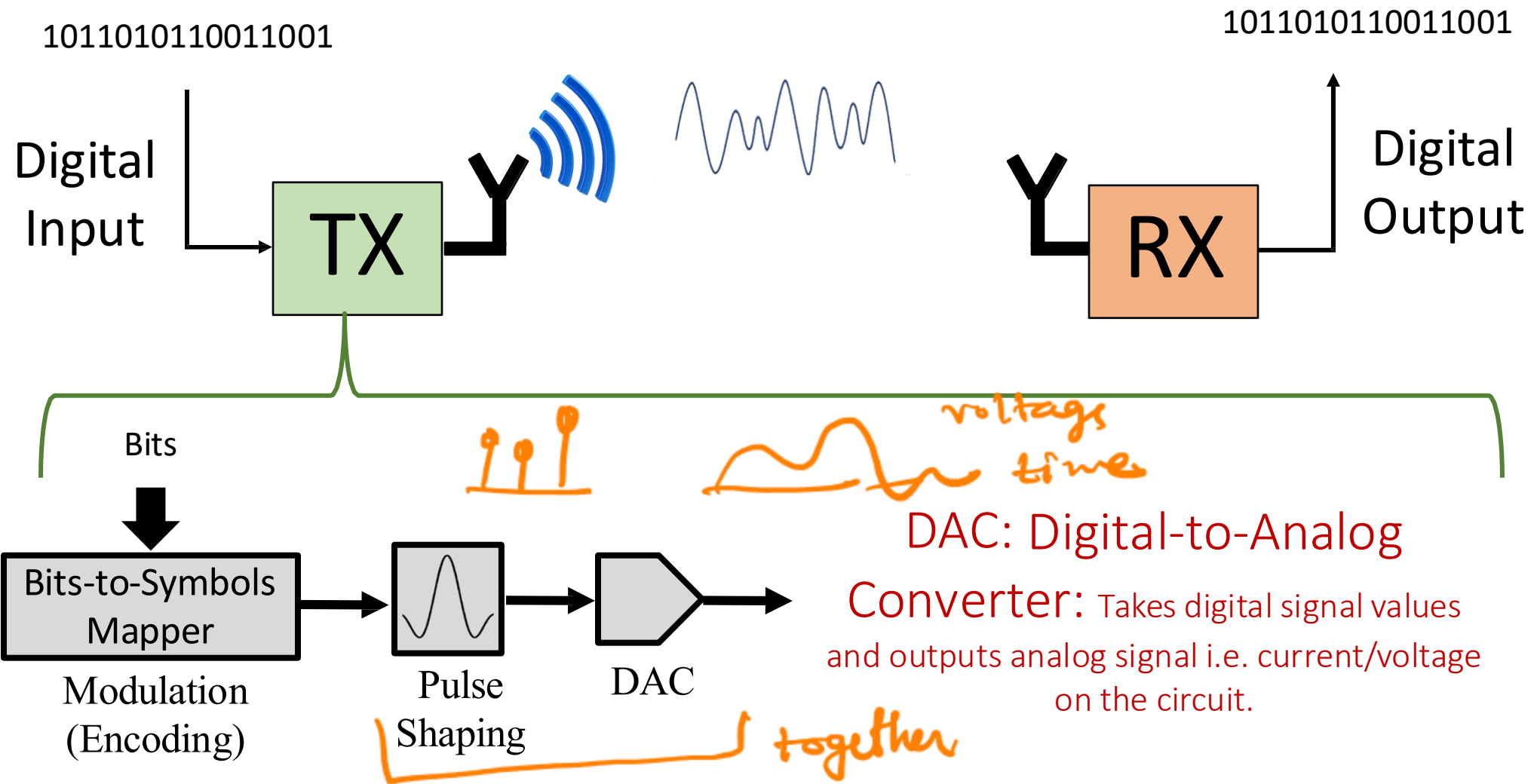
Wireless Communication System



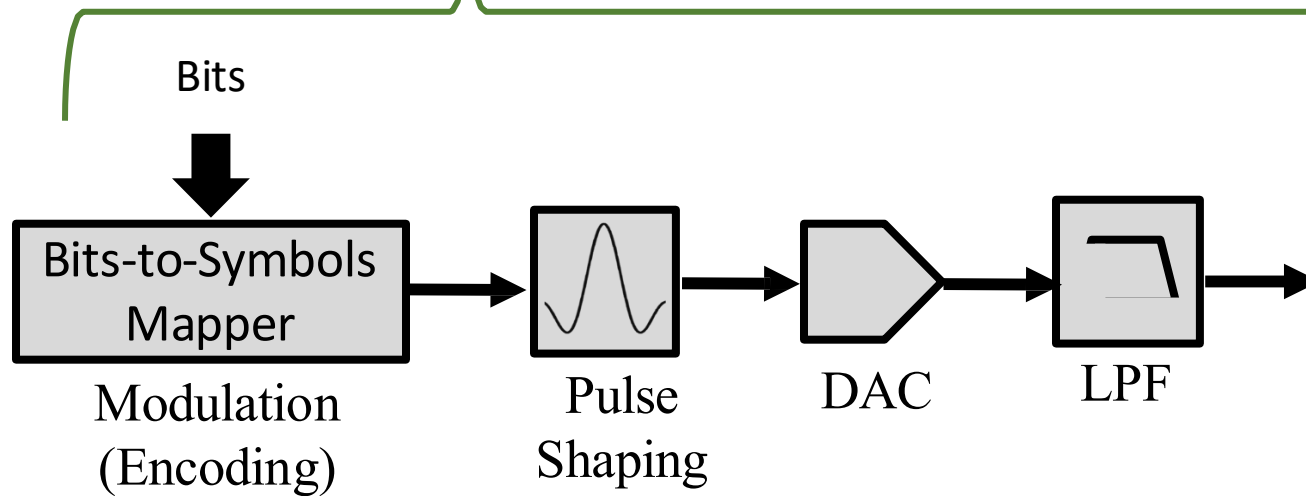
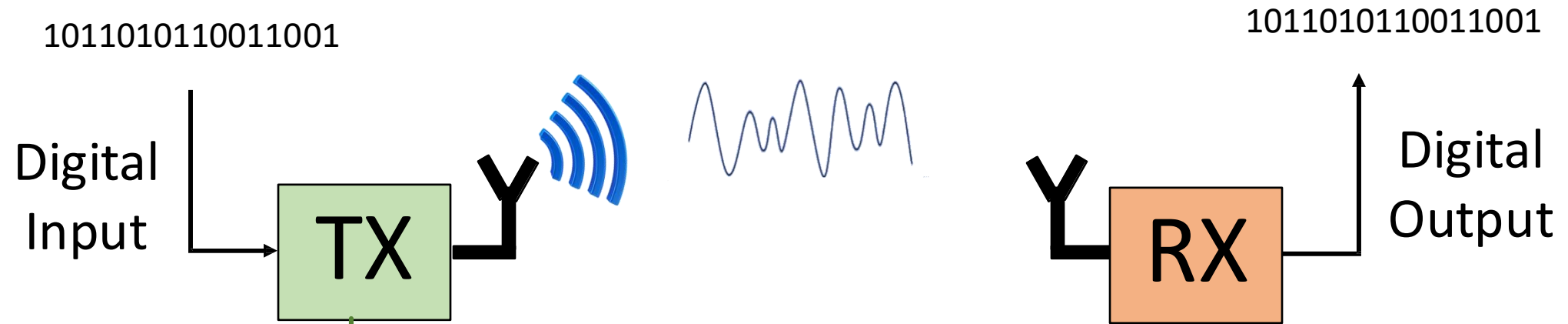
Wireless Communication System



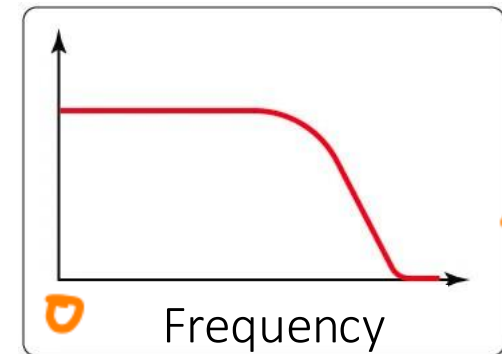
Wireless Communication System



Wireless Communication System



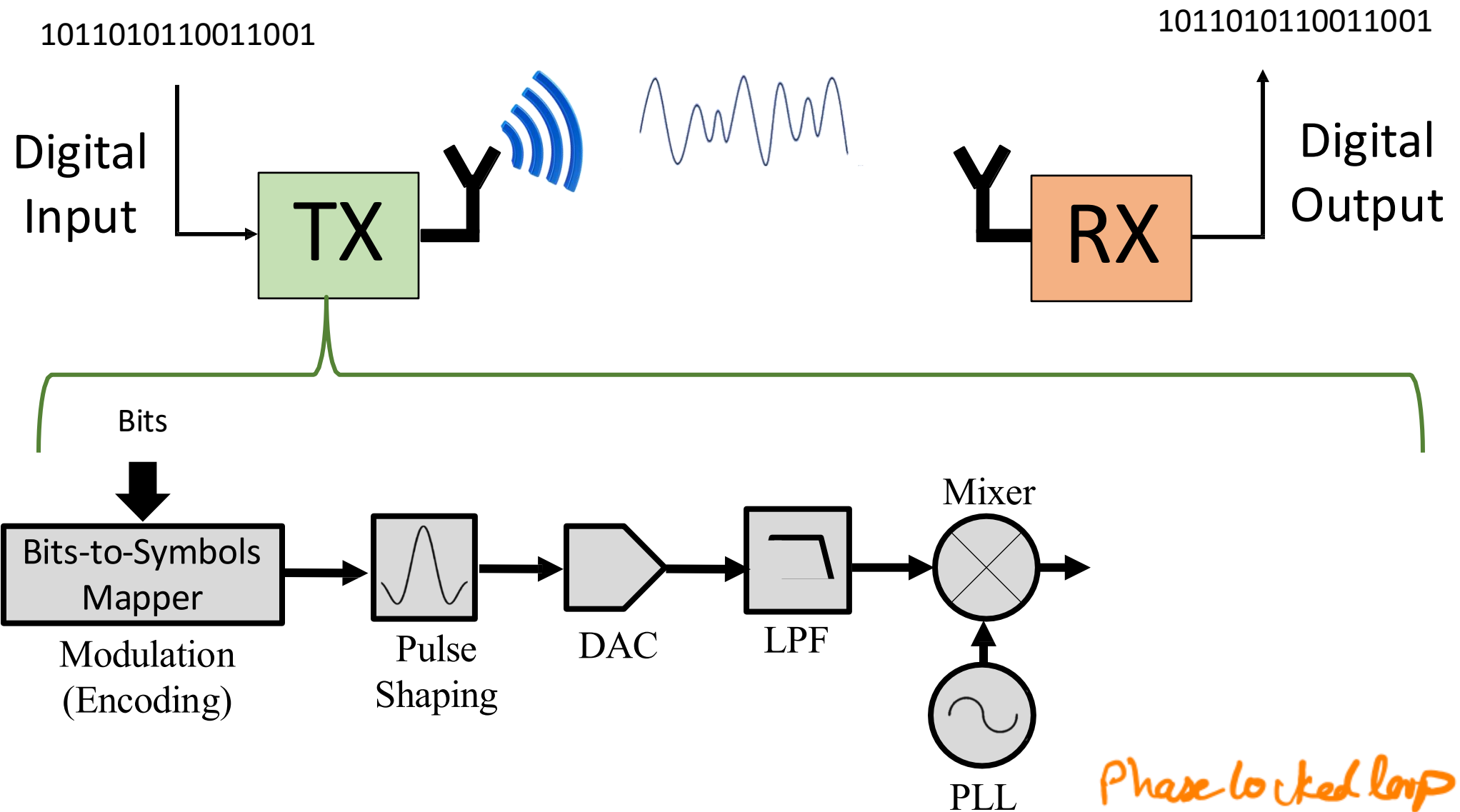
LPF: Low Pass Filter
Removes High Frequency Signals



$$\lambda = \frac{c}{f}$$

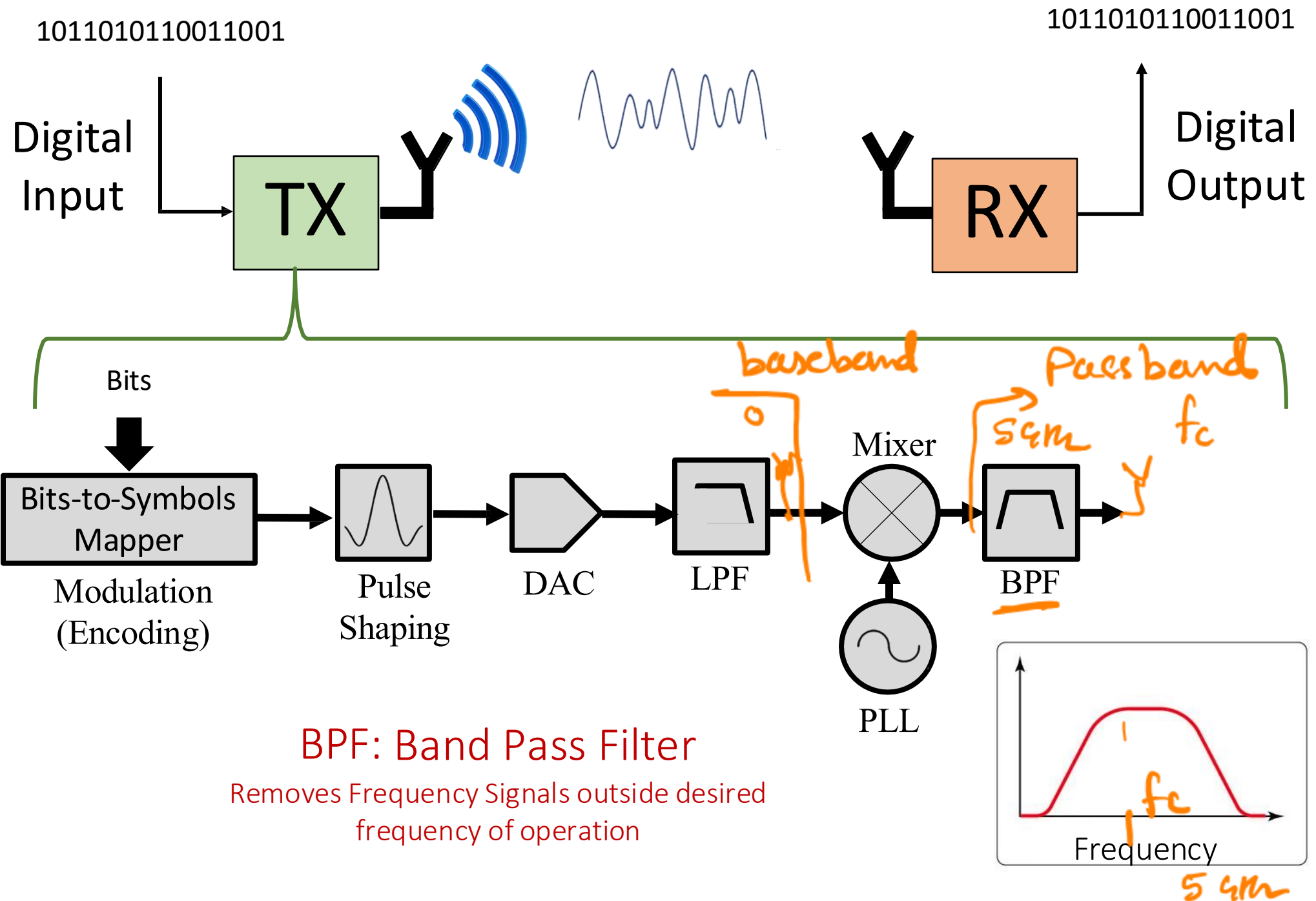
B 20MHz

Wireless Communication System

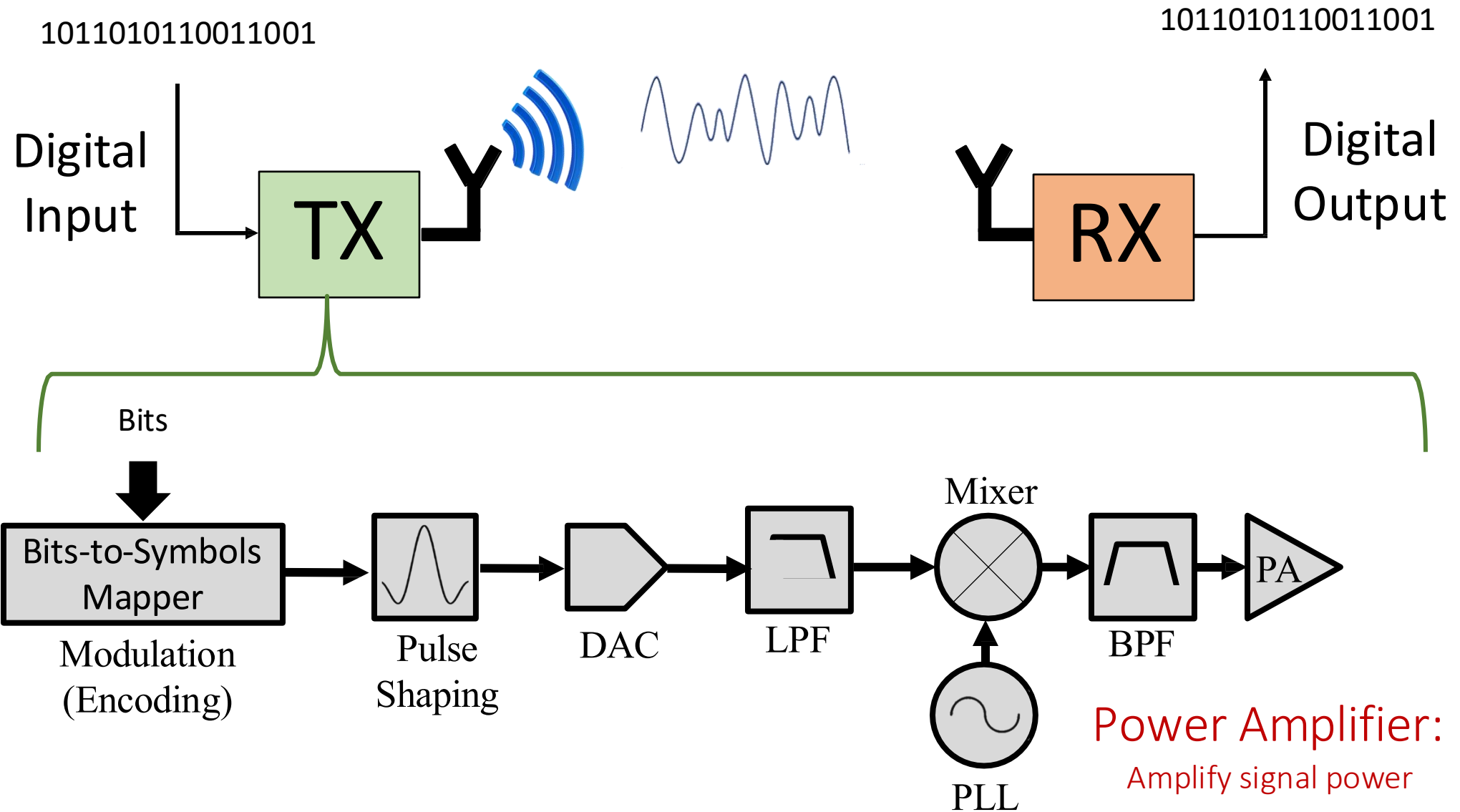


Mixer/PLL: Move signal to desired frequency of operation

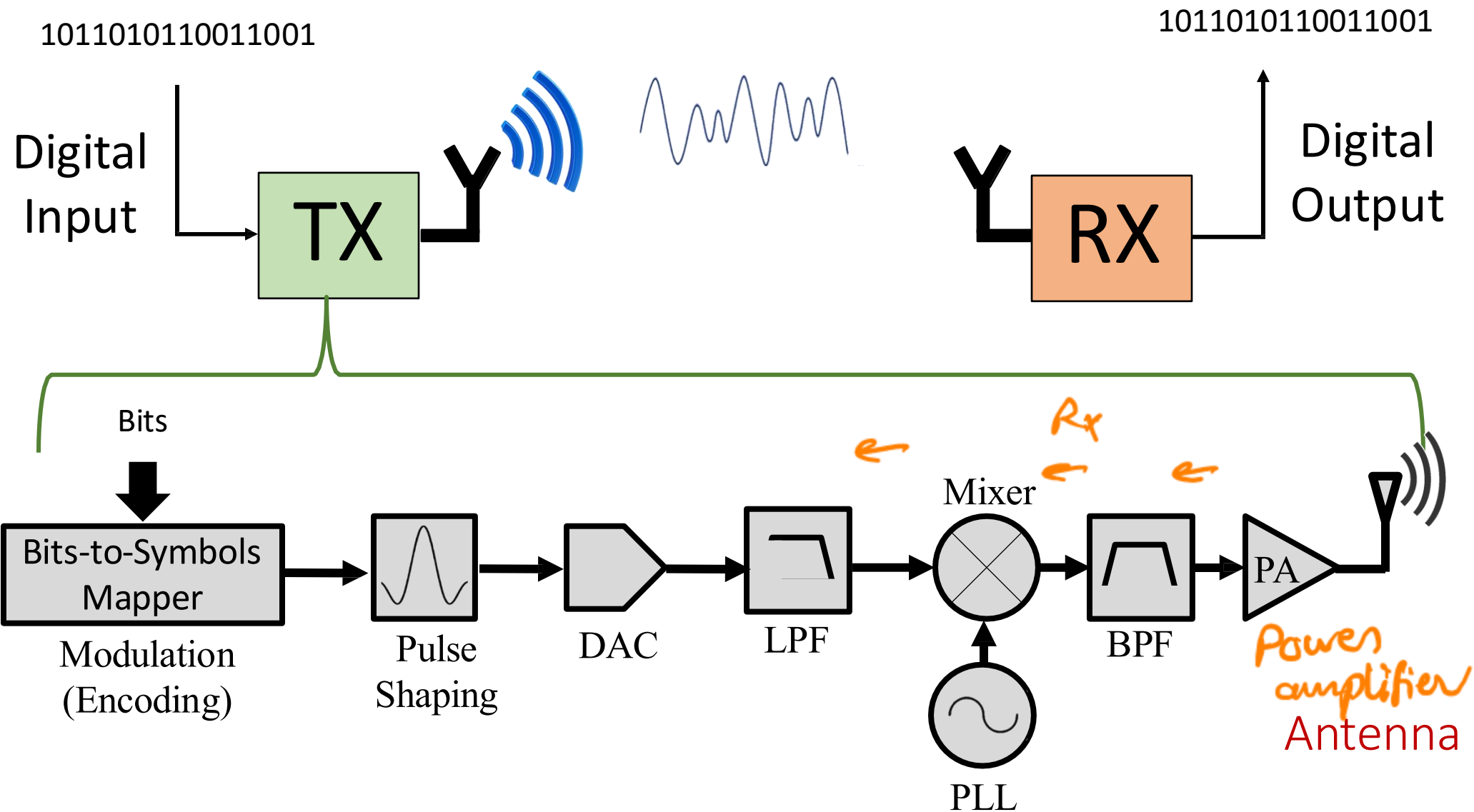
Wireless Communication System



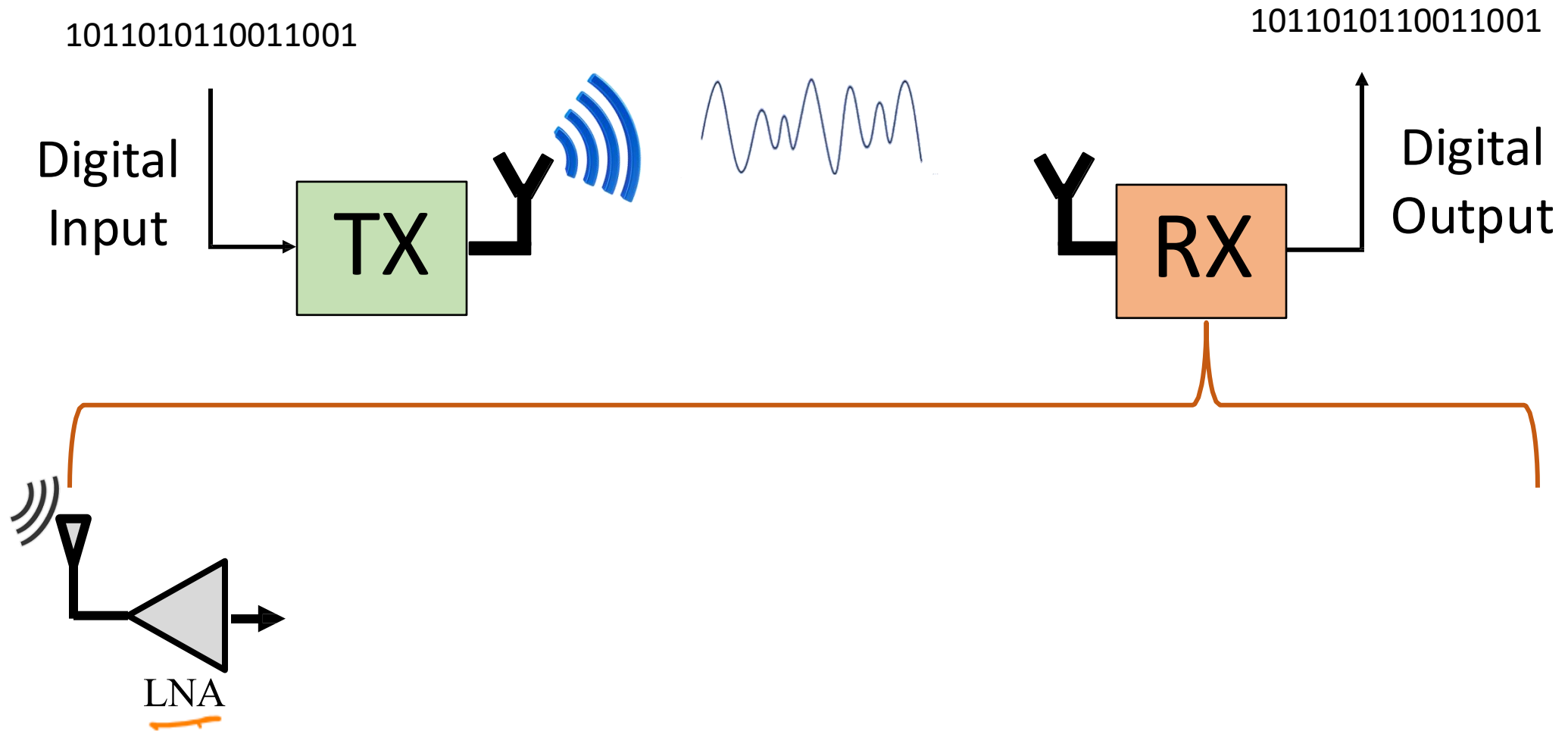
Wireless Communication System



Wireless Communication System

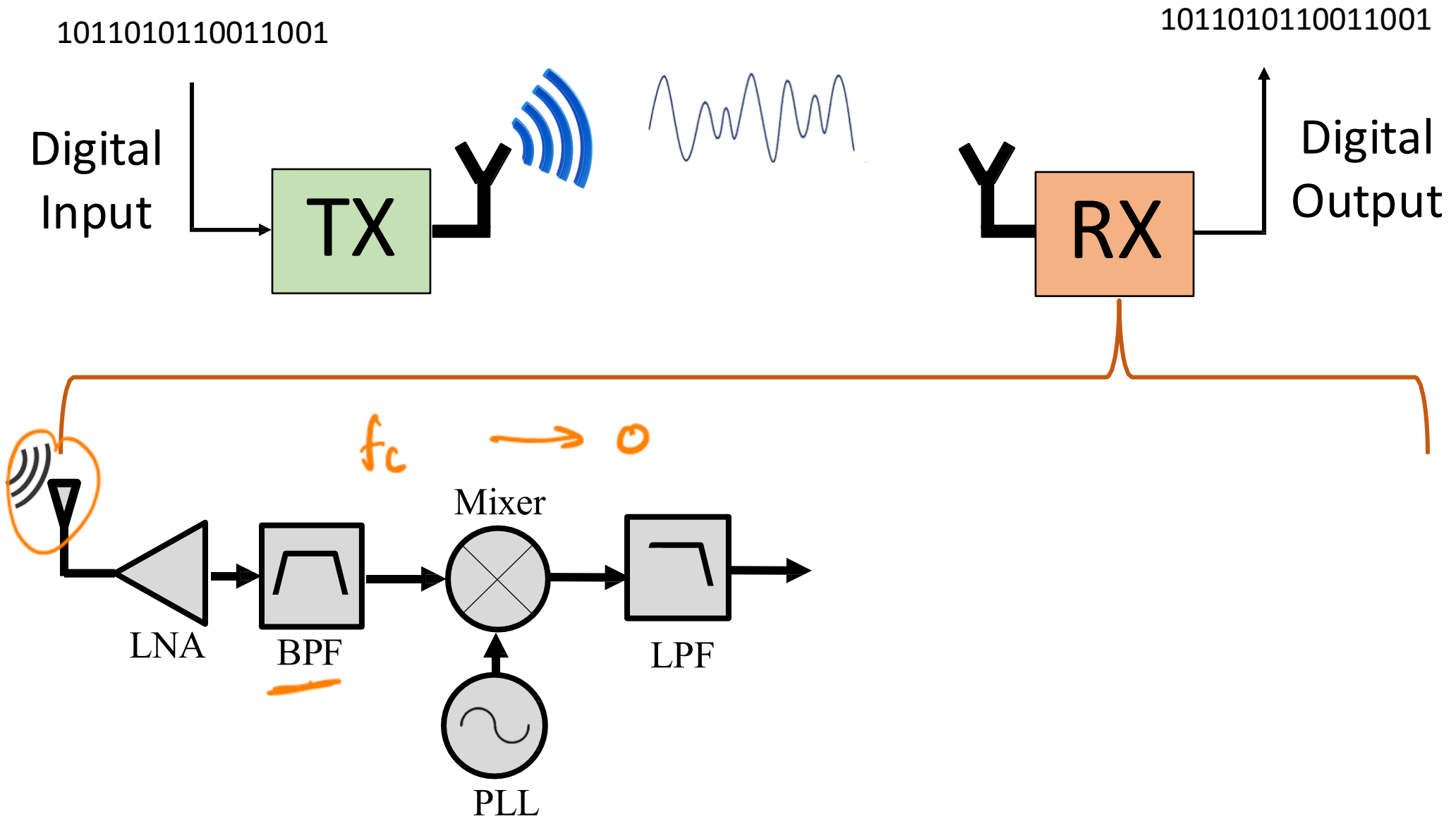


Wireless Communication System

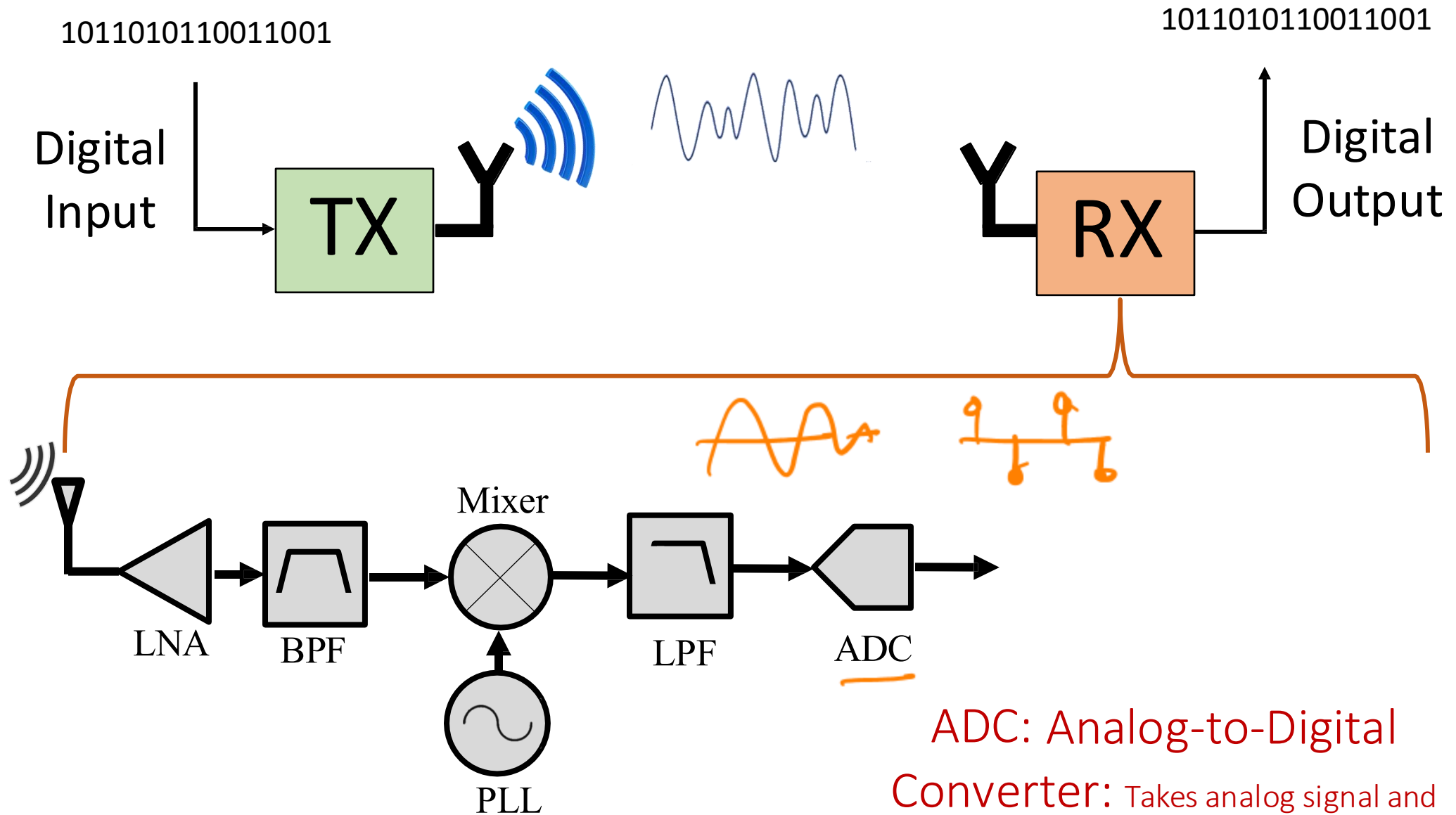


Low Noise Amplifier:
Amplify signal power with minimal
noise

Wireless Communication System

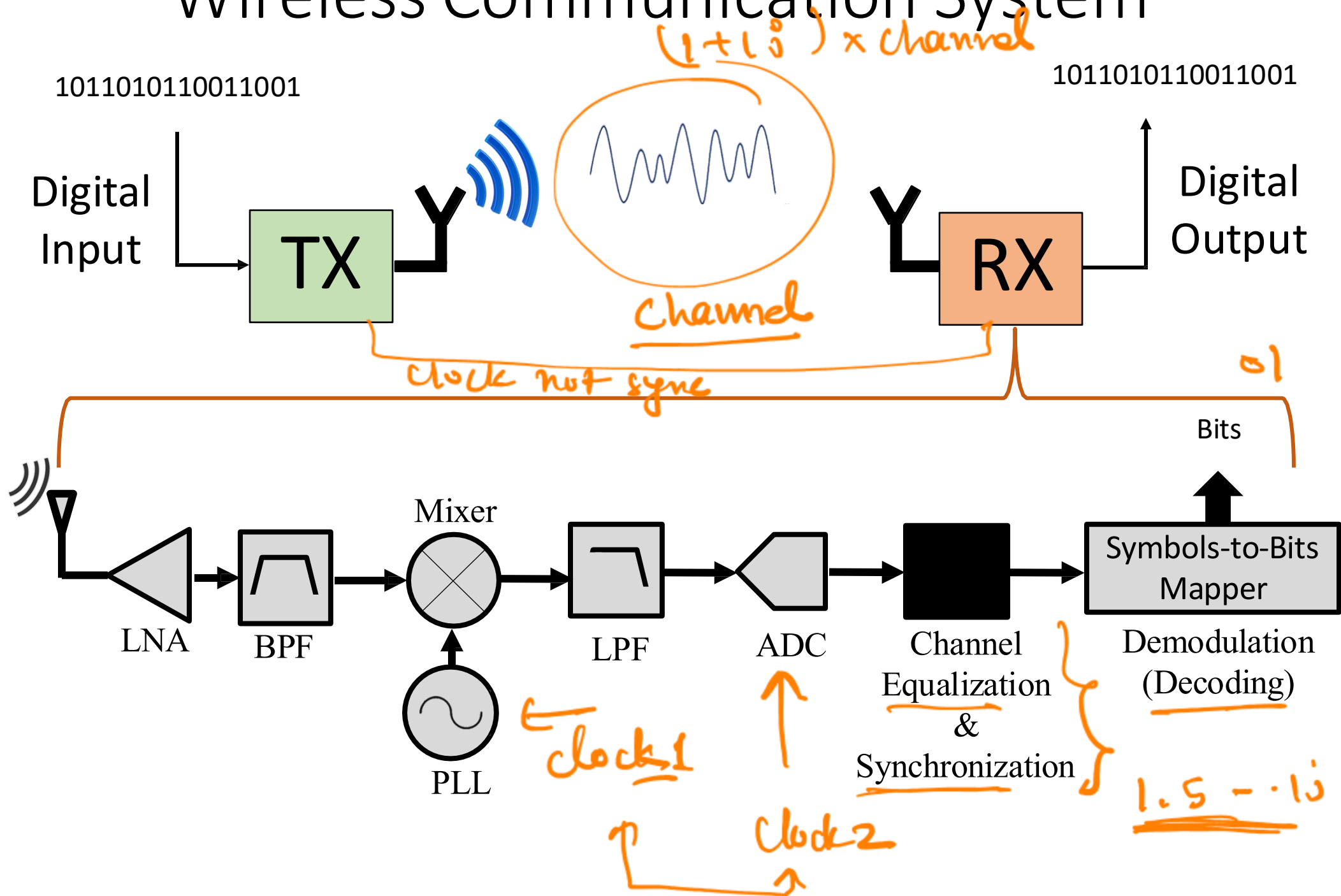


Wireless Communication System



ADC: Analog-to-Digital Converter: Takes analog signal and digitizes it, i.e. samples and quantizes the signal values.

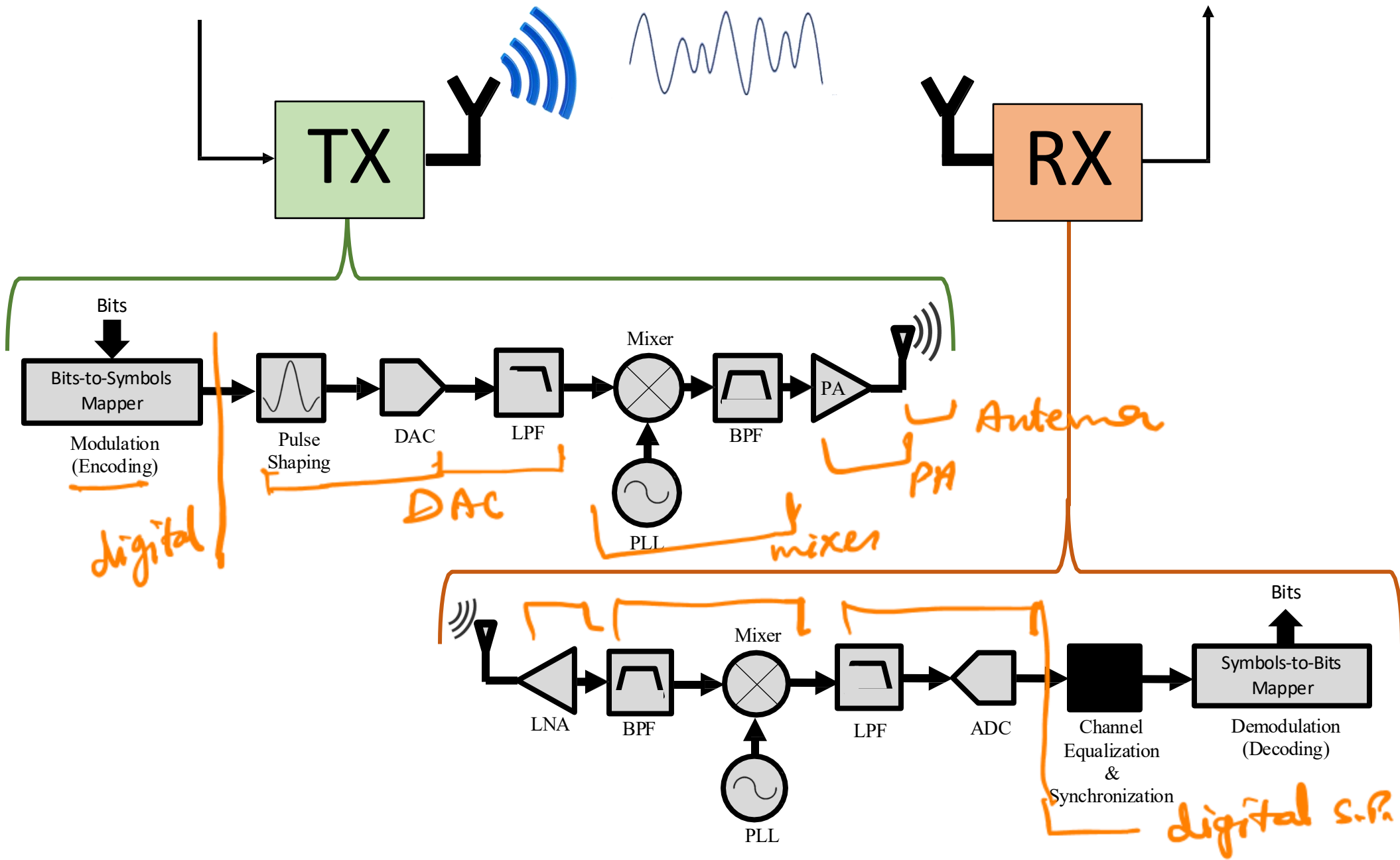
Wireless Communication System



Wireless Communication System

1011010110011001

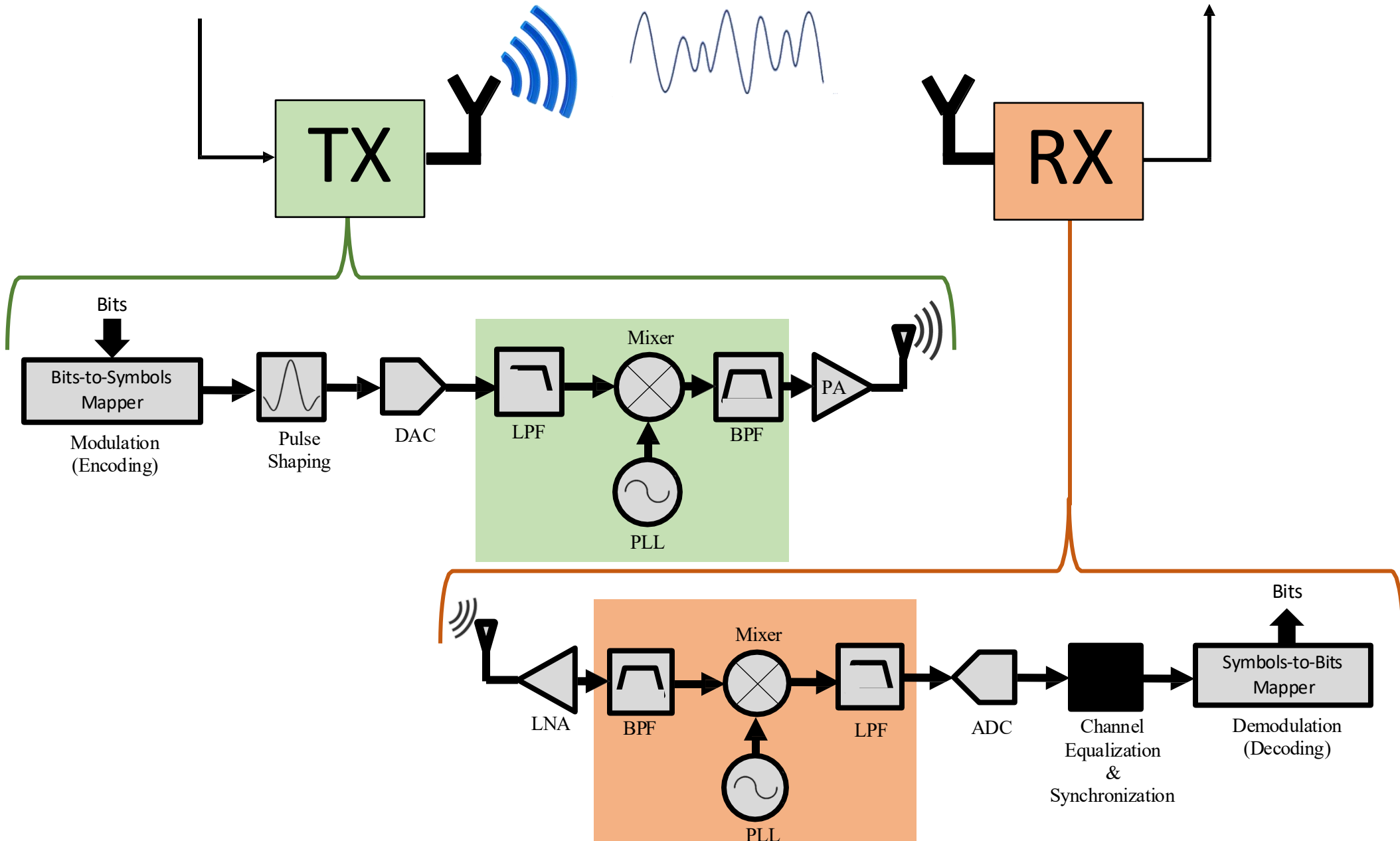
1011010110011001



Wireless Communication System

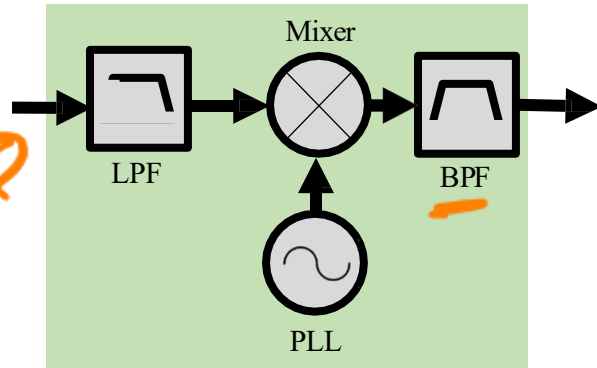
1011010110011001

1011010110011001

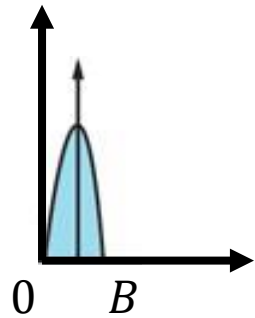


Up/Down Conversion

Transmitter
Up-conversion



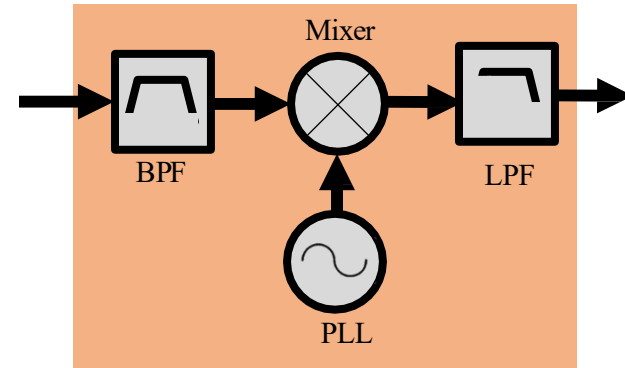
Baseband



Passband

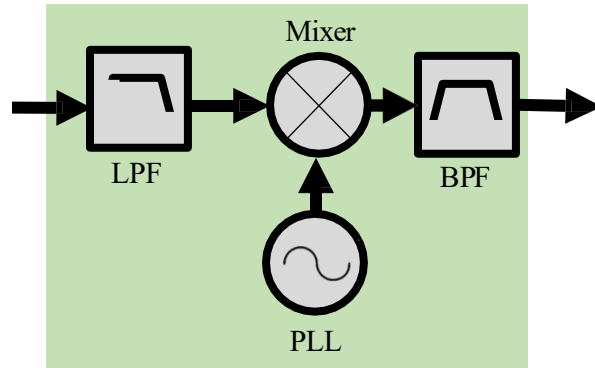


Receiver
Down-conversion

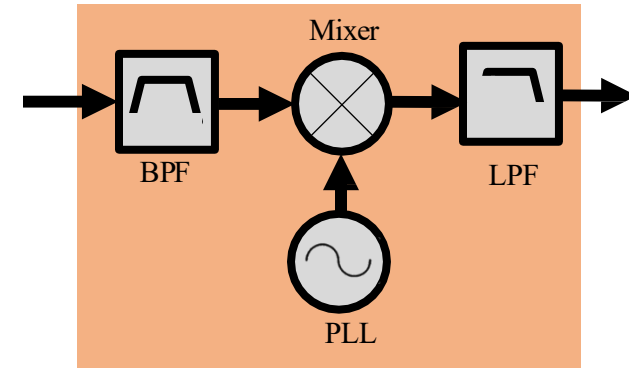


Up/Down Conversion

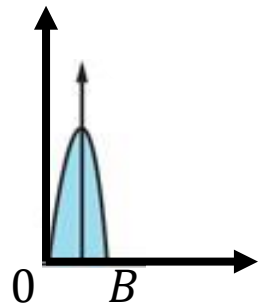
Transmitter
Up-conversion



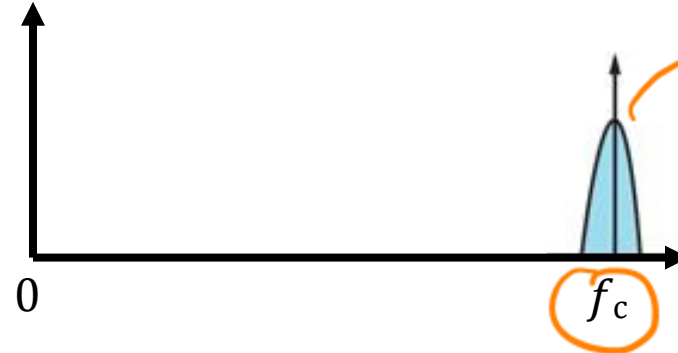
Receiver
Down-conversion



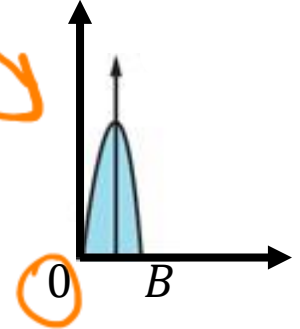
Baseband



Passband



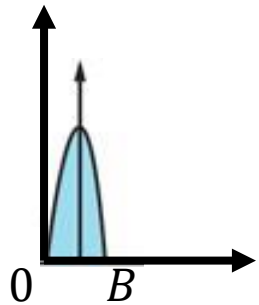
Baseband



Why do we need to up/down conversion?

Up/Down Conversion

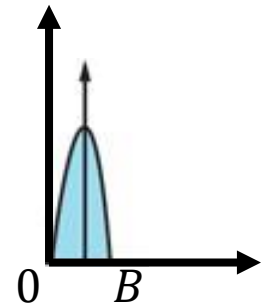
Baseband



Passband

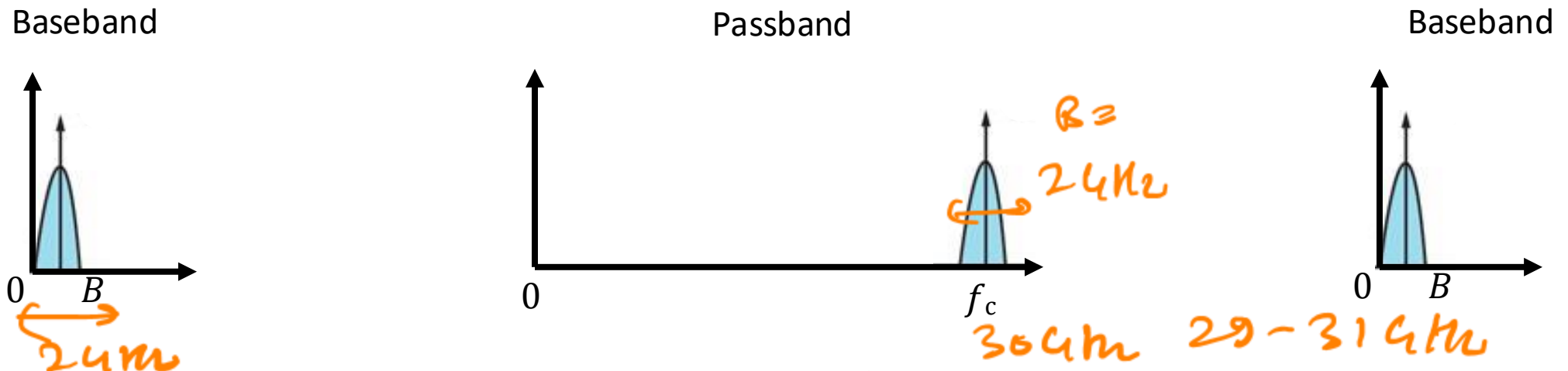


Baseband



Why do we need to up/down conversion?

Up/Down Conversion



Why do we need to up/down conversion?

■ Why not transmit everything at lower frequencies?

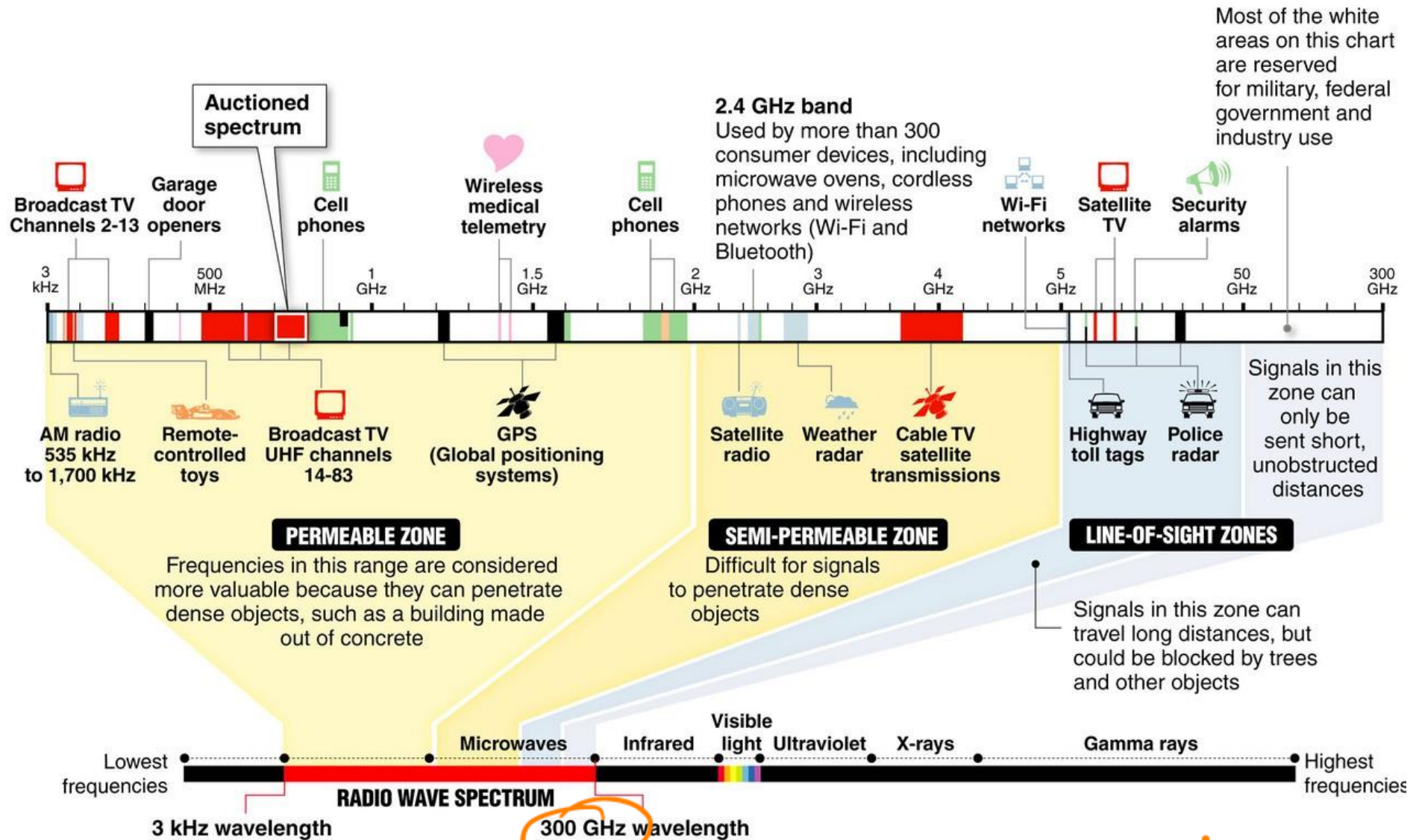
Not enough bandwidth:

Data rate $\propto$ bandwidth

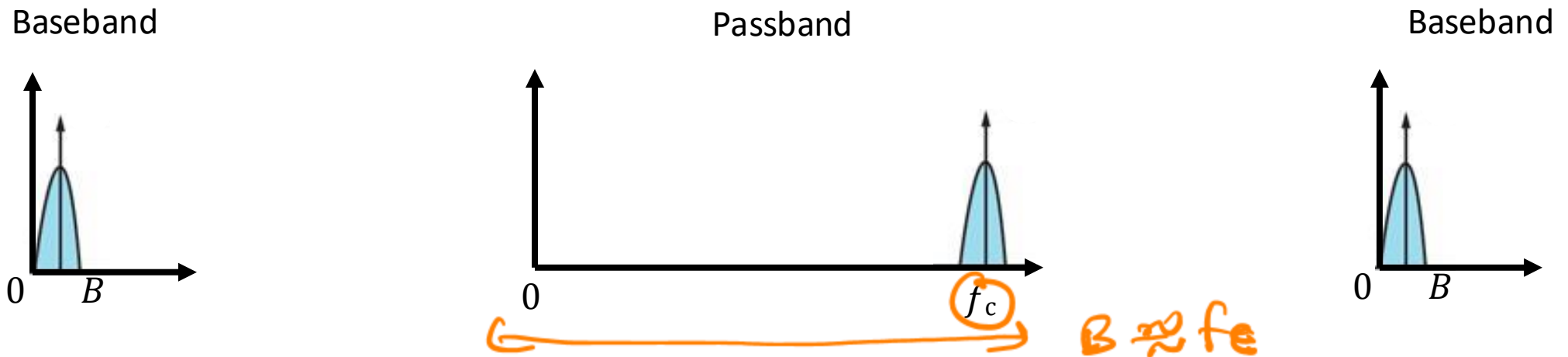
Different Technologies

Antenna size $\propto$ wavelength

The Wireless Spectrum



Up/Down Conversion



Why do we need to up/down conversion?

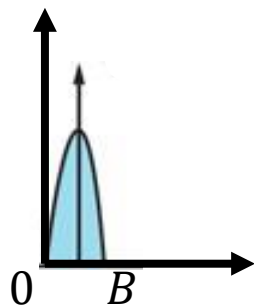
- Cannot transmit everything at low frequencies.
- Why not transmit & receive directly at high frequency?

Nyquist!

Nyquist Theorem

To recover the signal properly, we need to sample at twice the highest frequency, i.e. $2f_{\max}$

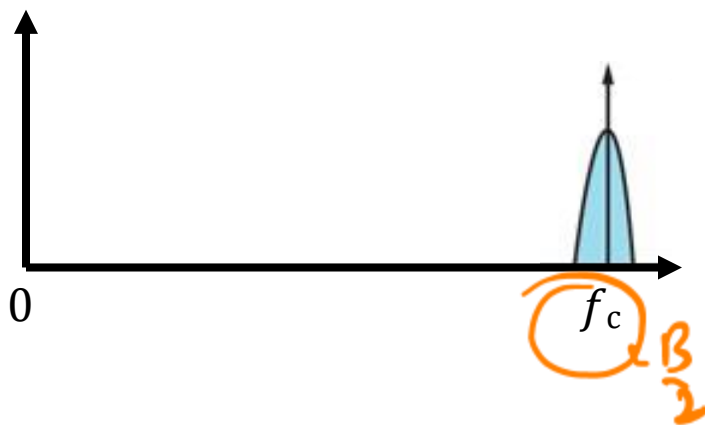
Baseband



➡ Sample at $2B$

e.g. WiFi 802.11b ≈ 40 MS/s

Passband



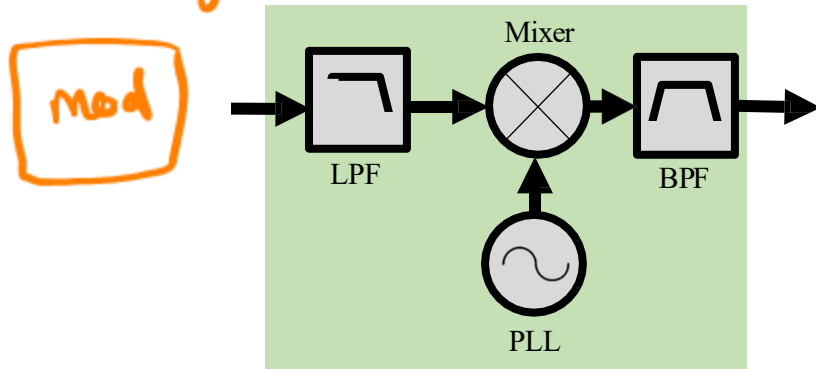
➡ Sample at $2f_c + B$

e.g. WiFi 802.11b ≈ 5 GS/s

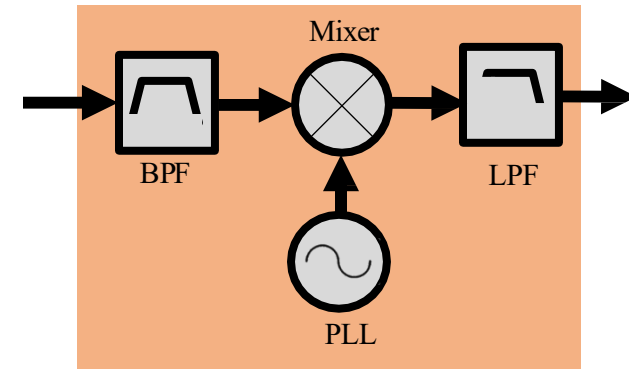
Up/Down Conversion

0101 $s(t)$ symbol
mod

Transmitter
Up-conversion



Receiver
Down-conversion

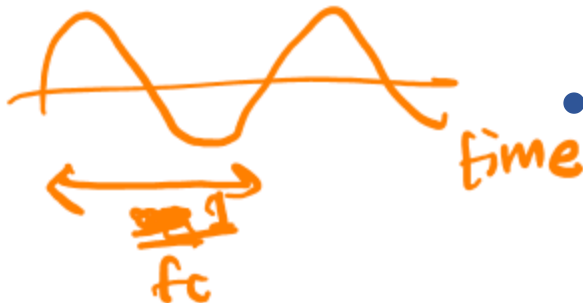


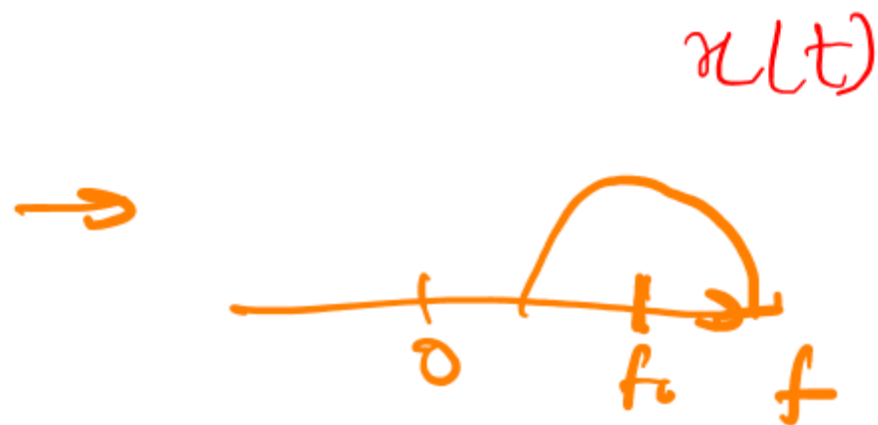
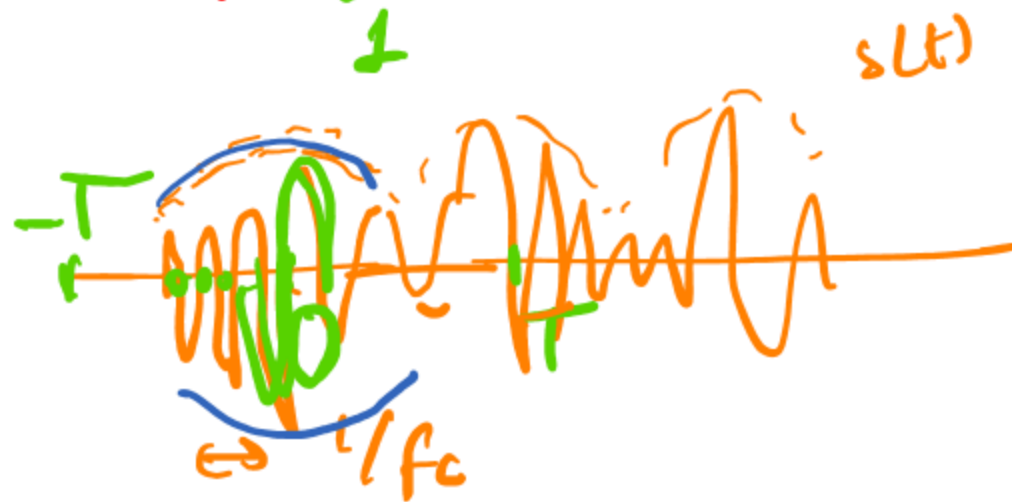
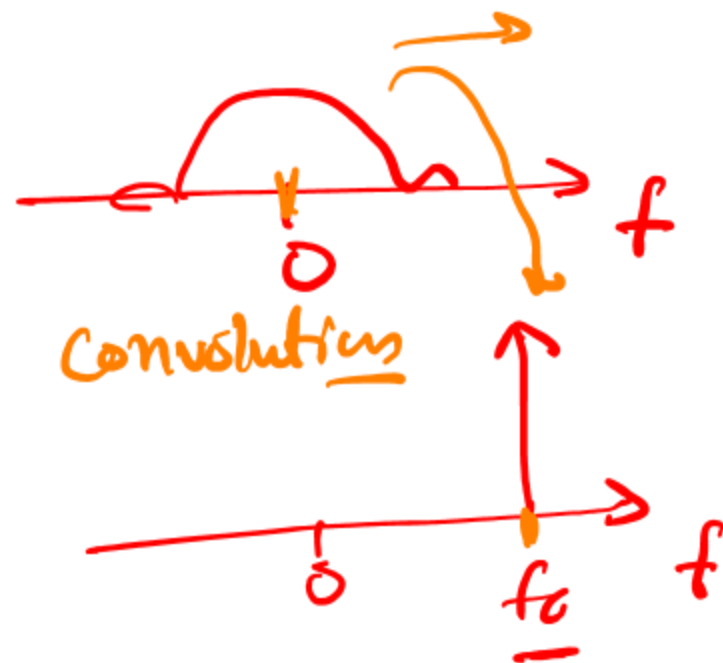
How to we do up/down conversion?

$$\underline{s(t)} \rightarrow \times \rightarrow \underline{s(t) \cos(2\pi f_c t)}$$

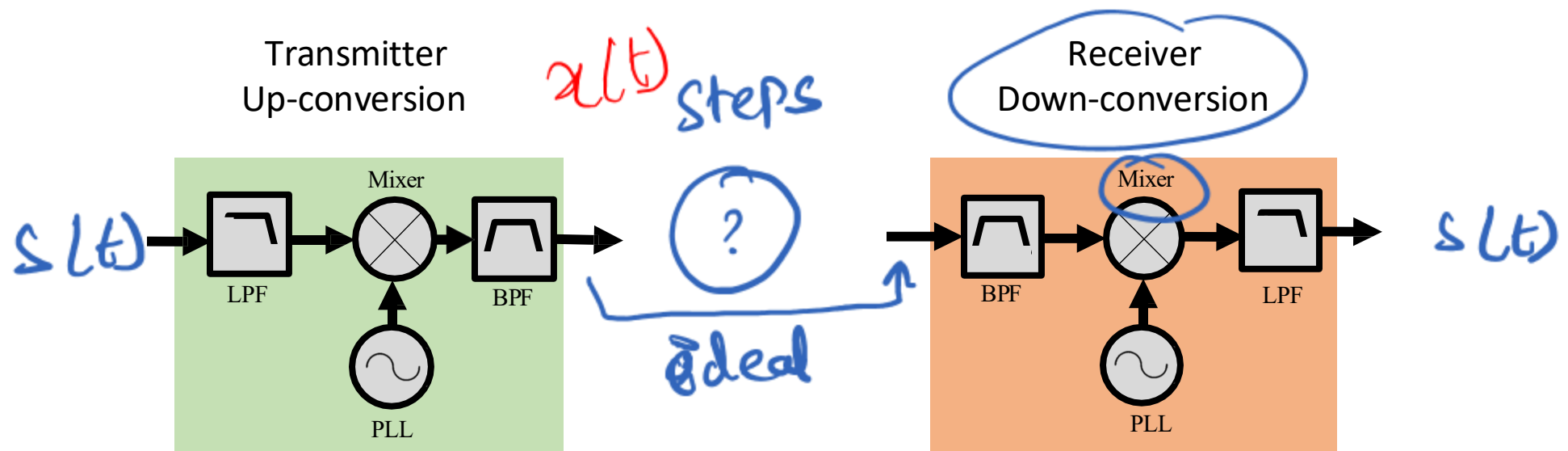
$$\cos(2\pi f_c t)$$

- PLL (Phased Lock Loop): create the cosine wave signal $\cos(2\pi f_c t)$
- Mixer: multiplies the two signals





Up/Down Conversion



How to we do up/down conversion?

$$\begin{aligned}
 &\underbrace{s(t)} \xrightarrow{\cos(2\pi f_c t)} \underbrace{s(t) \cos(2\pi f_c t)}_{x(t)} \xrightarrow{\cos(2\pi f_c t)} \underbrace{s(t) \cos^2(2\pi f_c t)} \\
 &\hspace{10em} \boxed{2 \cos^2 \theta = 1 + \cos 2\theta}
 \end{aligned}$$

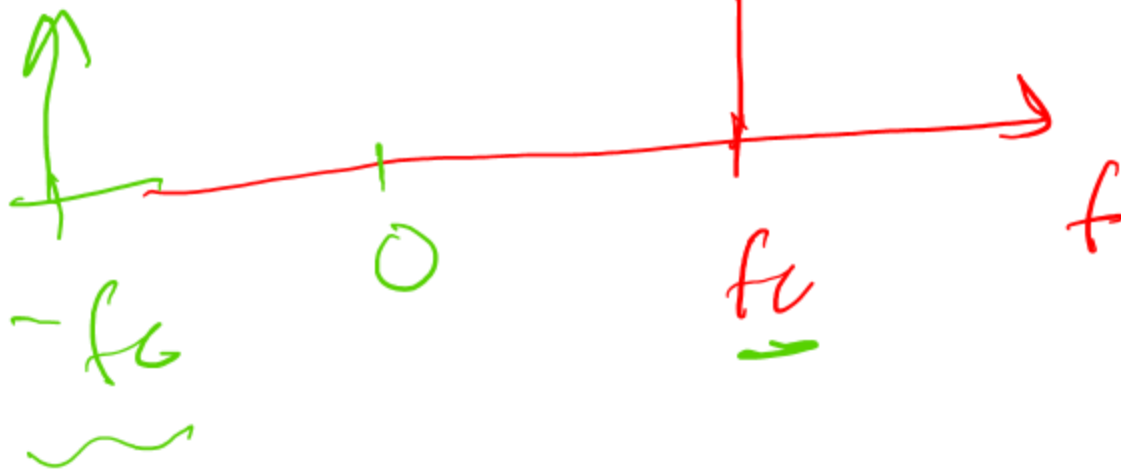
$$\underbrace{s(t) \cos^2(2\pi f_c t)} = \underbrace{s(t)} \left(\frac{1}{2} + \frac{1}{2} \cos(2\pi \underbrace{2f_c t}) \right)$$

Rx chain $X(f)$



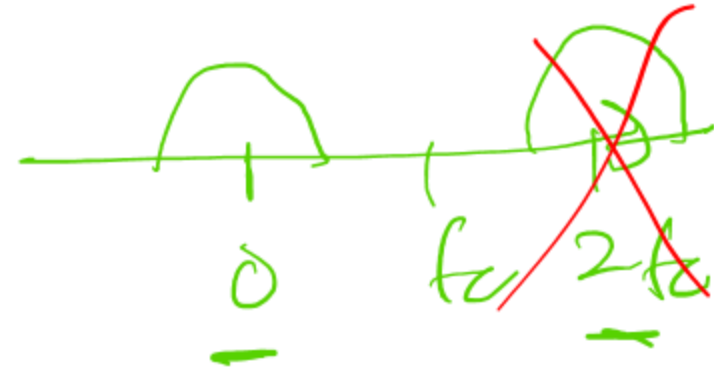
(conv.)

$\cos 2\pi f_c t$



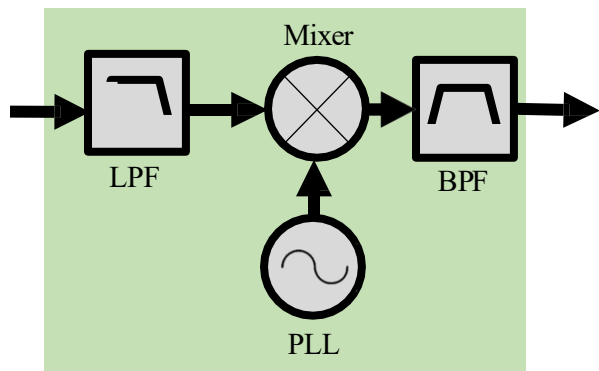
desired

not
desired

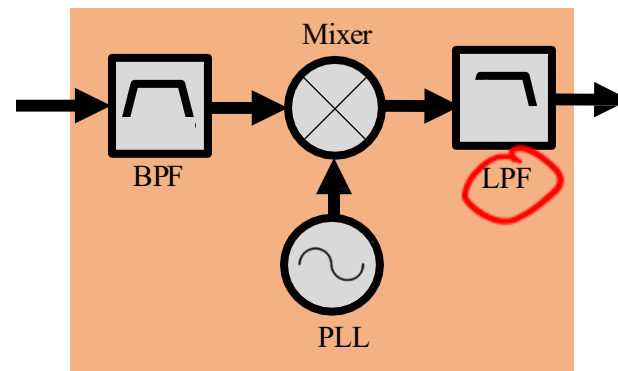


Up/Down Conversion

Transmitter
Up-conversion



Receiver
Down-conversion

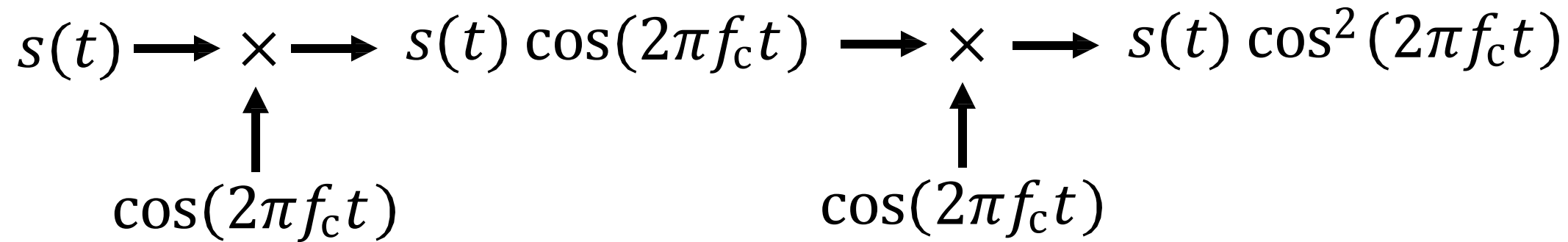


How to we do up/down conversion?

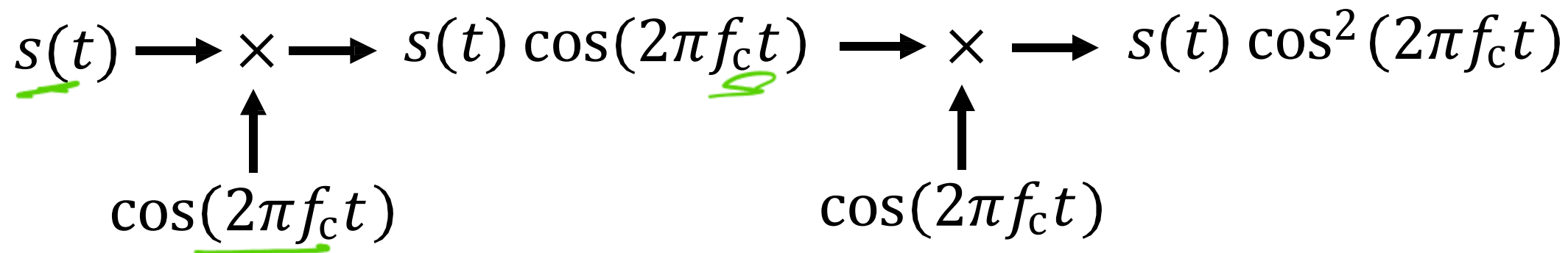
$$s(t) \xrightarrow{\uparrow \frac{1}{\sqrt{2}} \cos(2\pi f_c t)} \times \rightarrow s(t) \cos(2\pi f_c t) \xrightarrow{\uparrow \frac{1}{\sqrt{2}} \cos(2\pi f_c t)} \times \rightarrow s(t) \cos^2(2\pi f_c t)$$

$$s(t) \cos^2(2\pi f_c t) = s(t) \left(\frac{1}{2} + \frac{1}{2} \cos(4\pi f_c t) \right) \xrightarrow{\text{LPF Low Pass Filter}} \underline{\underline{\frac{1}{2} s(t)}}$$

Up/Down Conversion



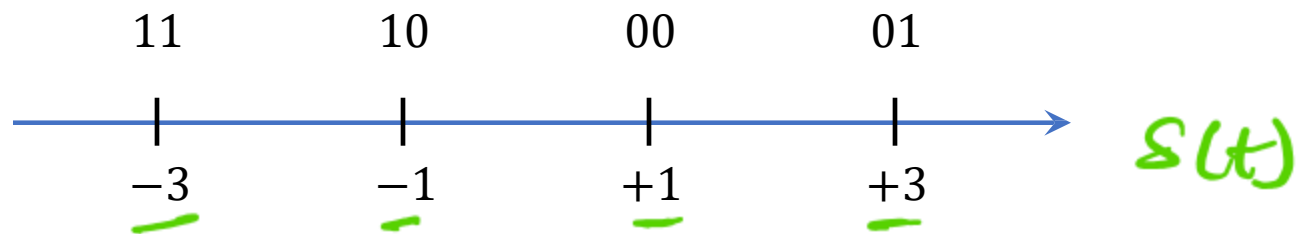
Up/Down Conversion



Consider using PAM: Pulse Amplitude Modulation

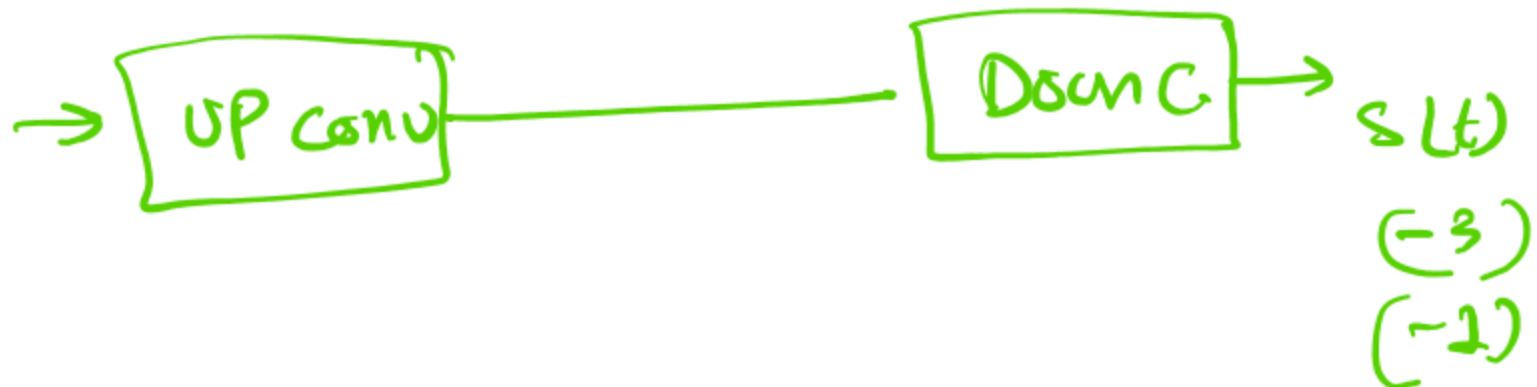
$2^2 = 4$

4 PAM:
2 bits/symbol

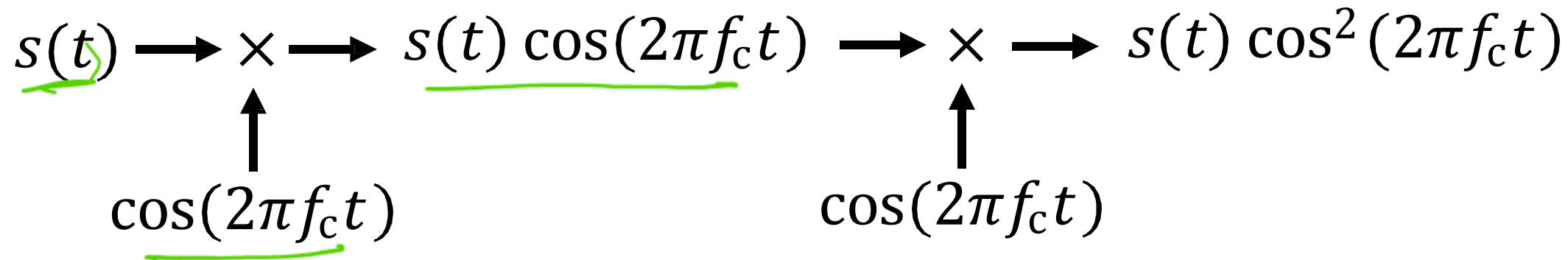


red

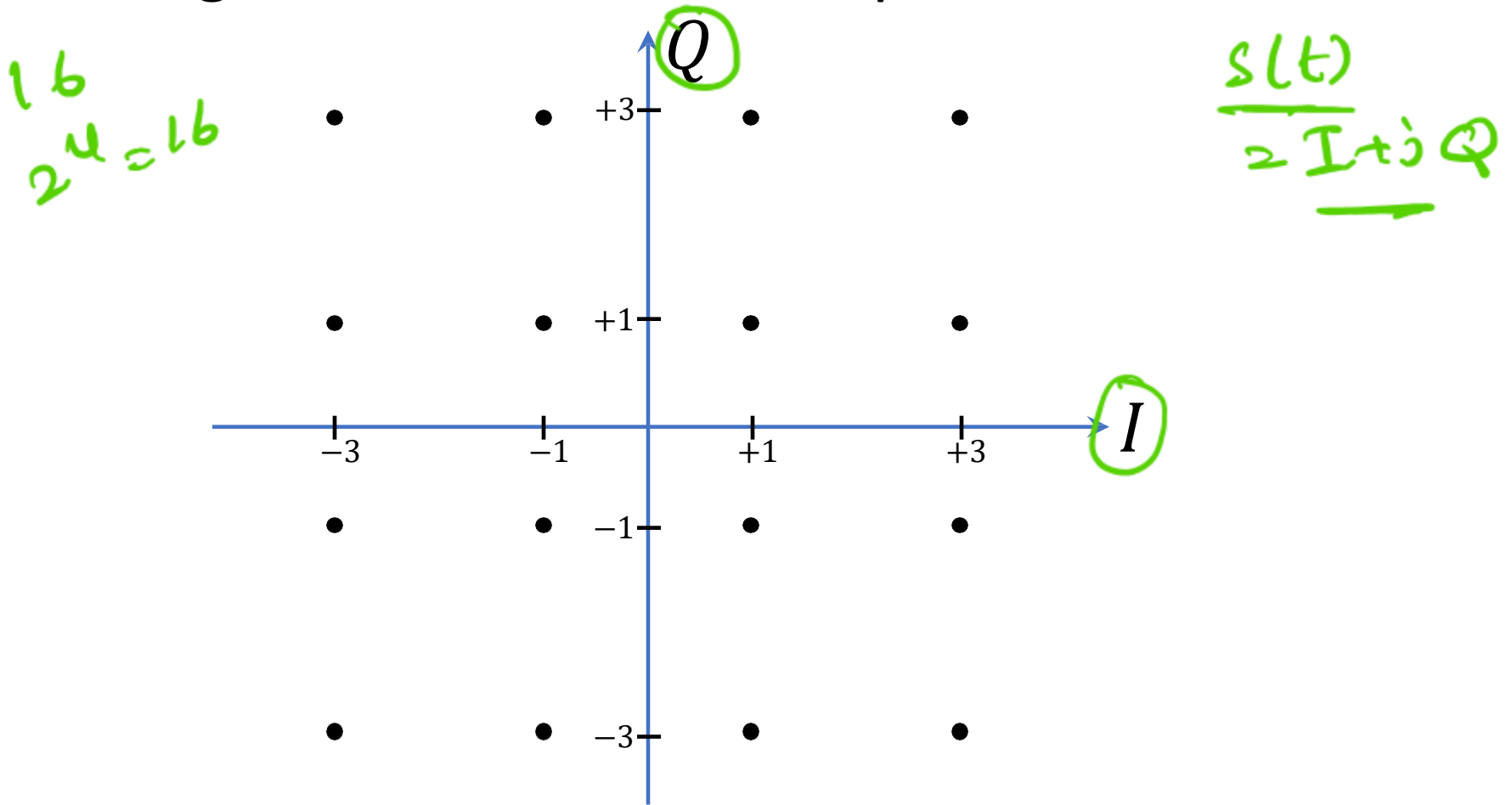
$s(t)$
(-3)
(-1)



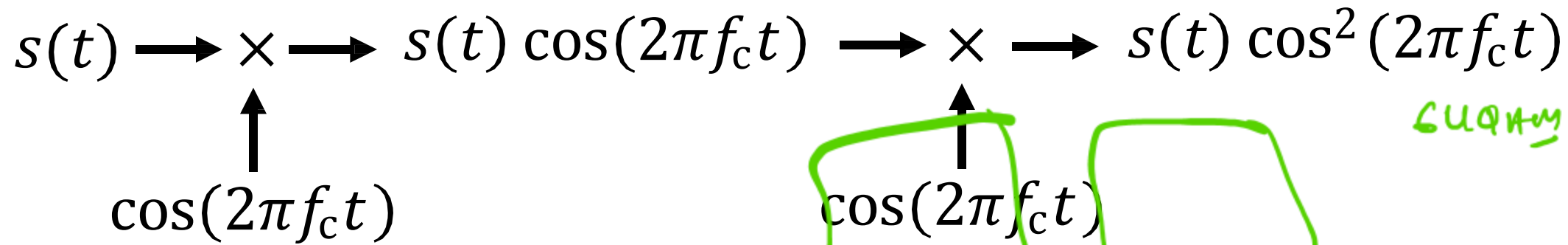
Up/Down Conversion



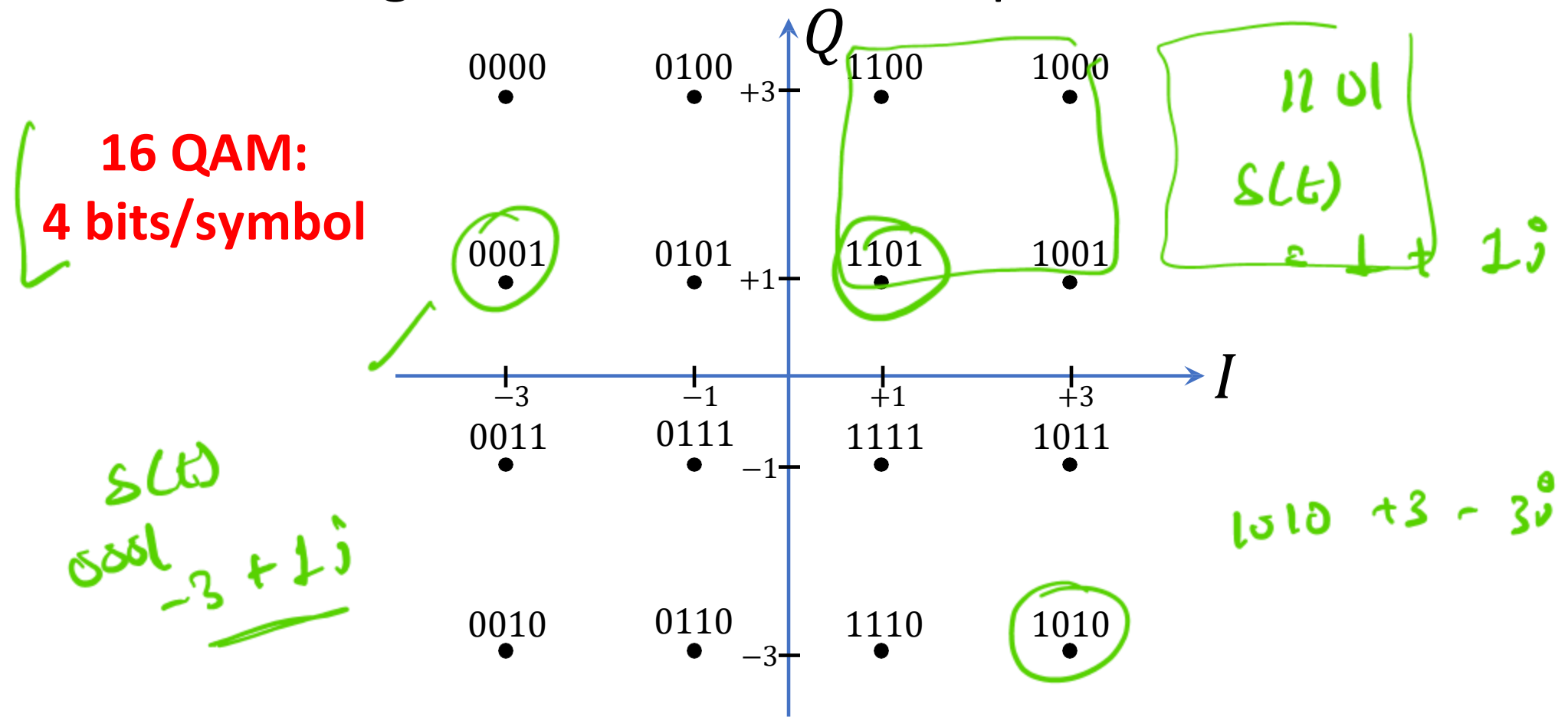
Consider using QAM: Quadrature Amplitude Modulation



Up/Down Conversion



Consider using QAM: Quadrature Amplitude Modulation



Up/Down Conversion

$$s(t) \xrightarrow{\uparrow \cos(2\pi f_c t)} \times \longrightarrow \underline{s(t) \cos(2\pi f_c t)} \xrightarrow{\uparrow \cos(2\pi f_c t)} \times \longrightarrow s(t) \cos^2(2\pi f_c t)$$

Consider using QAM: Quadrature Amplitude Modulation

$$s(t) \text{ is complex: } \underline{s(t) = I + jQ}$$

Up Conversion

$$\underline{s(t)} \longrightarrow \times \longrightarrow s(t) \cos(2\pi f_c t)$$

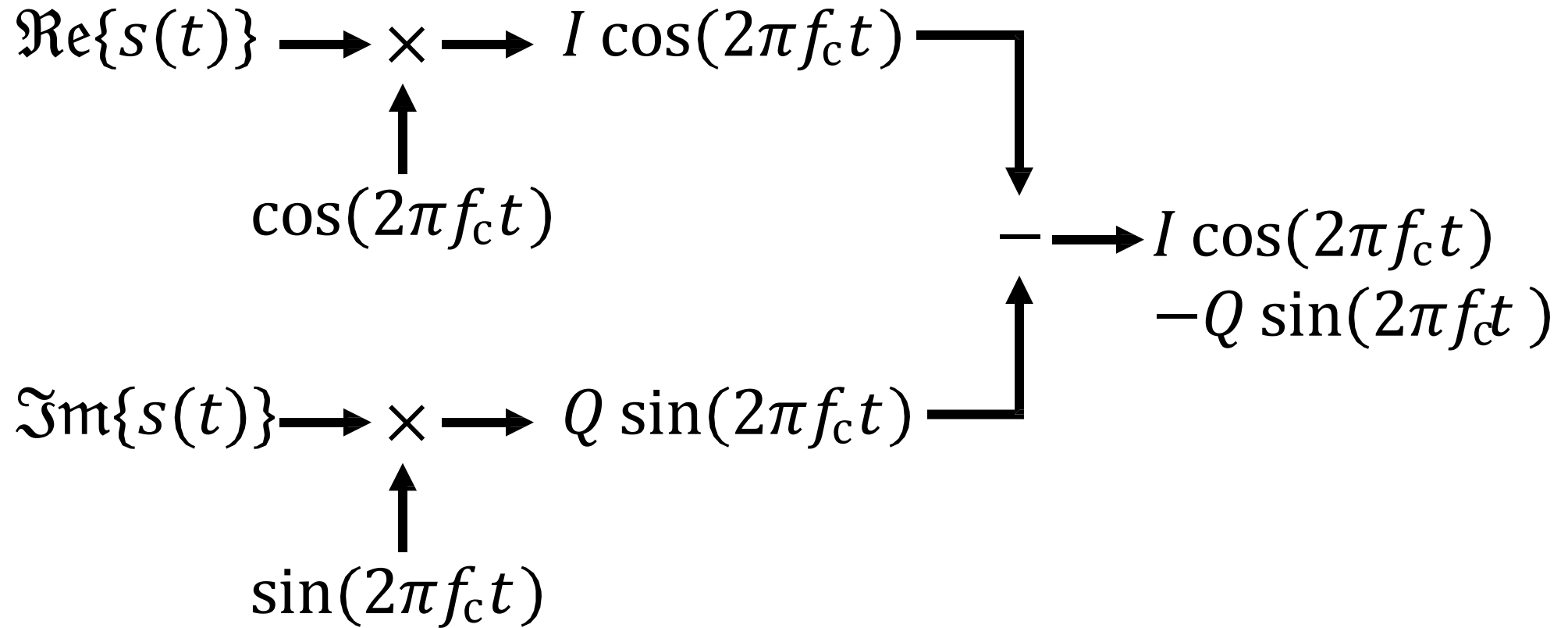
$\uparrow$
 $\cos(2\pi f_c t)$

Traditional real chain

Consider using QAM: Quadrature Amplitude Modulation

$$s(t) \text{ is complex: } s(t) = I + jQ$$

Complex Up Conversion



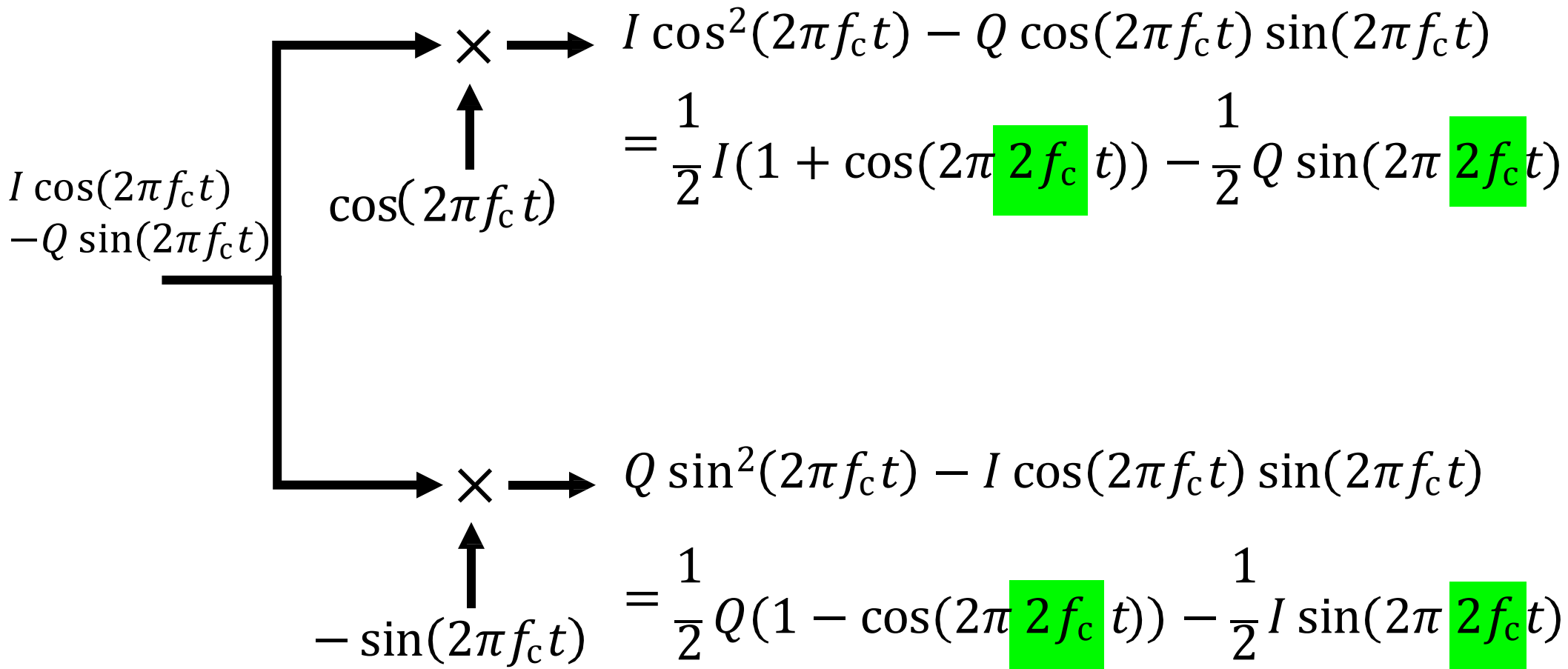
Consider using QAM: Quadrature Amplitude Modulation

$$s(t) \text{ is complex: } s(t) = I + jQ$$

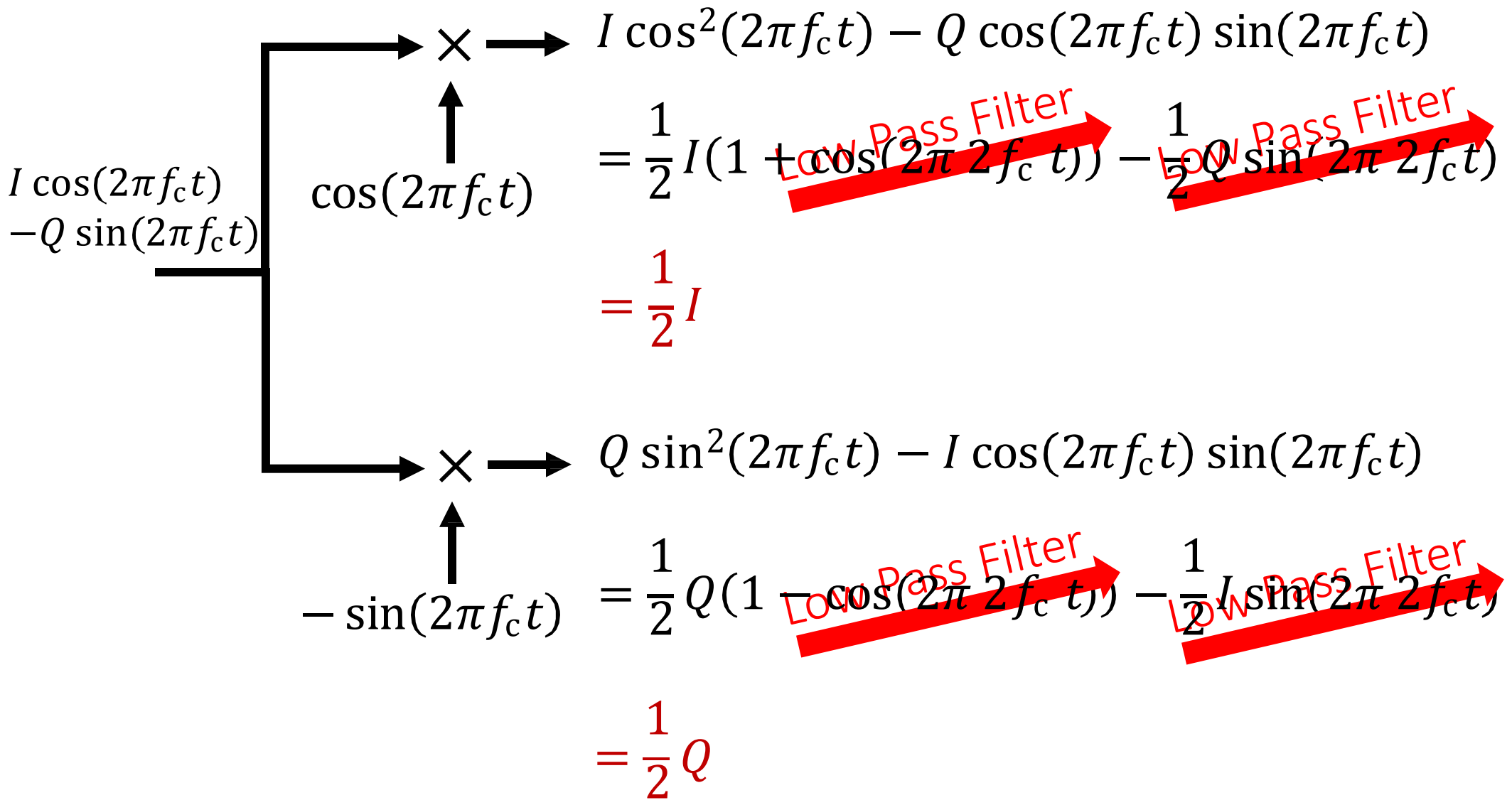
Complex Up Conversion

Sine and Cosine are
Orthogonal

Complex Down Conversion



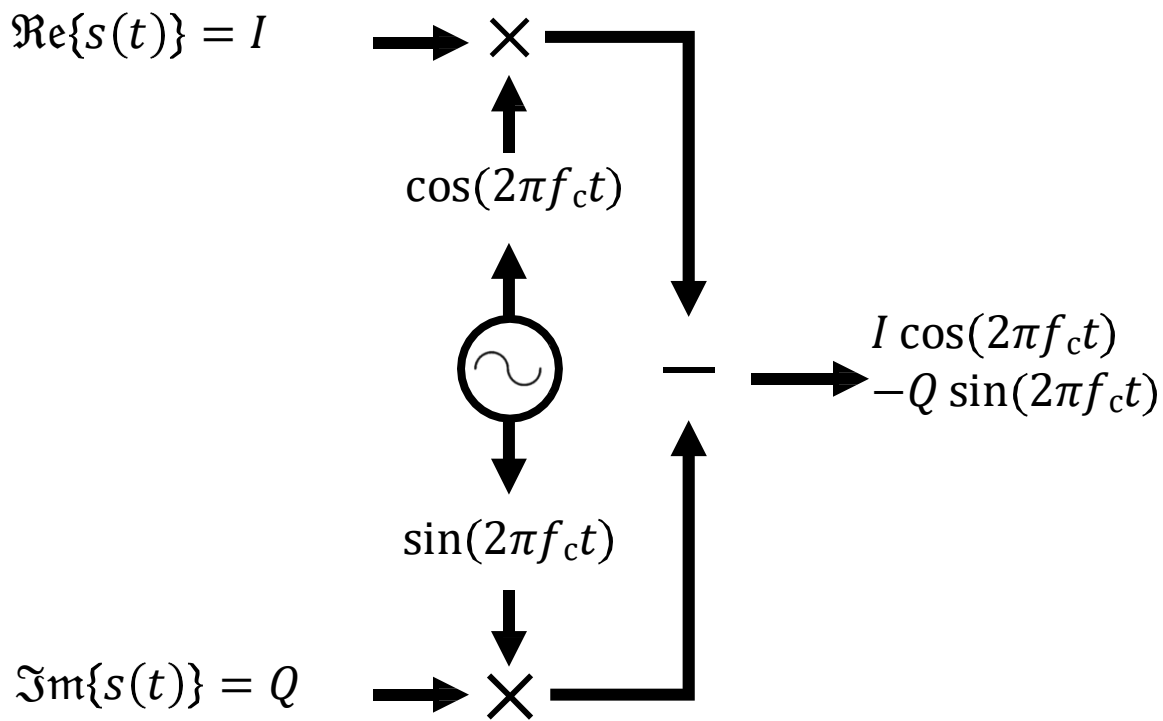
Complex Down Conversion



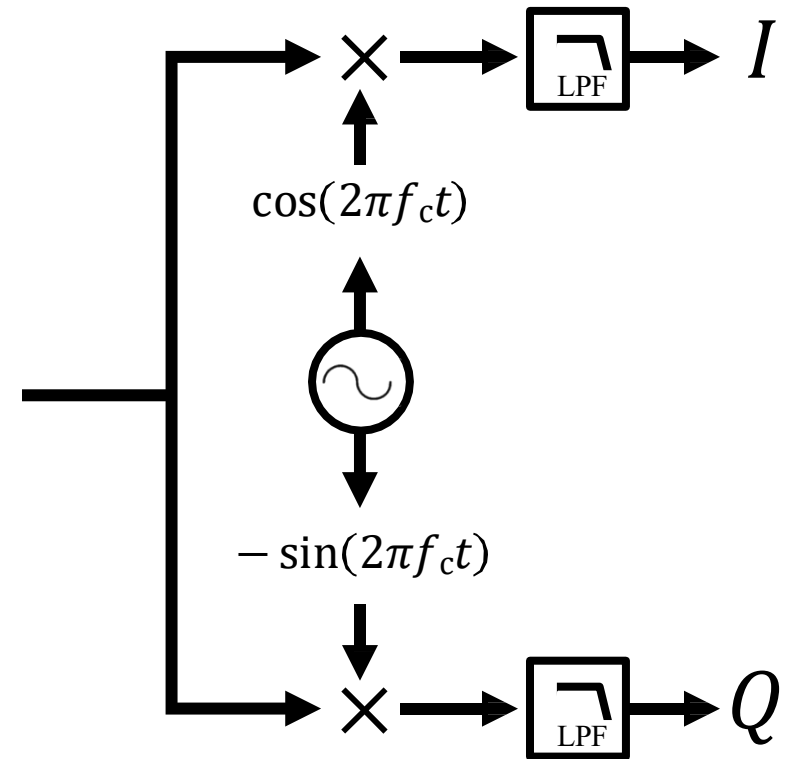
Complex Down Conversion

Up/Down Conversion

Transmitter



Receiver



$$s(t) \times e^{j2\pi f_c t} \rightarrow \Re\{s(t)e^{j2\pi f_c t}\} \rightarrow \times e^{-j2\pi f_c t} \rightarrow s(t)$$

Up/Down Conversion

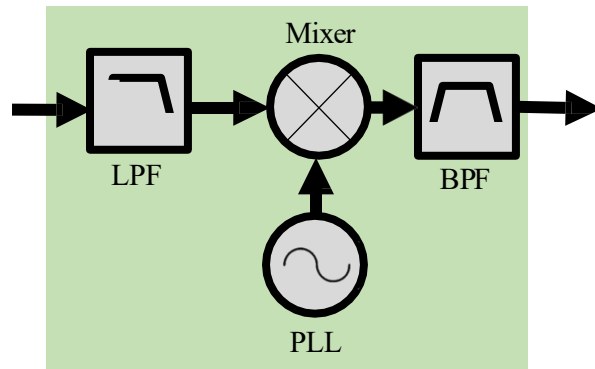
Transmitter

Receiver

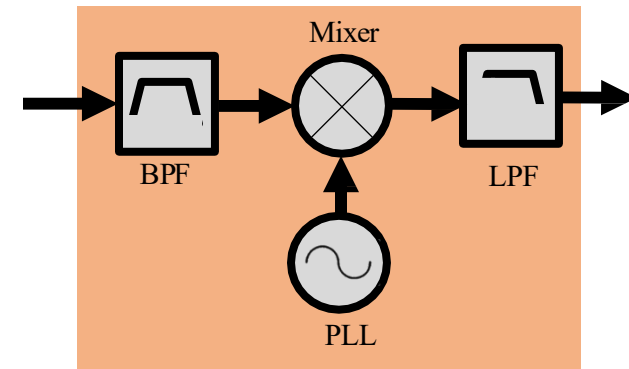
$$s(t) \times e^{j2\pi f_c t} \rightarrow \Re\{s(t)e^{j2\pi f_c t}\} \rightarrow \times e^{-j2\pi f_c t} \rightarrow s(t)$$

Up/Down Conversion

Transmitter
Up-conversion

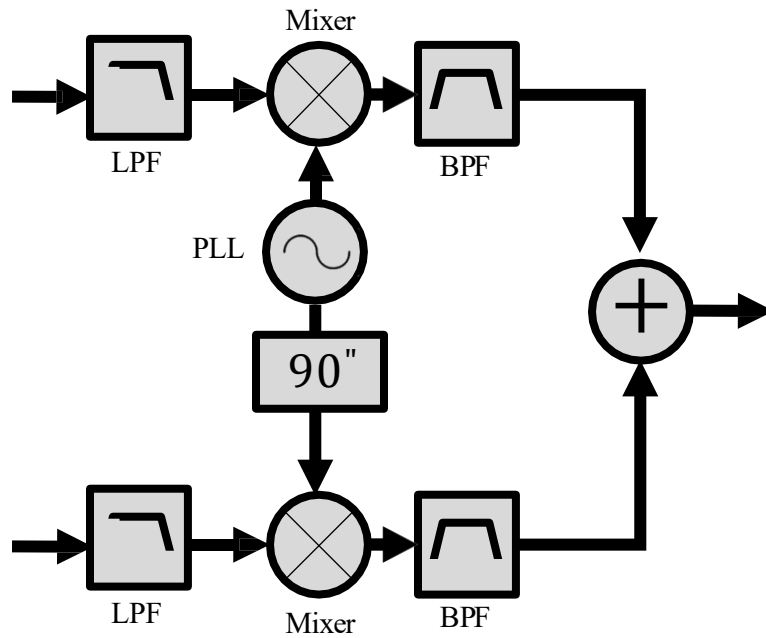


Receiver
Down-conversion

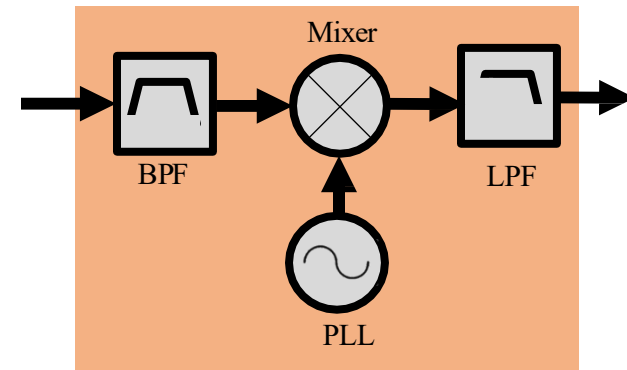


Up/Down Conversion

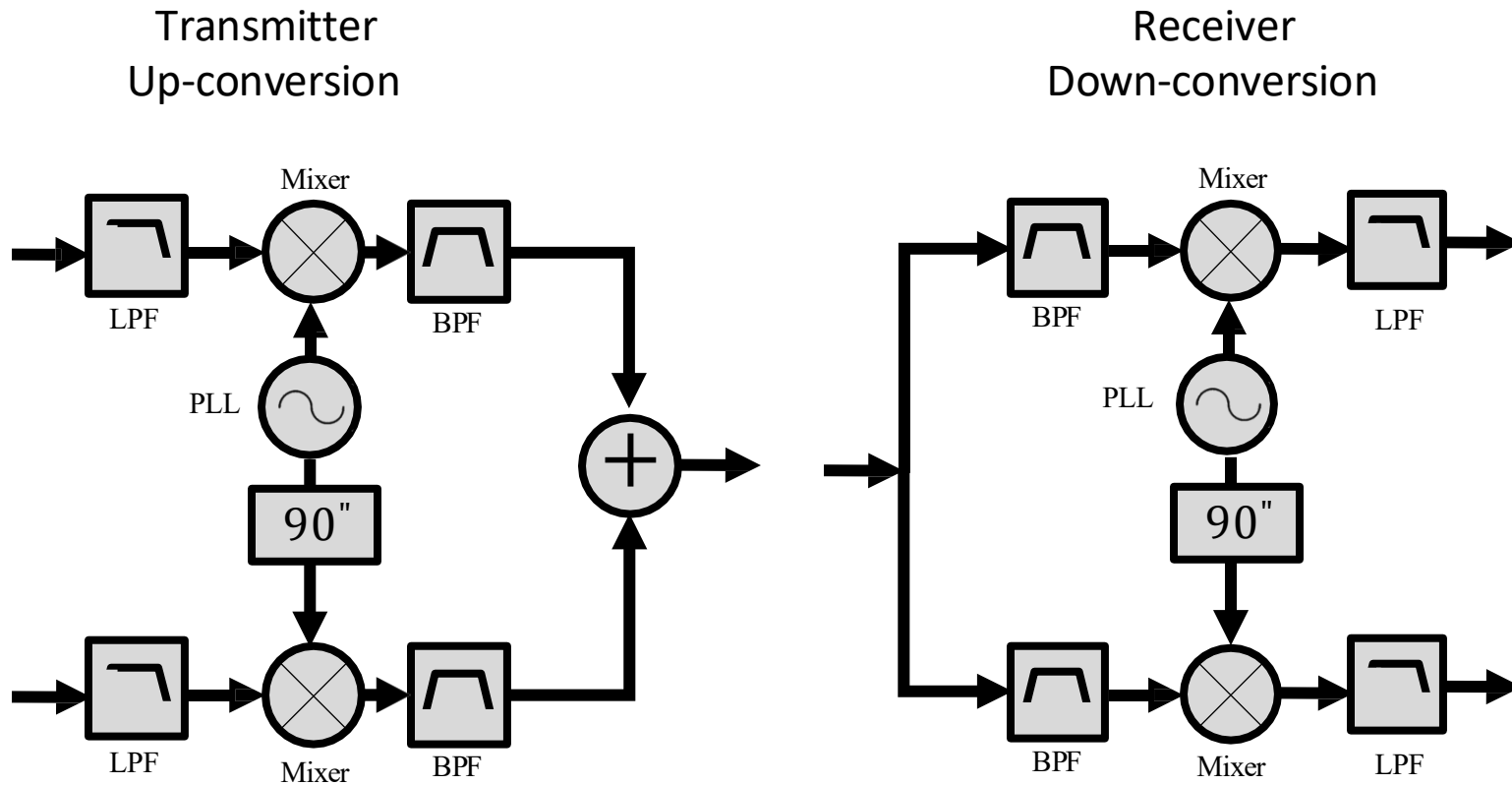
Transmitter
Up-conversion



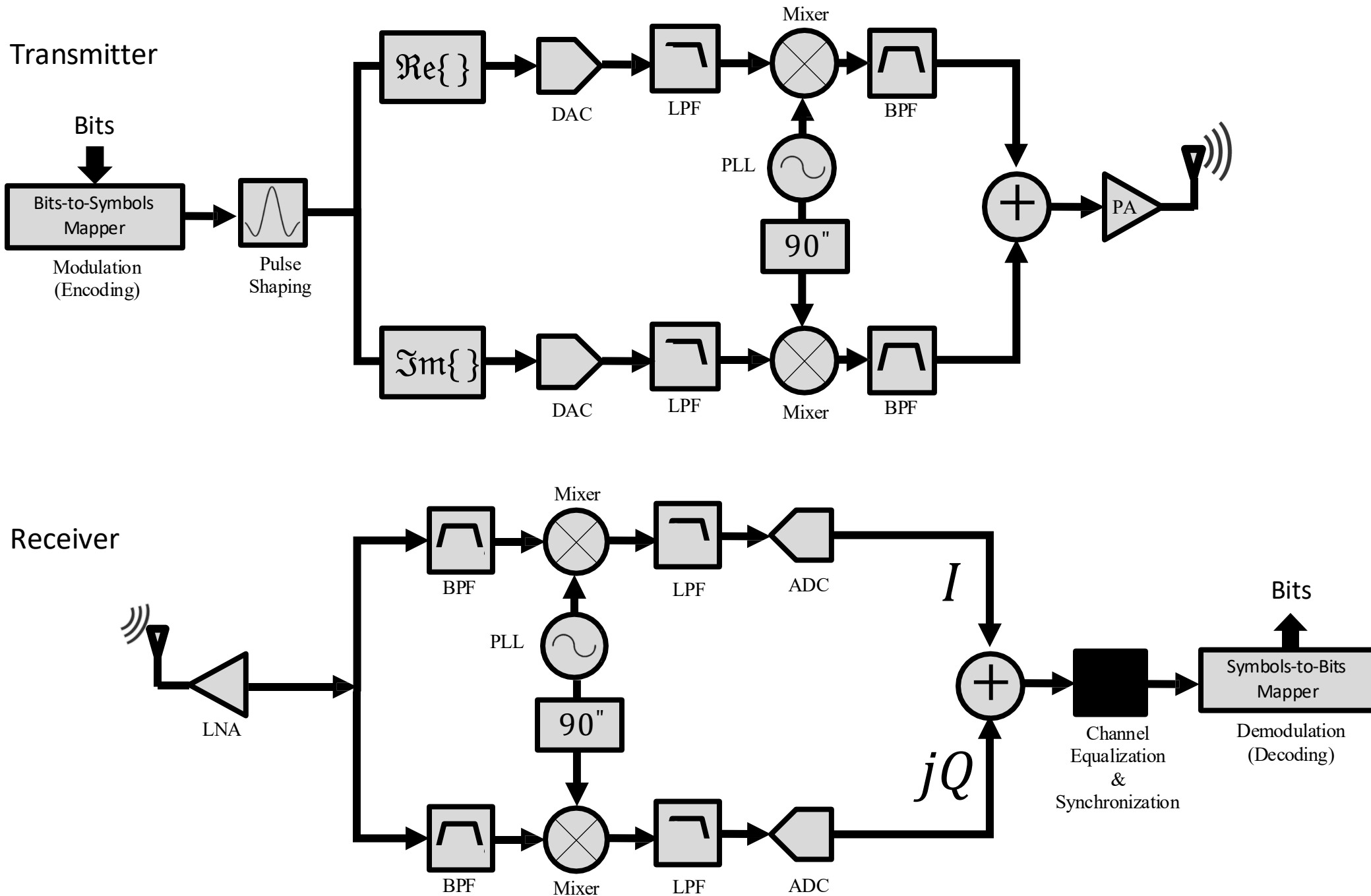
Receiver
Down-conversion



Up/Down Conversion

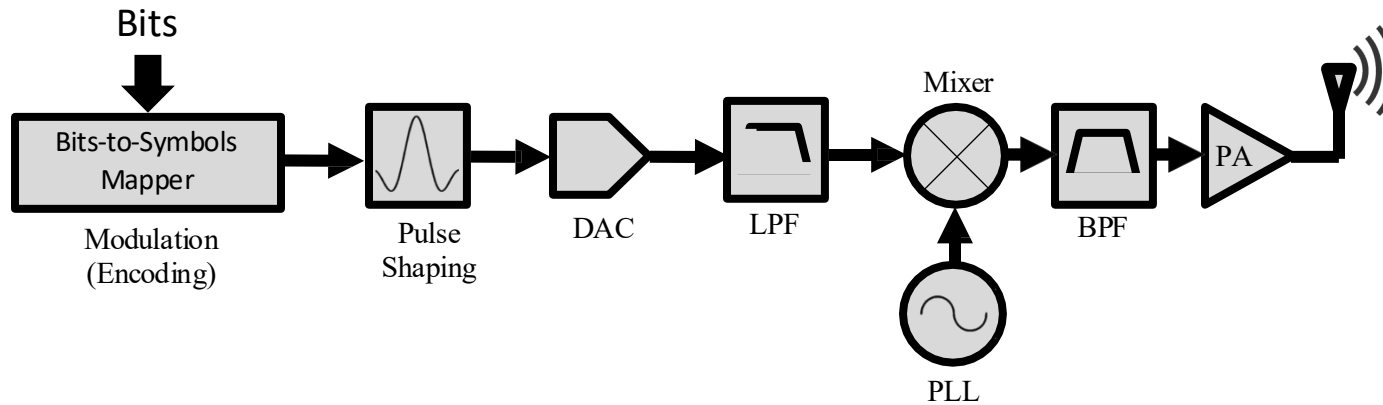


Wireless Communication System

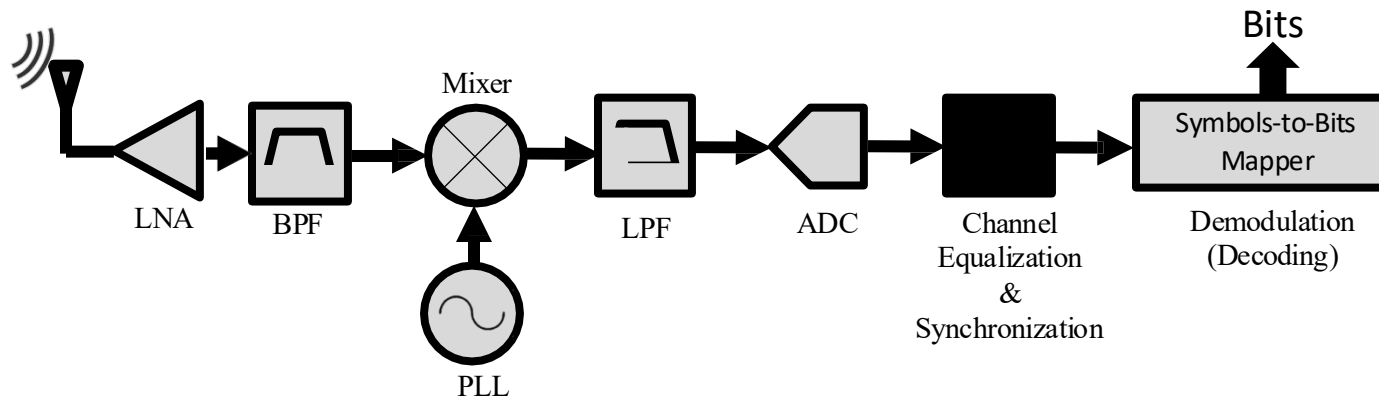


Wireless Communication System

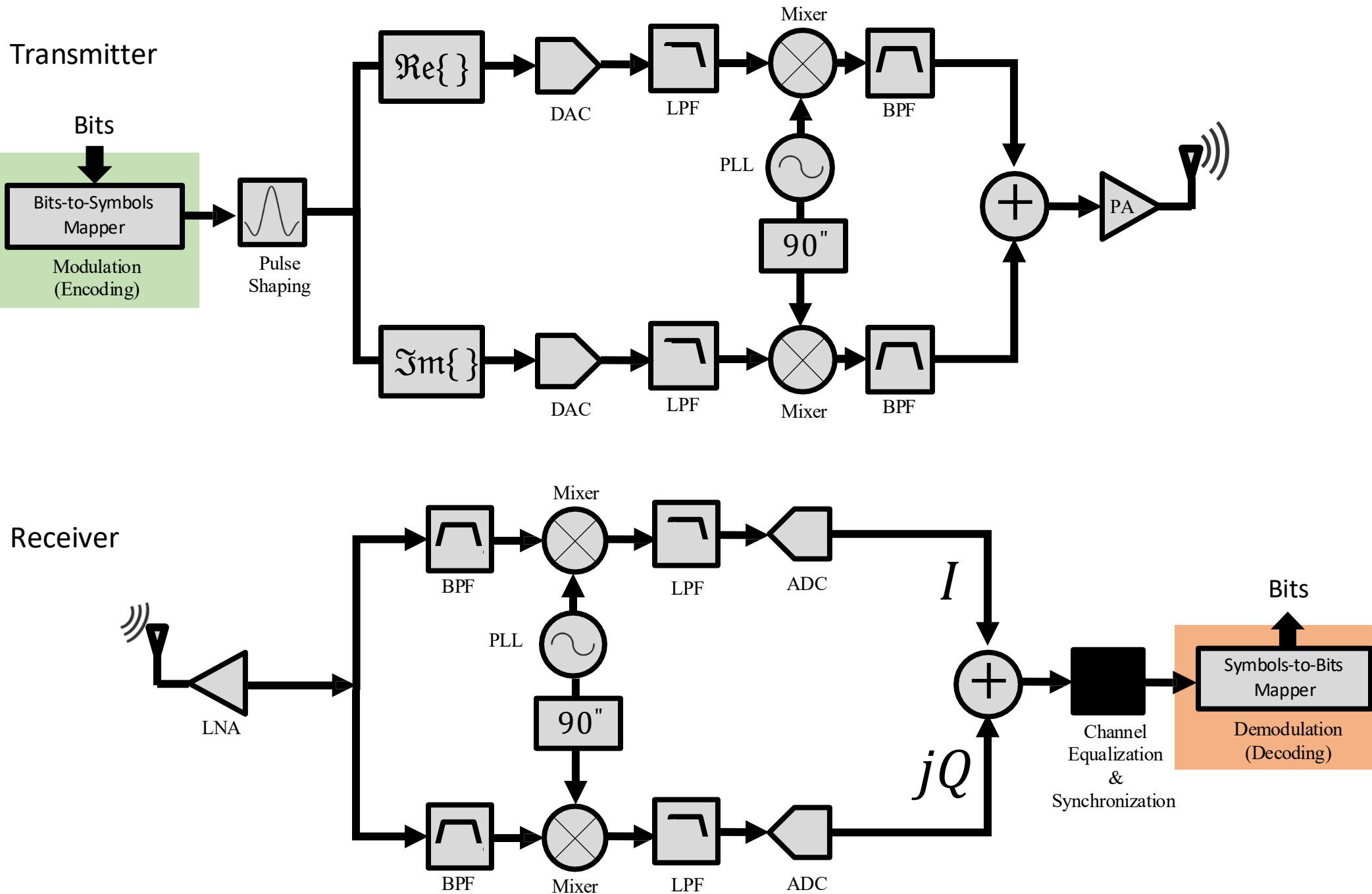
Transmitter



Receiver

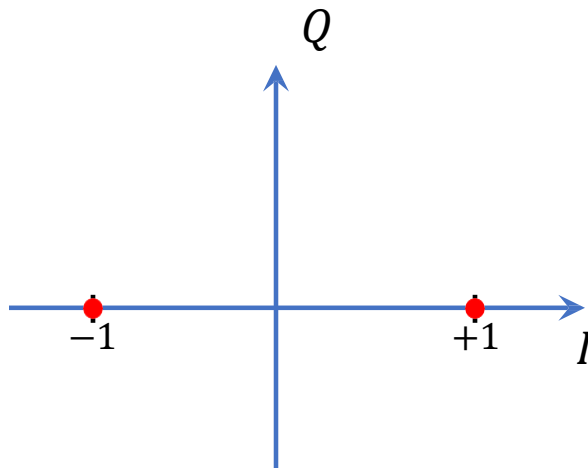


Wireless Communication System



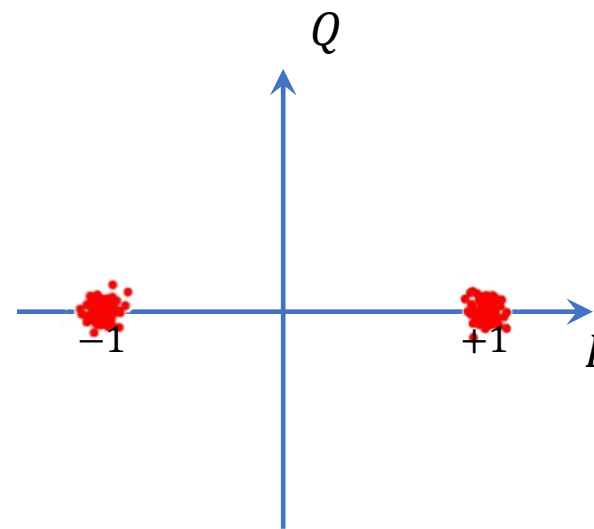
BPSK Modulation

$$\begin{aligned} 0 &\rightarrow -1 \\ 1 &\rightarrow +1 \end{aligned}$$



$$x(t)$$

Transmitted Constellation

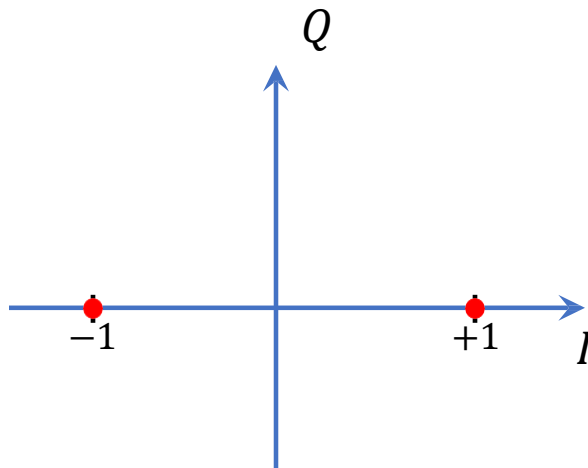


$$y(t) = x(t) + v(t)$$

Received Constellation

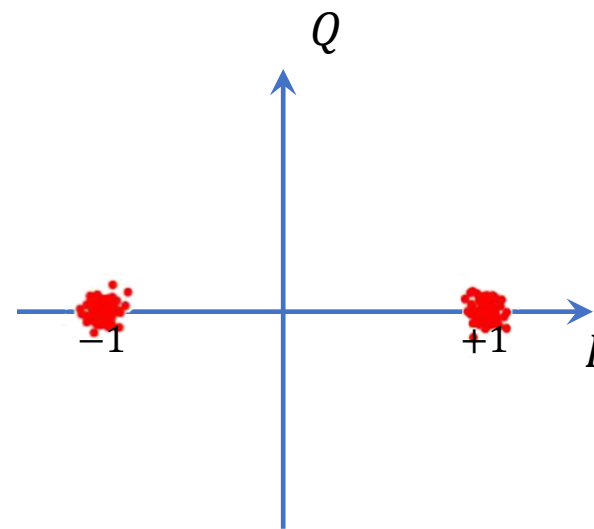
BPSK Modulation

$0 \rightarrow -1$
 $1 \rightarrow +1$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

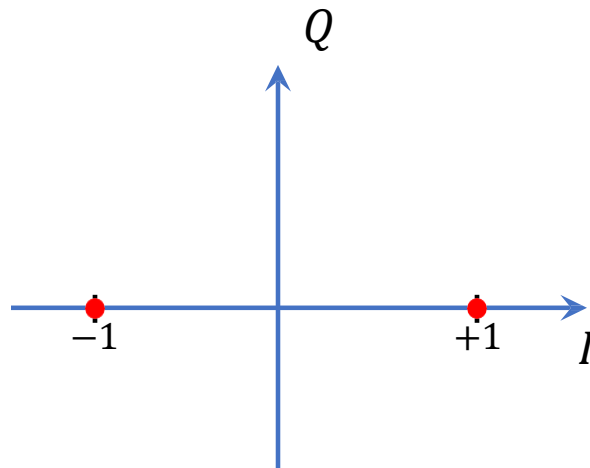
Received Constellation

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E[|x(t)|^2]}{E[|v(t)|^2]} = \frac{E_s}{N_0}$$

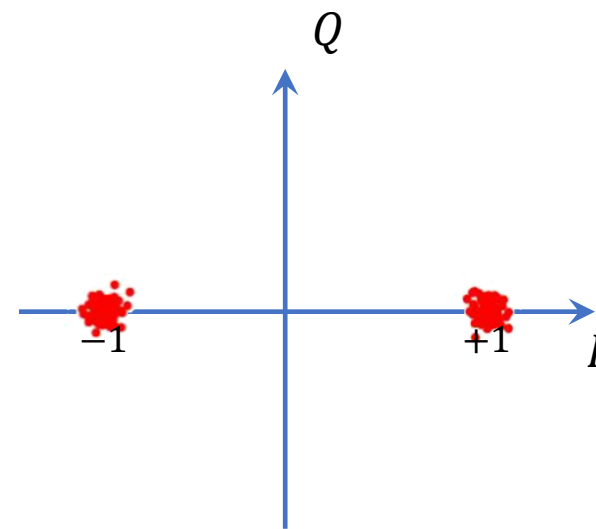
BPSK Modulation

$$\begin{aligned} 0 &\rightarrow -1 \\ 1 &\rightarrow +1 \end{aligned}$$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

Received Constellation

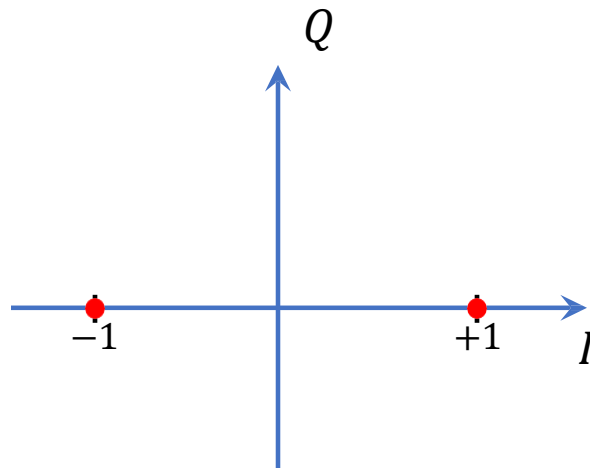
AWGN

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E_s}{N_0} = 25 \text{ dB}$$

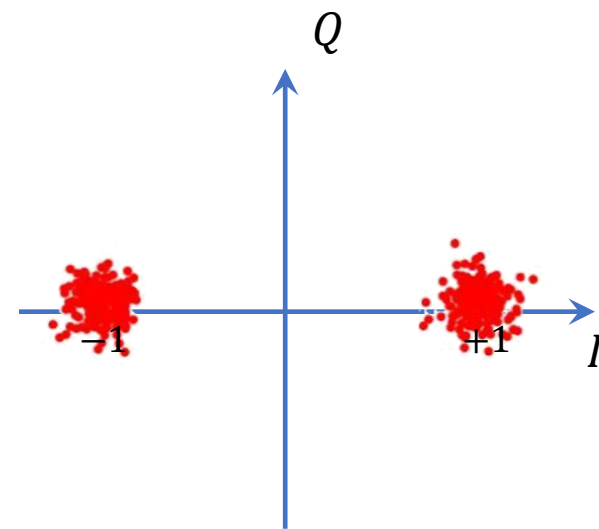
BPSK Modulation

$$\begin{aligned} 0 &\rightarrow -1 \\ 1 &\rightarrow +1 \end{aligned}$$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

Received Constellation

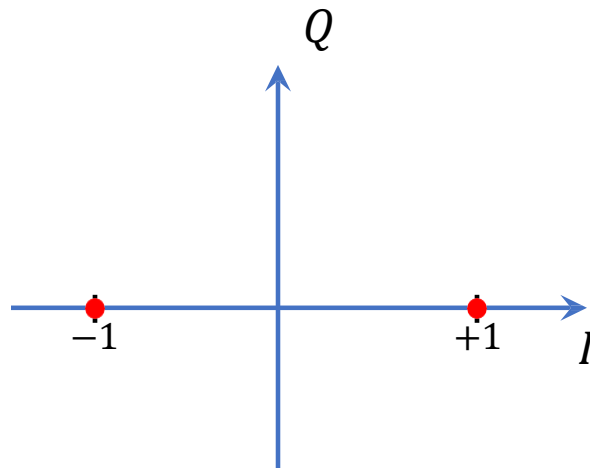
AWGN

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E_s}{N_0} = 19 \text{ dB}$$

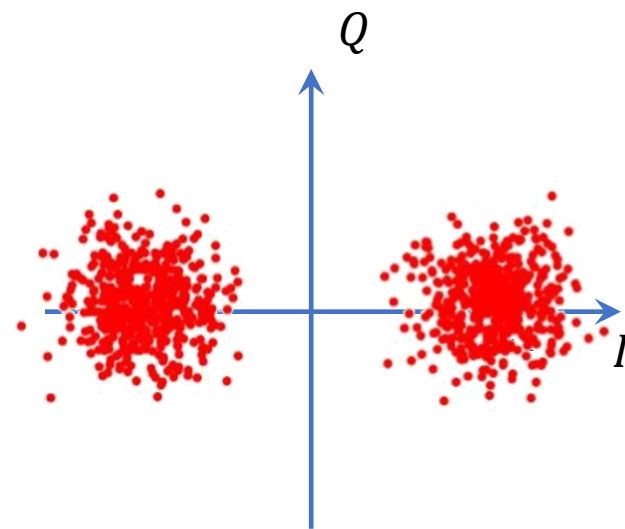
BPSK Modulation

$$\begin{aligned} 0 &\rightarrow -1 \\ 1 &\rightarrow +1 \end{aligned}$$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

Received Constellation

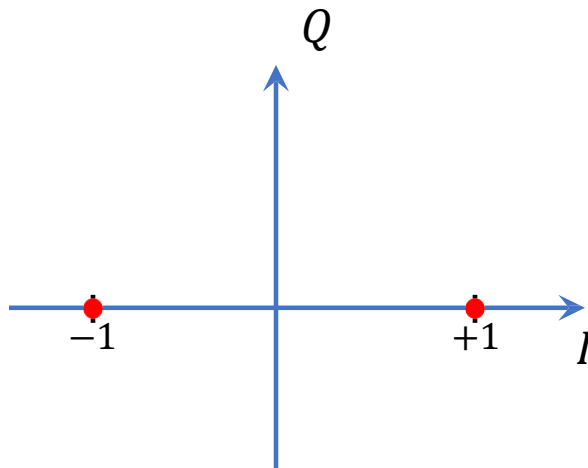
AWGN

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E_s}{N_0} = 13 \text{ dB}$$

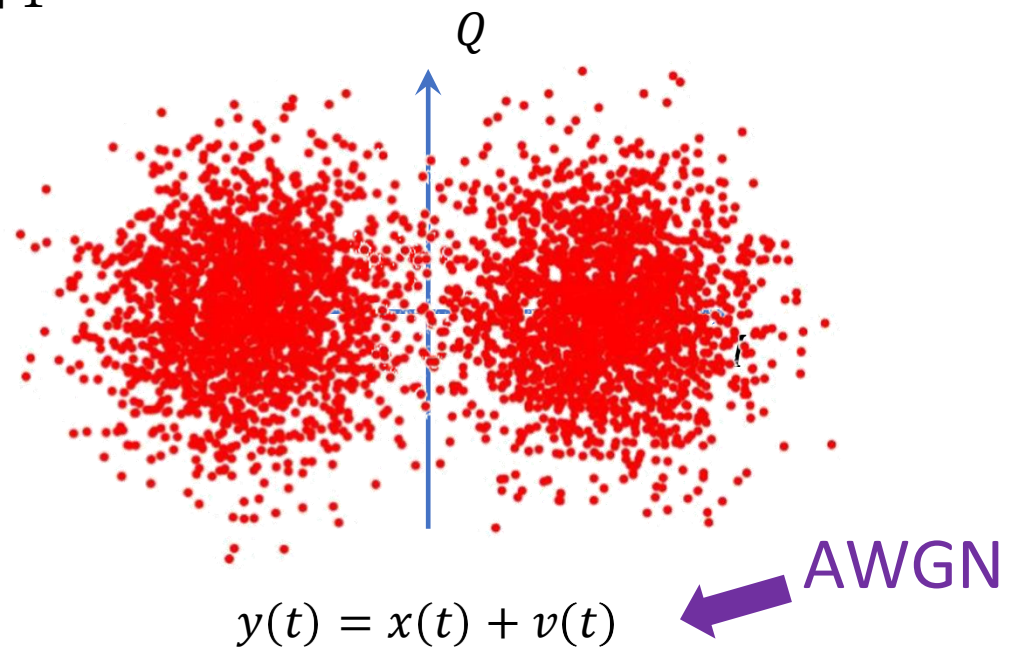
BPSK Modulation

$$\begin{aligned} 0 &\rightarrow -1 \\ 1 &\rightarrow +1 \end{aligned}$$



$x(t)$

Transmitted Constellation



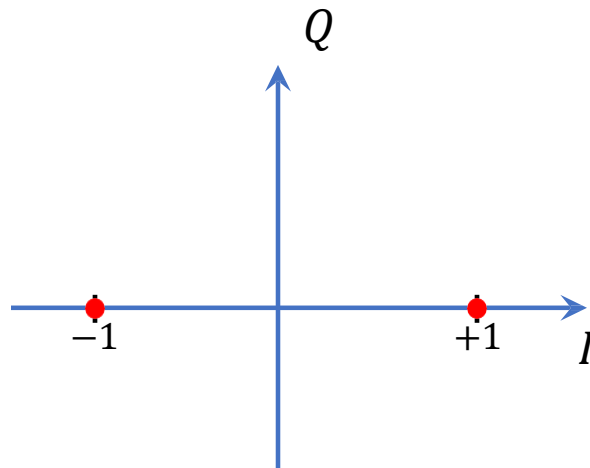
Received Constellation

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E_s}{N_0} = 7 \text{ dB}$$

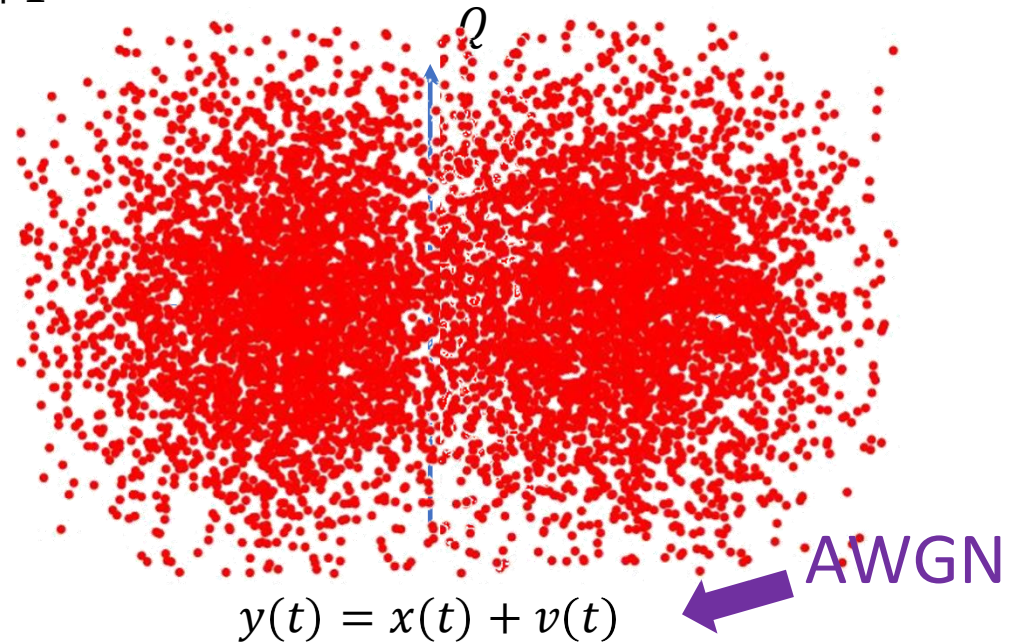
BPSK Modulation

$0 \rightarrow -1$
 $1 \rightarrow +1$



$x(t)$

Transmitted Constellation

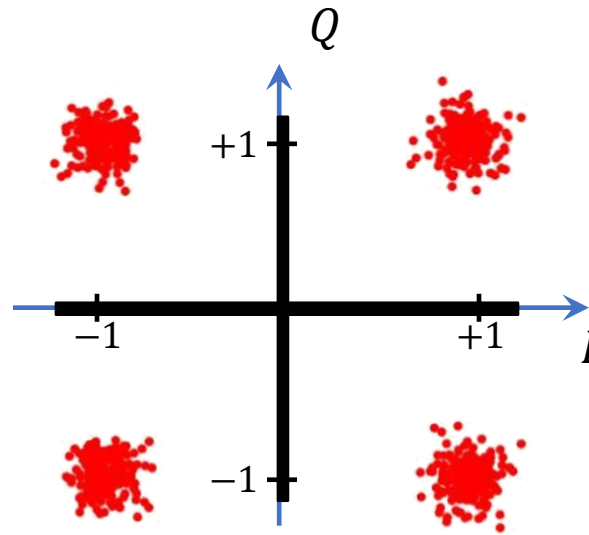


Received Constellation

How well we can decode depends on SNR?

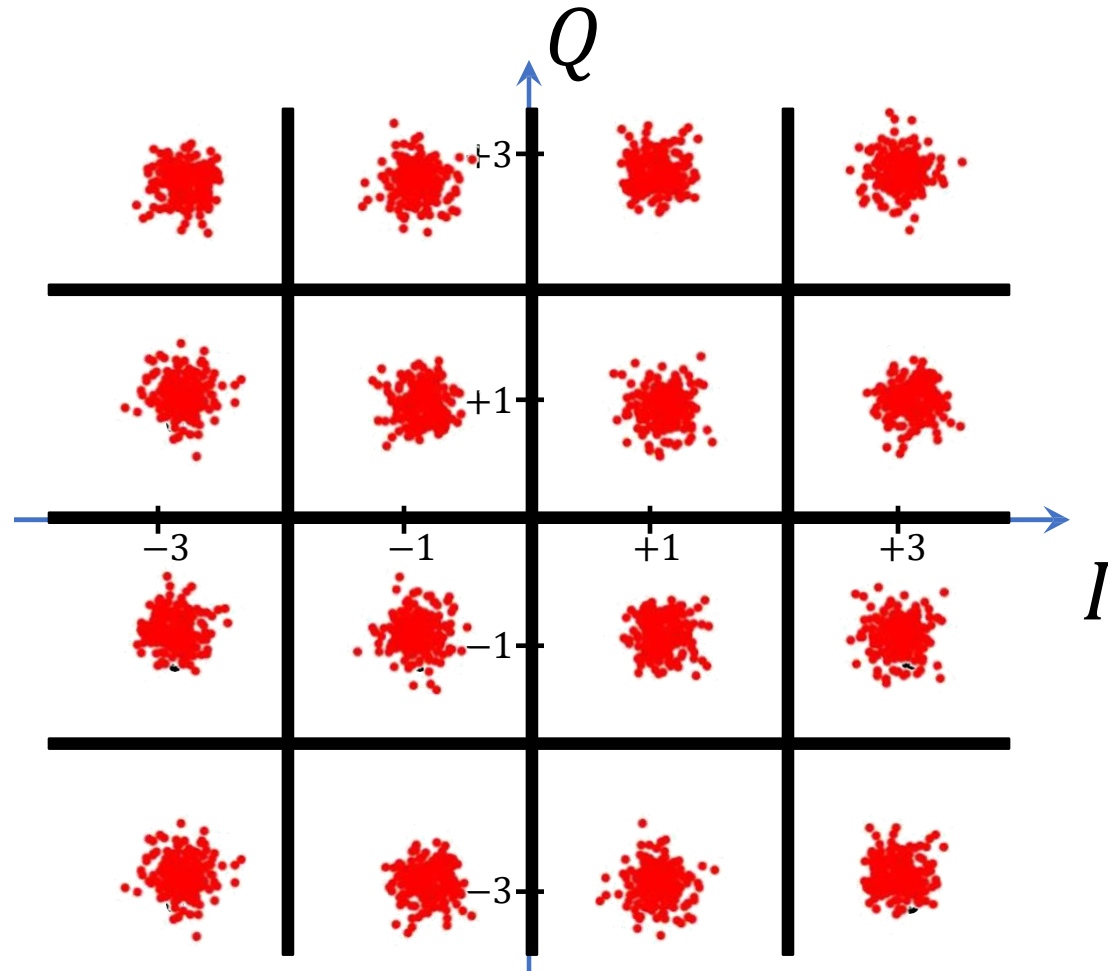
$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E_s}{N_0} = 3.5 \text{ dB}$$

4-QAM



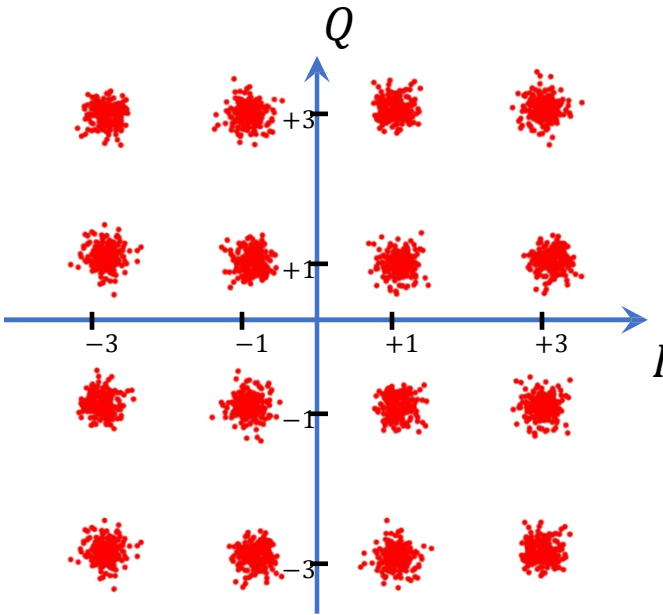
Encodes 2 bits per symbol resulting in Higher Data Rate

16-QAM

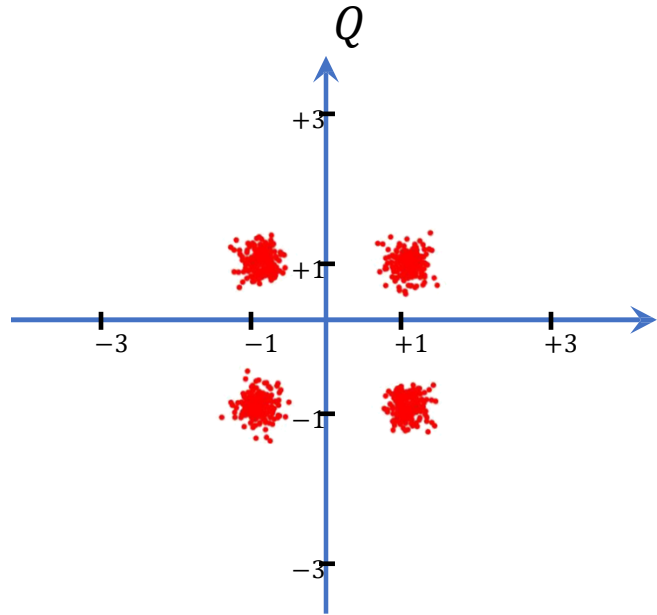


Encodes 4 bits per symbol resulting in Higher Data Rate

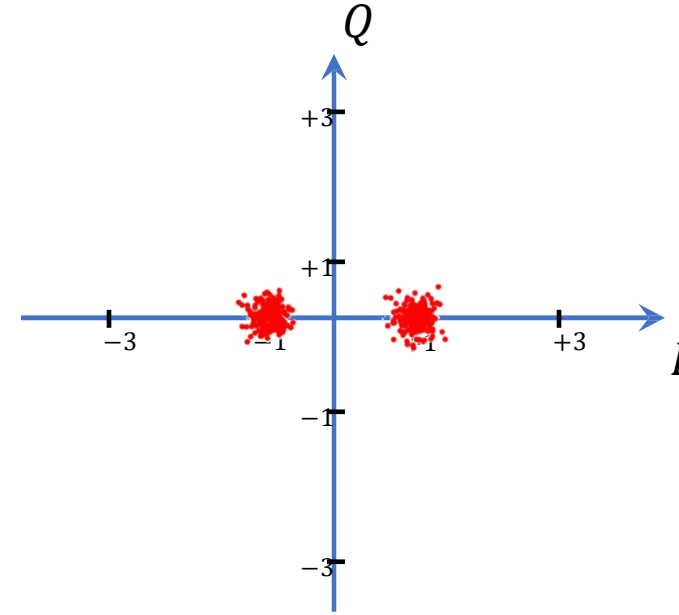
16-QAM



4-QAM



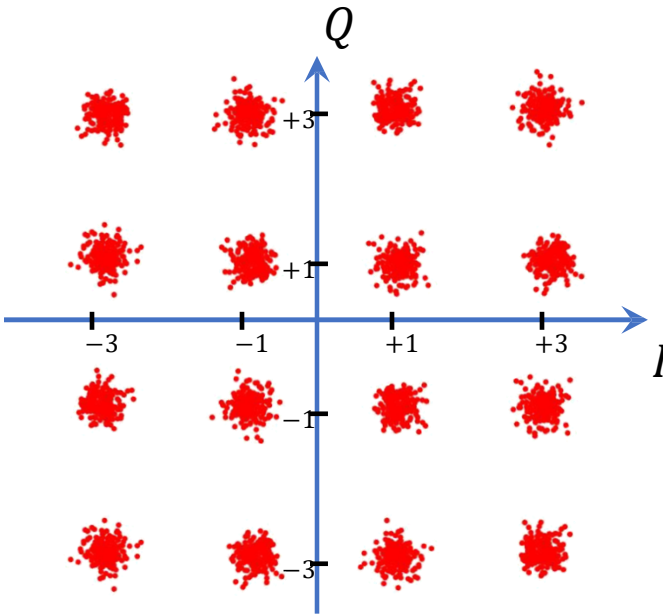
BPSK



Is the SNR the same in these 3 constellations?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E[|x(t)|^2]}{E[|v(t)|^2]} = \frac{E_s}{N_0}$$

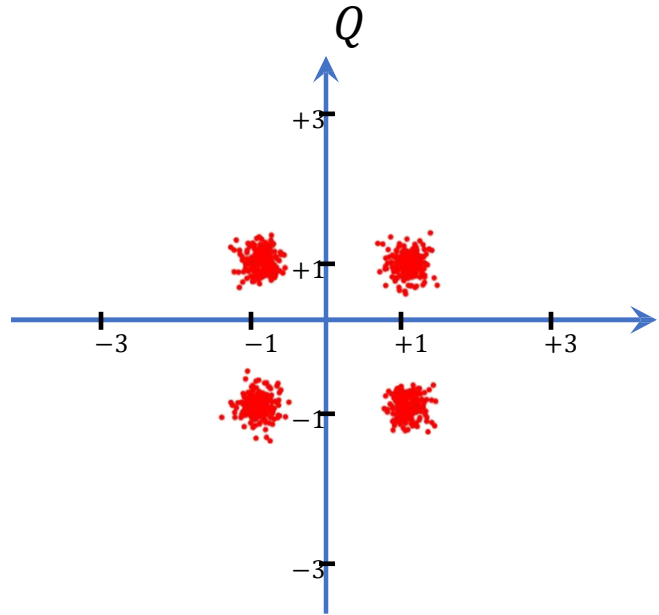
16-QAM



$$SNR = \frac{(4 \times 2 + 8 \times 10 + 4 \times 18) / 16}{N_0}$$

$$= \frac{10}{N_0}$$

4-QAM

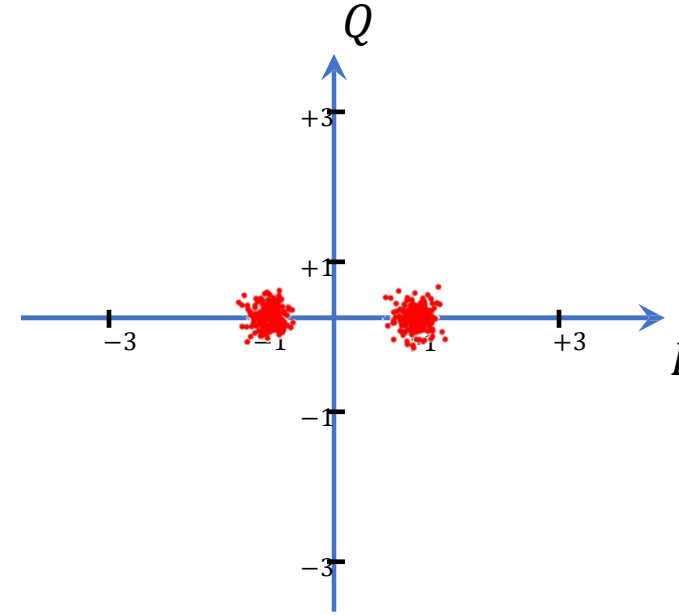


$$SNR = \frac{E[|x(t)|^2]}{N_0}$$

$$SNR = \frac{(2 + 2 + 2 + 2) / 4}{N_0}$$

$$= \frac{2}{N_0}$$

BPSK



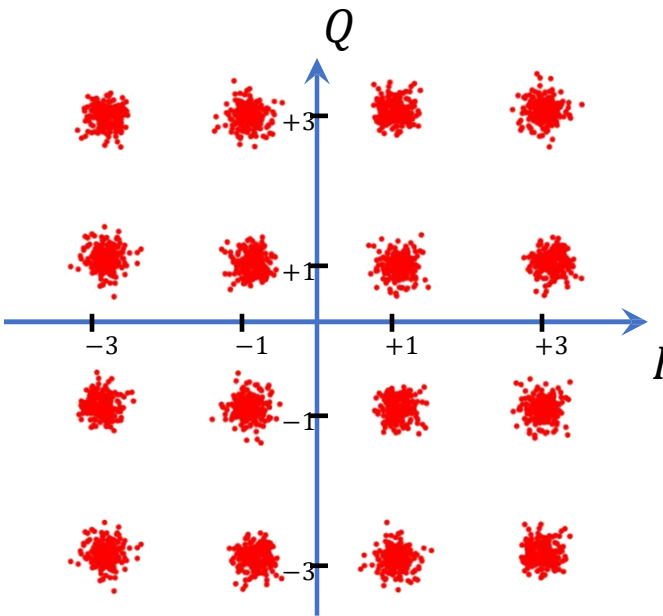
$$SNR = \frac{(1 + 1) / 2}{N_0}$$

$$= \frac{1}{N_0}$$

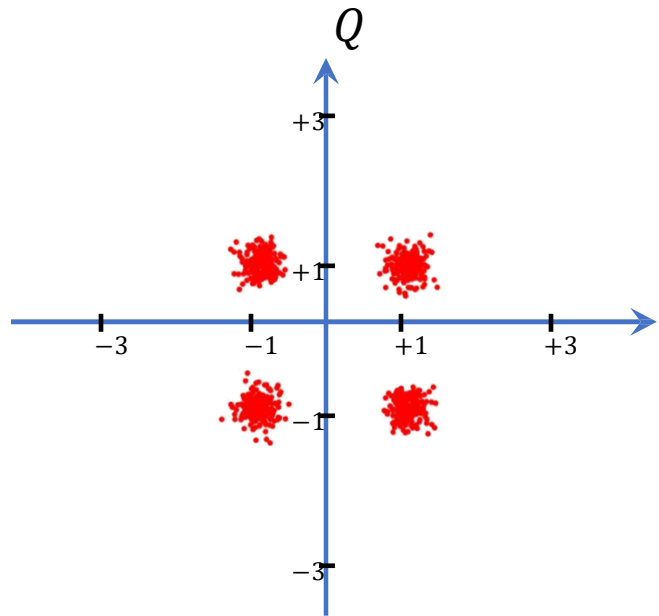
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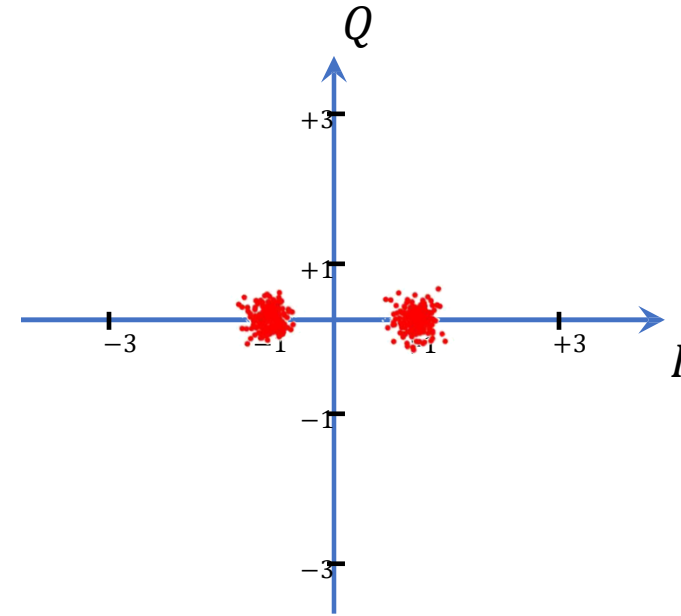
16-QAM



4-QAM



BPSK



Transmit power is not the same!

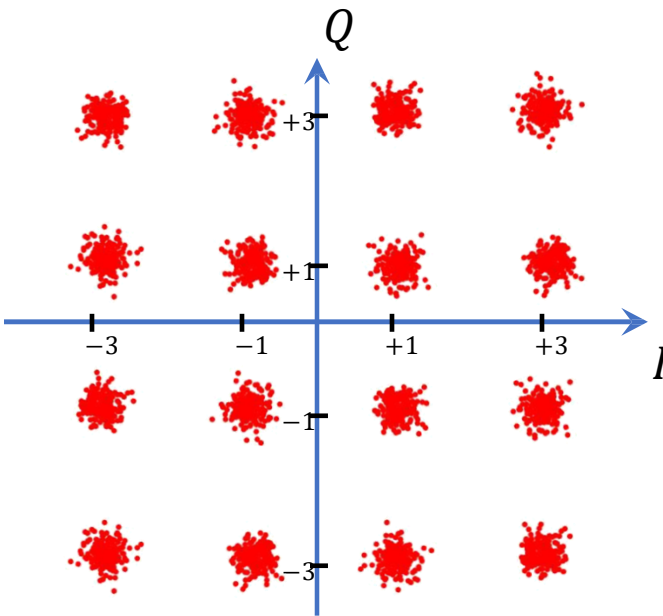
$$SNR = \frac{(4 \times 2 + 8 \times 10 + 4 \times 18)}{N_0}$$

$$= \frac{10}{N_0}$$

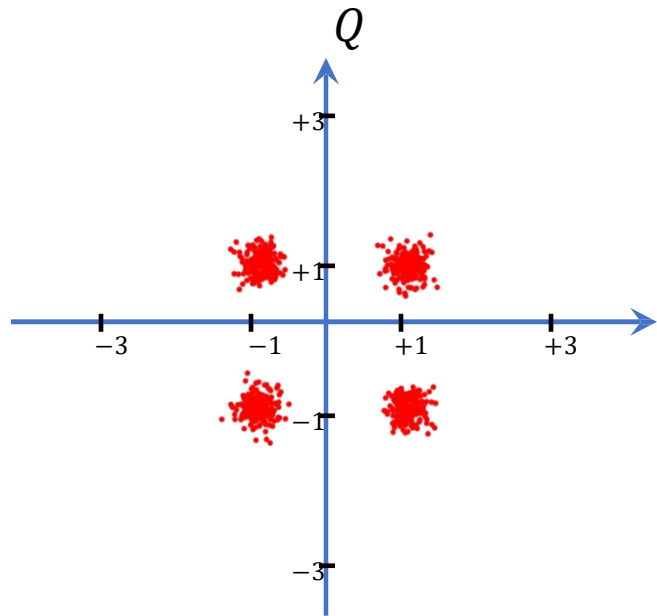
Must not increase transmit power to use higher order modulation!

$$\frac{1}{2} \frac{1}{N_0}$$

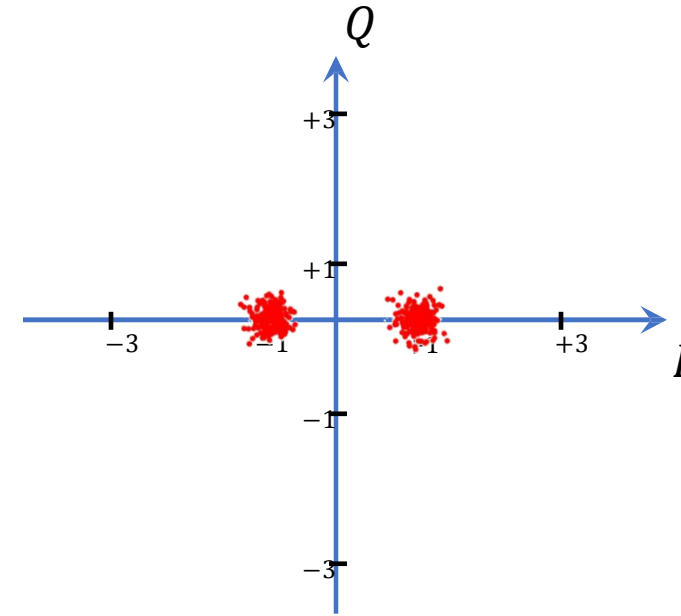
16-QAM



4-QAM



BPSK



Normalize to maintain constant transmit power.

$$SNR = \frac{(4 \times 2 + 8 \times 10 + 4 \times 18) / 16}{N_0}$$

$$= \frac{10}{N_0}$$

>

$$SNR = \frac{(2 + 2 + 2 + 2) / 4}{N_0}$$

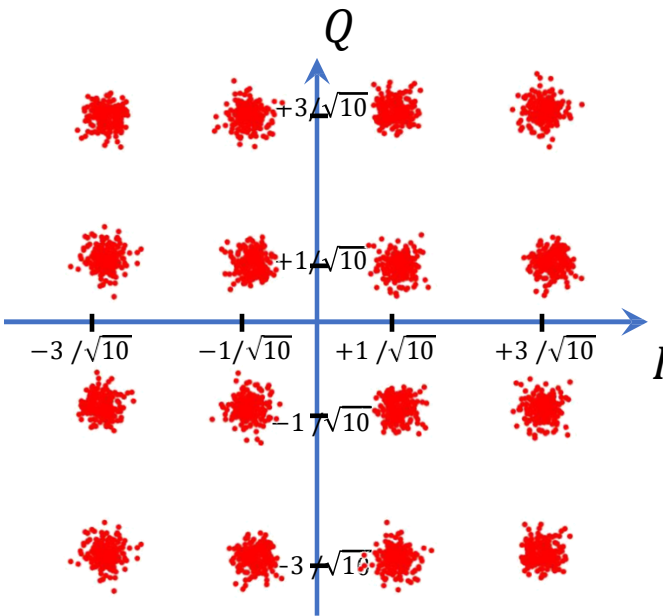
$$= \frac{2}{N_0}$$

>

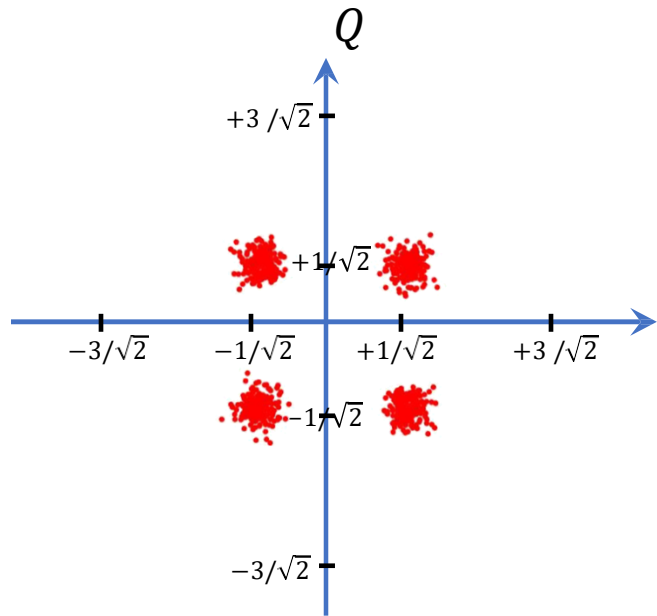
$$SNR = \frac{(1 + 1) / 2}{N_0}$$

$$= \frac{1}{N_0}$$

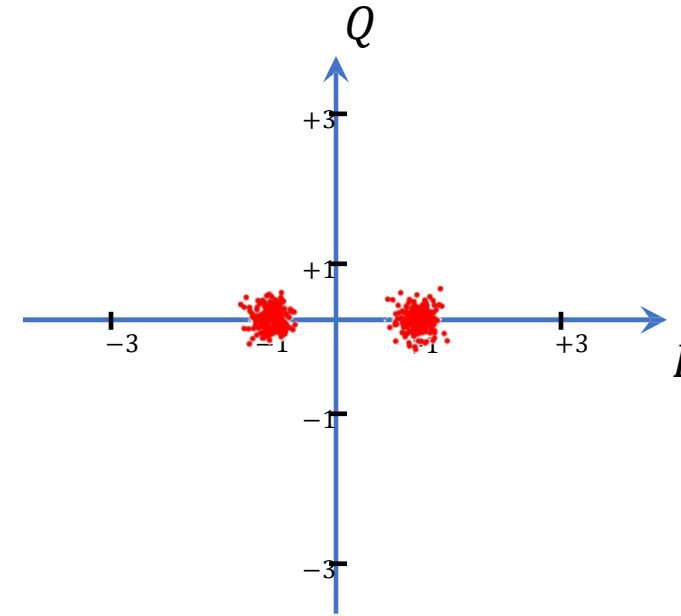
16-QAM



4-QAM



BPSK



Normalize to maintain constant transmit power.

$$SNR = \frac{(4 \times 2/10 + 8 \times 1 + 4 \times 18/10)/16}{N_0}$$

$$= \frac{1}{N_0}$$

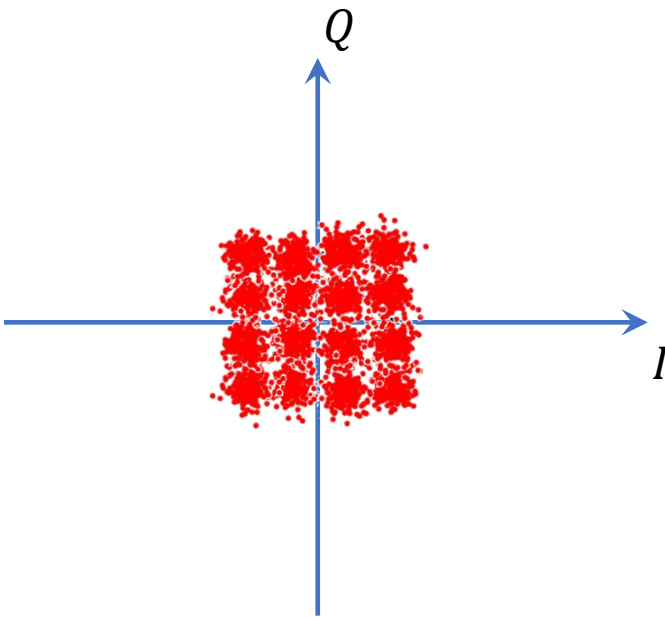
$$SNR = \frac{(1 + 1 + 1 + 1)/4}{N_0}$$

$$= \frac{1}{N_0}$$

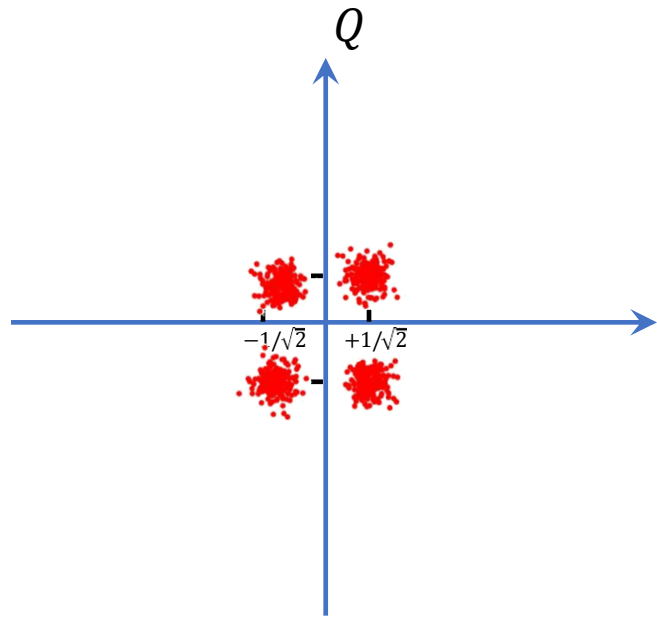
$$SNR = \frac{(1 + 1)/2}{N_0}$$

$$= \frac{1}{N_0}$$

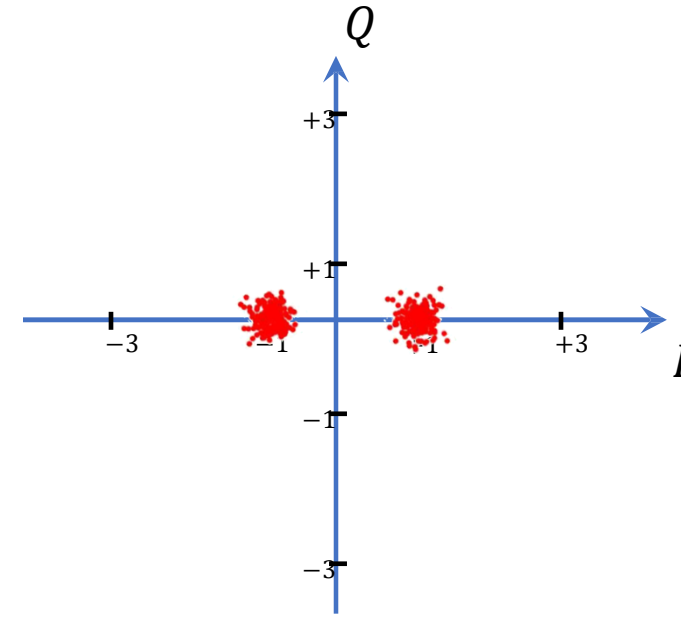
16-QAM



4-QAM



BPSK



Normalize to maintain constant transmit power.

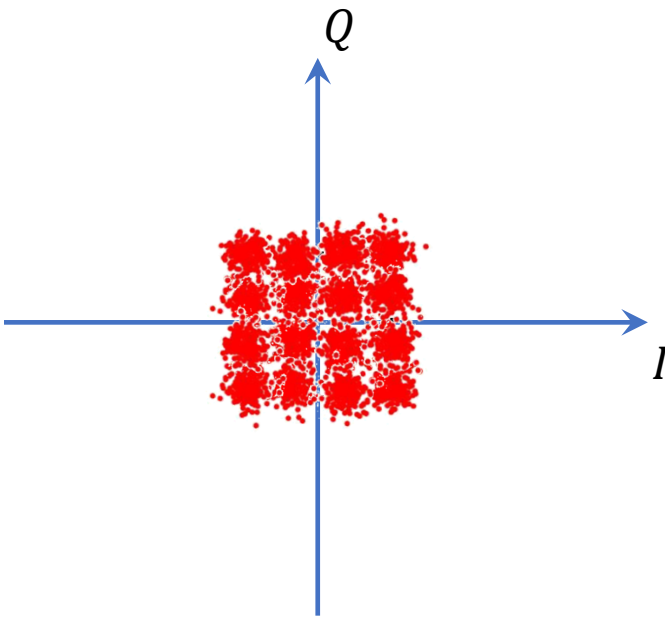
$$SNR = \frac{1}{N_0}$$

$$SNR = \frac{1}{N_0}$$

$$SNR = \frac{1}{N_0}$$

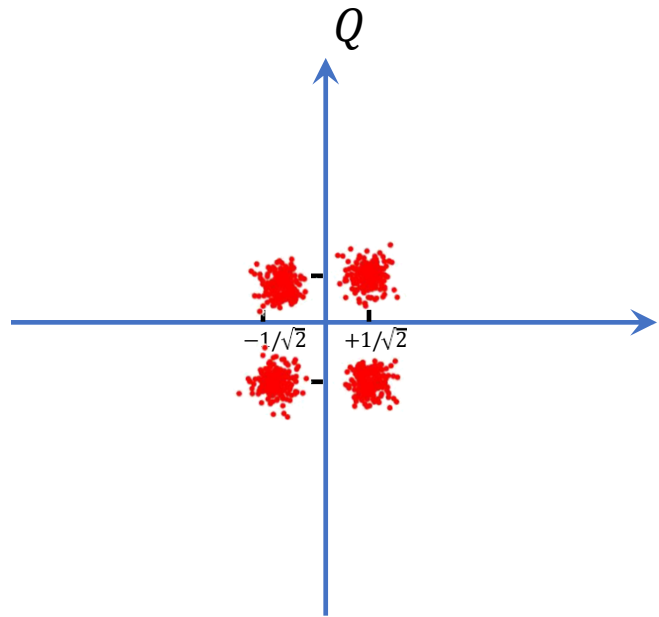
Need Higher SNR to Decode
Higher Order Modulation.

16-QAM



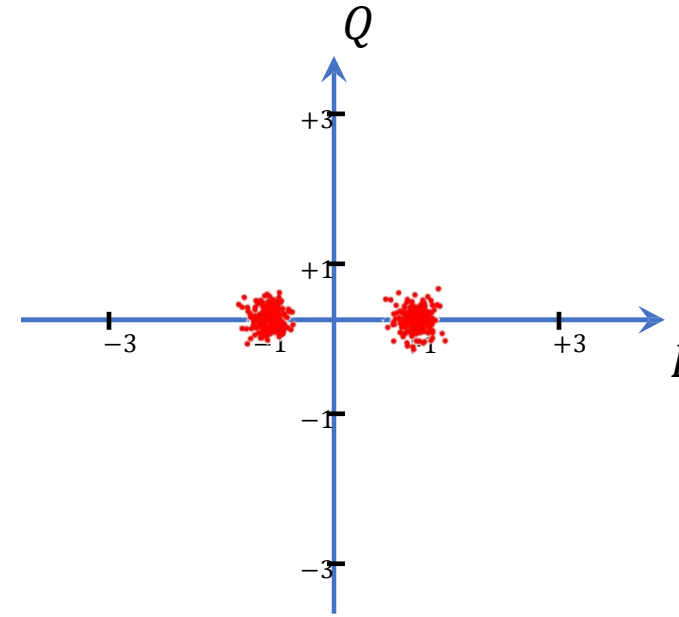
4 bits/symbol

4-QAM



2 bits/symbol

BPSK



1 bit/symbol

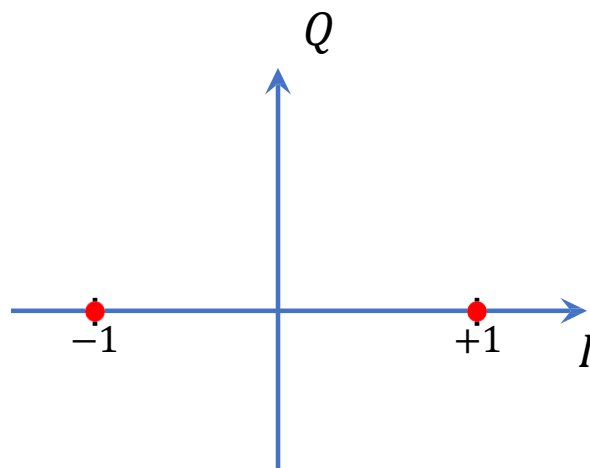
Higher Order Modulation:

- Needs higher SNR to decode correctly.
- Achieves higher Bite Rate

Given an SNR, choose highest order modulation that guarantees minimal Bite Error Rate (BER)

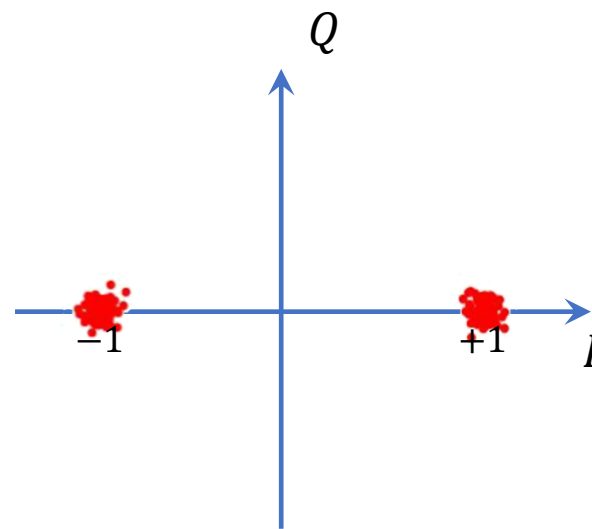
Decoding under noise

$$\begin{aligned} 0 &\rightarrow -1 \\ 1 &\rightarrow +1 \end{aligned}$$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

Received Constellation

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E[|x(t)|^2]}{E[|v(t)|^2]} = \frac{E_s}{N_0}$$

Decoding under noise

Assume Data: $y_i \sim \mathcal{N}(\theta, \sigma^2)$

MLE (maximum likelihood):

$$\hat{\theta}_{MLE} = \arg \max_{\theta} p(\mathbf{y} \mid \theta)$$

Chooses the parameter that makes the observed data most likely.

MAP (maximum a posteriori):

$$\hat{\theta}_{MAP} = \arg \max_{\theta} p(\theta \mid \mathbf{y}) = \arg \max_{\theta} \underbrace{\log p(\mathbf{y} \mid \theta)}_{\text{likelihood}} + \underbrace{\log p(\theta)}_{\text{prior}}$$

Maximum Likelihood Estimator

Maximum A Posteriori Estimator

Decision Boundary

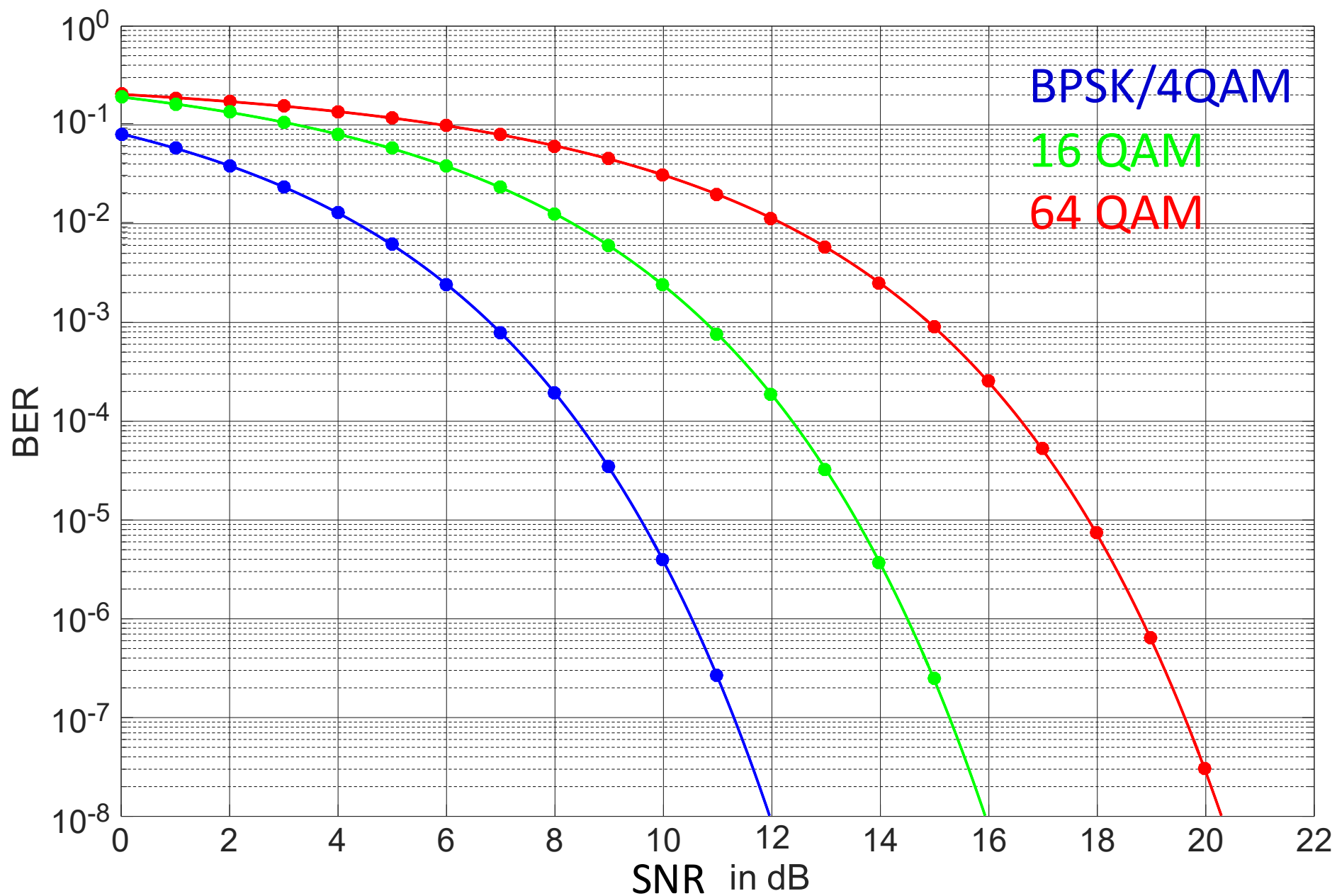
- Decide between BPSK symbols $s \in \{\pm A\}$ given y .
- **ML detector:** $\arg \max_m p(y | s_m) \rightarrow$ nearest-neighbor rule.
- **MAP detector:** $\arg \max_m p(y | s_m)P(s_m) \rightarrow$ **ML plus** prior on symbol probabilities.
- Binary antipodal example $y = s + \mathcal{N}(0, \sigma^2)$, $s \in \{\pm A\}$:
Decide $+A$ if

$$\underbrace{\ln \frac{p(y | +A)}{p(y | -A)}}_{(2Ay)/\sigma^2} \geq \ln \frac{P(-A)}{P(+A)} \Rightarrow y \geq \frac{\sigma^2}{2A} \ln \frac{P(-A)}{P(+A)}.$$

- With equal priors the threshold is 0 $\rightarrow$ converges to ML.

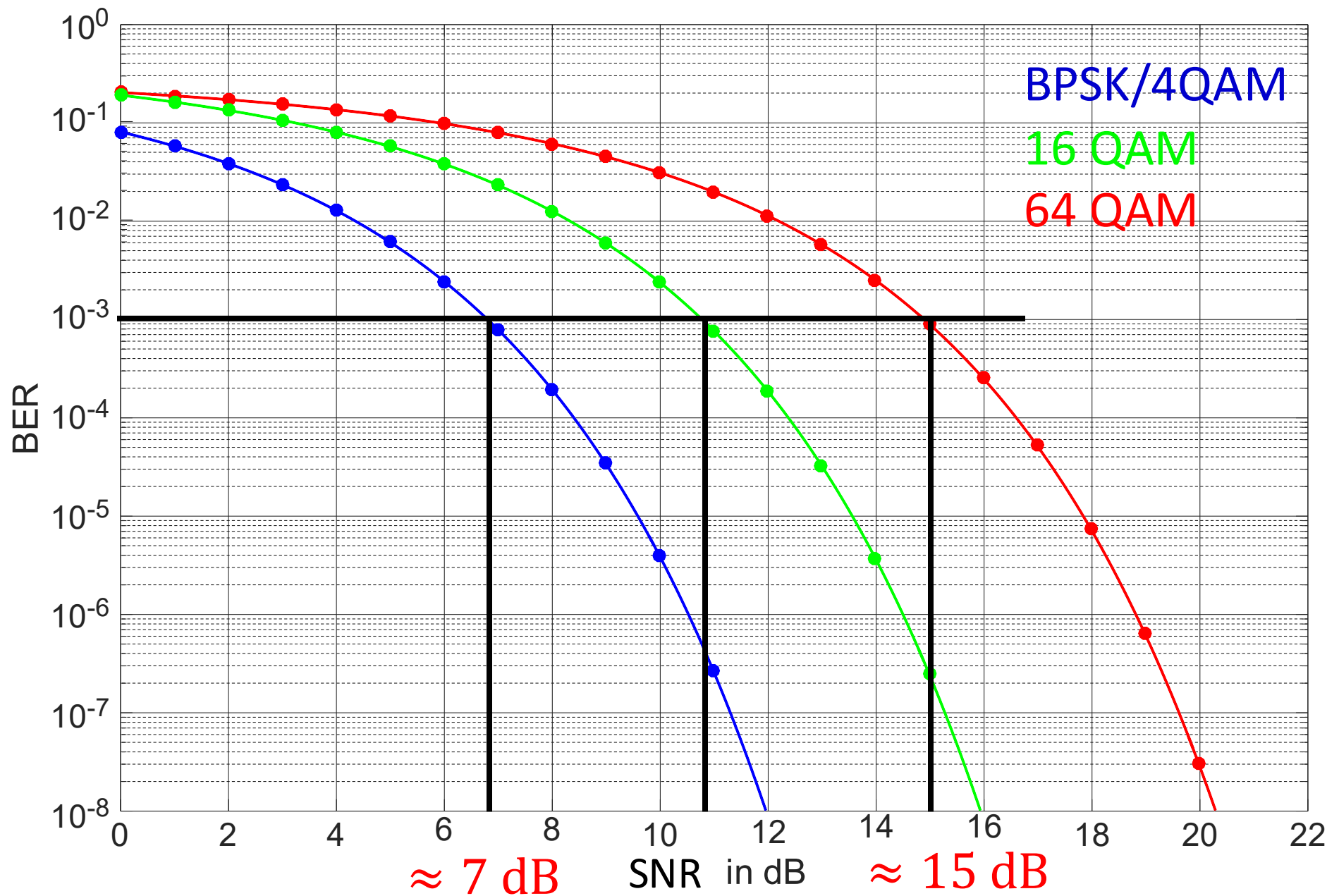
Decision Boundary

Bit Error Rate (BER)

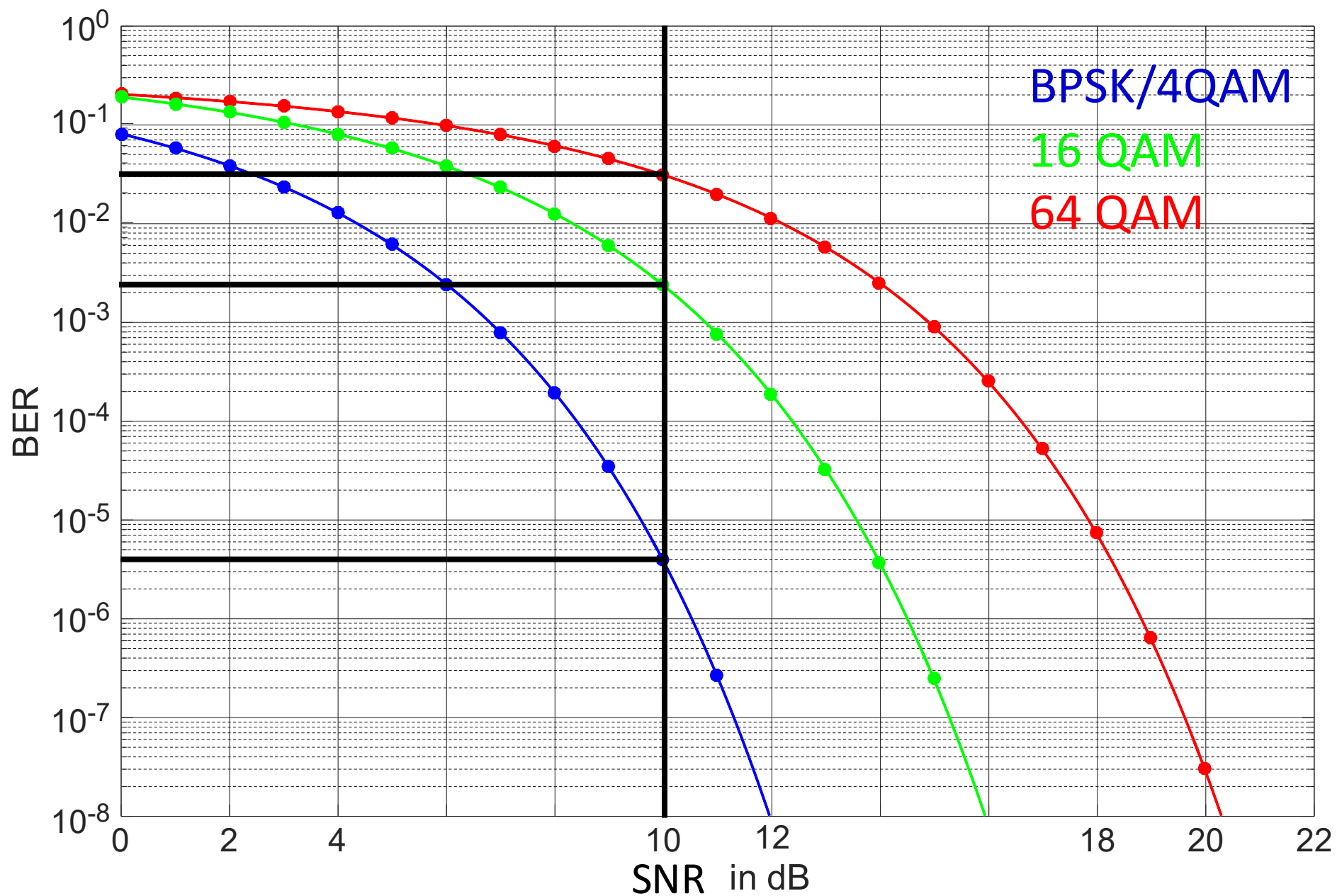


Bit Error Rate (BER)

Bit Error Rate (BER)

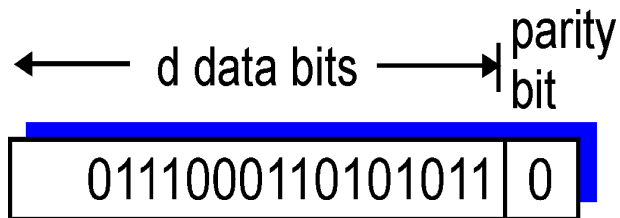


Bit Error Rate (BER)

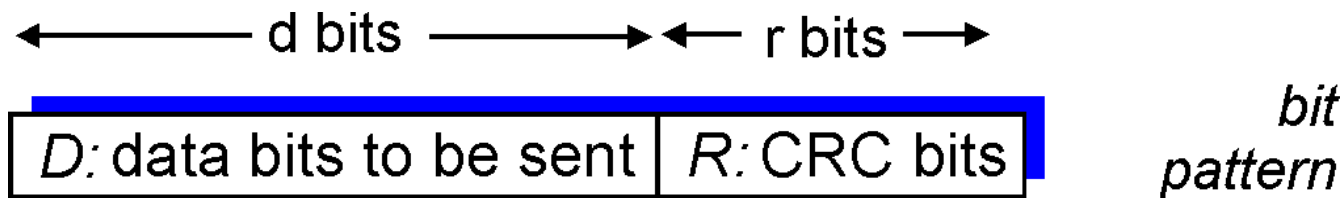


Error Detection

- Add Redundant bits in order to detect errors at receiver
- Checksums → Detect Errors
 - Parity Check



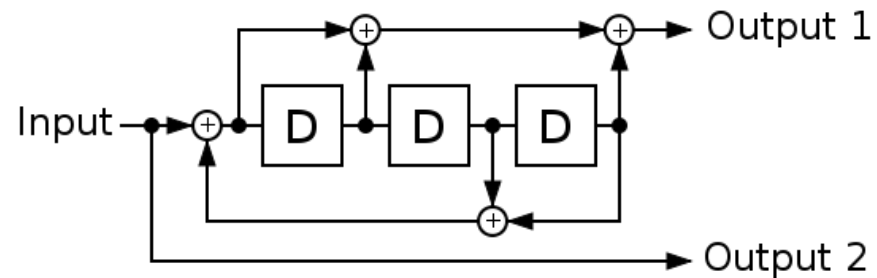
- CRC: Cyclic Redundancy Check



- If packet (frame, sequence of bits) fails the checksum test
 - Packet is dropped
 - Packet loss.

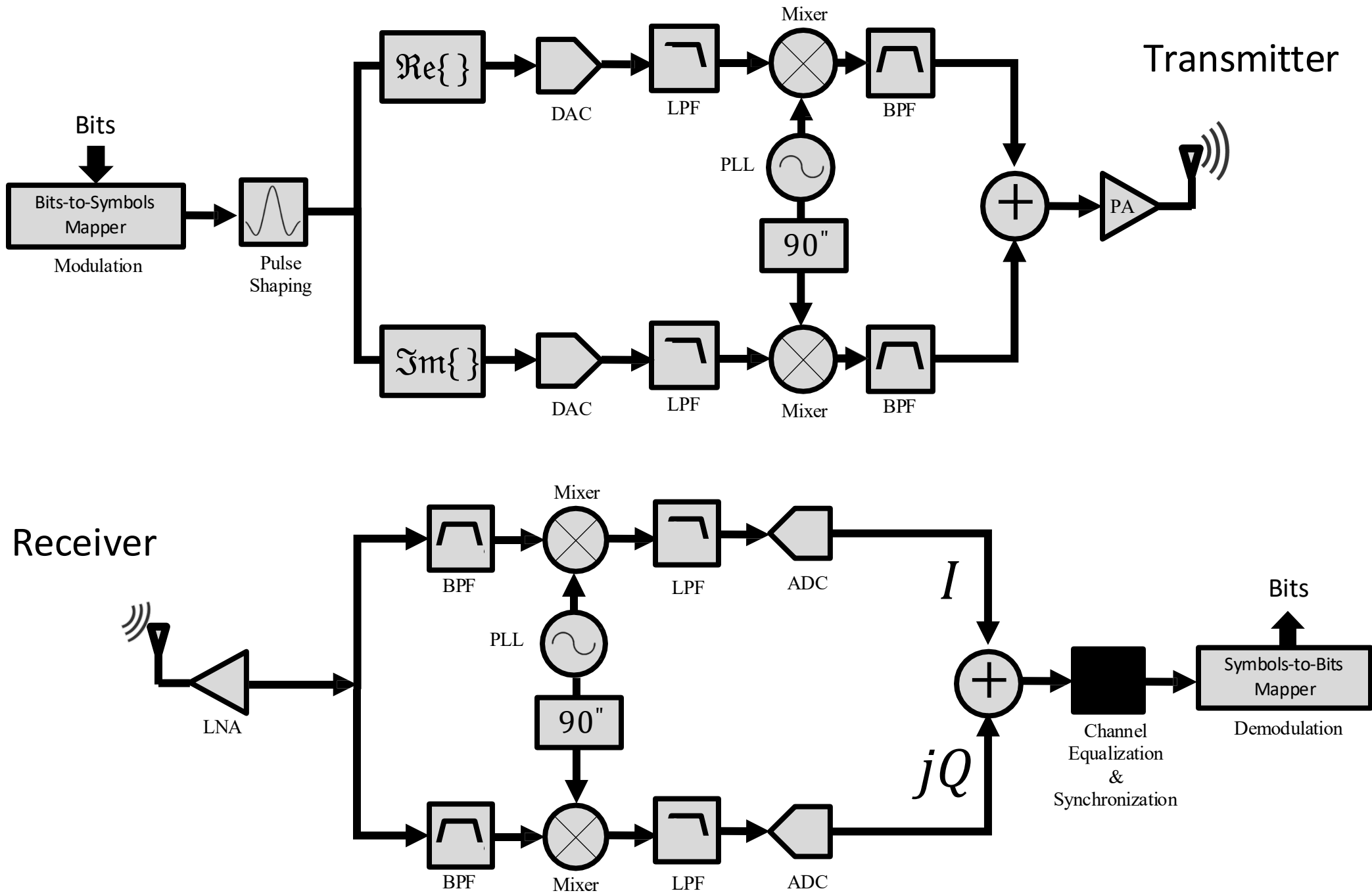
Coding: Forward Error Correction (FEC)

- Add Redundant bits in order to correct errors at receiver
- FEC: Forward Error Correction
 - Repetition Code
 - Convolutional codes
 - Reed Solomon codes
 - Turbo codes
 - Polar codes
 - LDPC codes



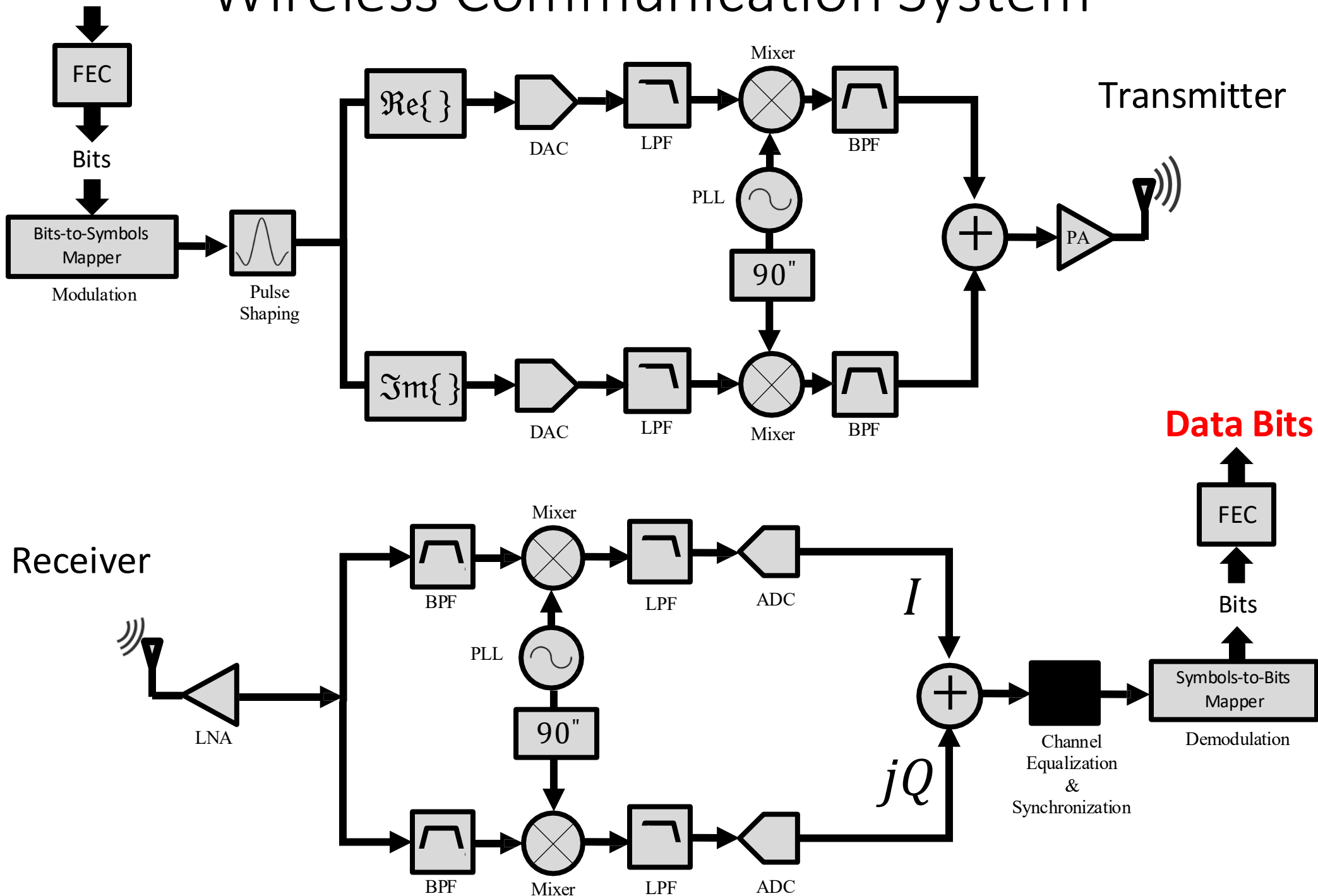
- Code Rate =
$$\frac{\# \text{ of data bits (uncoded)}}{\# \text{ of transmitted bits (coded)}}$$
 - E.g. coding rate 2/3, for every 2 data bits, I transmit 3 bits.

Wireless Communication System



Data Bits

Wireless Communication System



Wireless Data Rates

- *Data Rate*

- *Bandwidth: Samples/sec*
- *Modulation: Bits/sample*
- *Coding Rate: Data Bits/Coded Bits*

$$\text{Data Rate} = \text{Bandwidth} \times \text{Bits/sample} \times \text{Code Rate}$$

- *Capacity*

- *Maximum Achievable Data Rate*
- *Shannon Capacity Theorem:*

$$\text{Capacity} = \text{Bandwidth} \times \log_2(1 + \text{SNR})$$